FORMULA ELECTRIC BERKELEY

Aerodynamics

#100

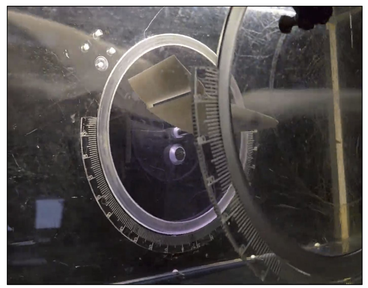
Aerodynamics Overview

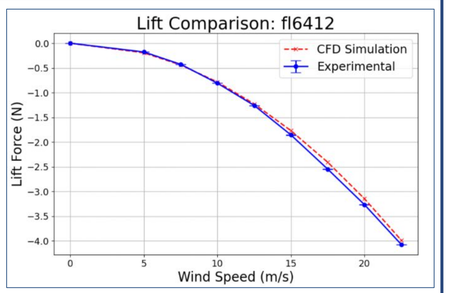
UC Berkeley

Overview

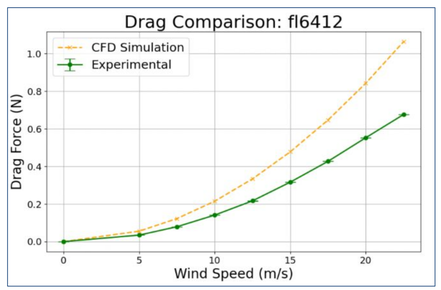
- **973 N** total downforce at **20 m/s**
- **2.72 Cl/Cd** ratio
- **3.97 Cl / 1.46 Cd** aero performance
- **21.3 kg** aero package mass
- **4.74 kg (18.2%)** aero package mass reduction
- Tunable **45–60% wheelbase CoP range**
- **5-element** Rear Wing **/ 3-element** Front Wing
- Foam-core wings reduced element mass by up to **61%**

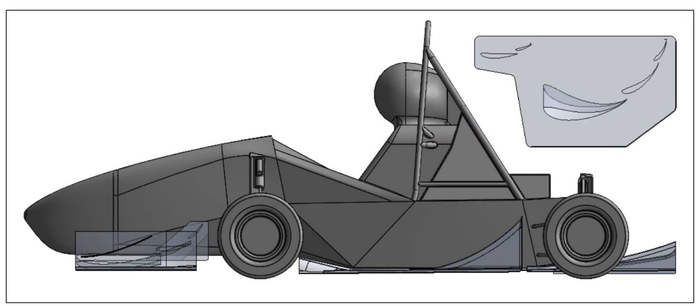
CFD and Turbulence Modeling





- CFD downforce prediction accurately matches wind tunnel validation data
- Boundary layer refinement and mesh independence studies for k-ω SST
- OptiSlang parametric optimization of chord, position, and AoA





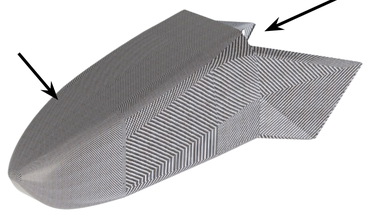
Full-car CFD driven aerodynamic development

Aerodynamic Development Areas

Bodywork

34% bodywork mass reduction

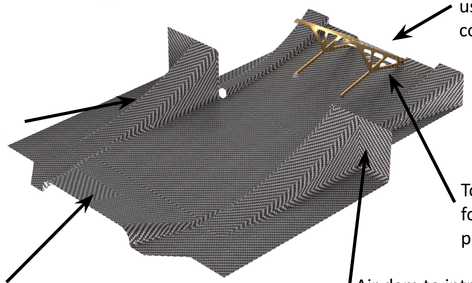
Integrated alignment slots and reduced mounting hardware



Undertray

7-channel diffuser using strakes for flow control

Side expansion tunnels produce local load at the center of the car



Towbar integration for diffuser performance

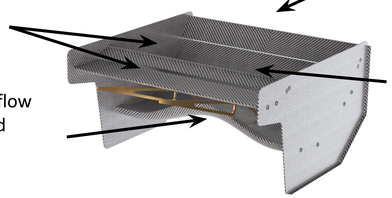
Inlet & diffuser permanently mounted

Air dam to introduce loss in front of rear wheel

Rear Wing

XPS foam-core wing construction

MSHD and NACA 6412 airfoil profiles

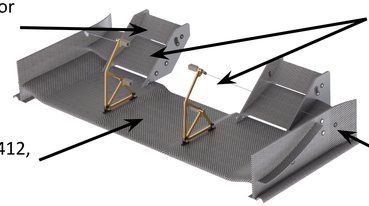


Added RE4 and RE5 to maximize downforce produced

Curved RE1 to prevent flow separation for improved aerodynamic efficiency

Front Wing

Adjustable FE3 AoA for CoP and outwash tuning

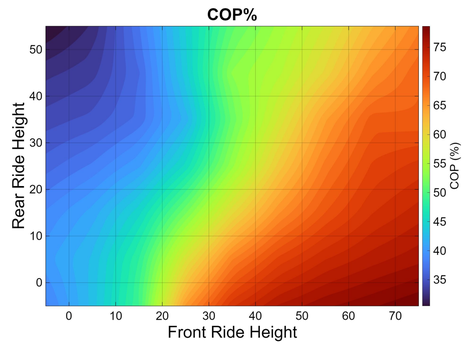


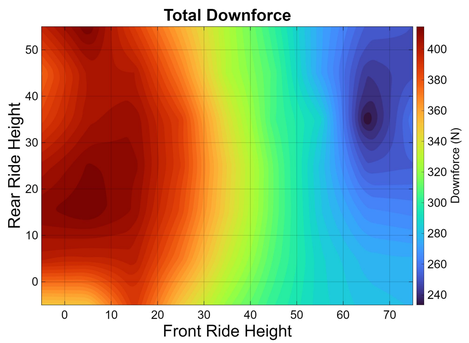
Pylons mounted outside of bodywork for easy accessibility

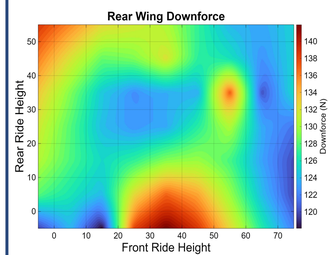
NACA 4412, 3405, 6412, 8412 airfoil profiles

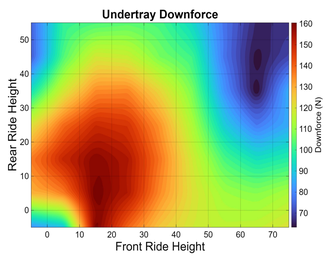
Outwashing endplate to drive wheel wake away from body

CFD Forces







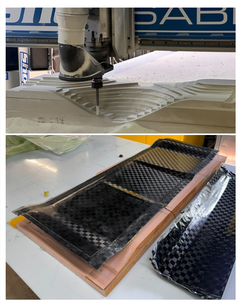


CFD Downforce Generation

Assembly	Downforce (N)	Drag (N)
Bodywork	5.6	143.4
Front Wing	313.8	39.8
Rear Wing	282	125.4
Undertray	383	50.6
Total Car	**973.2** **3.97 ClA**	**359.2** **1.46 CdA**

Manufacturing

CNC Routed PBLT Molds



Resin Infusion

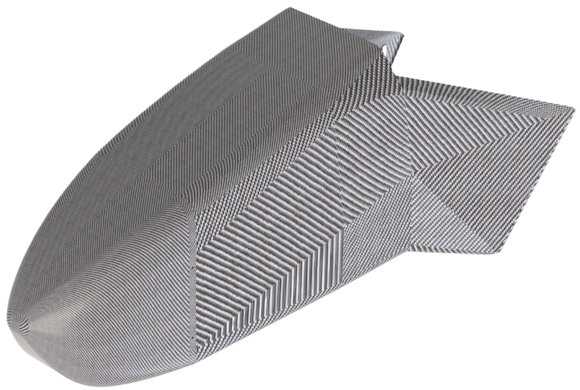
Carbon-wrapped XPS cores

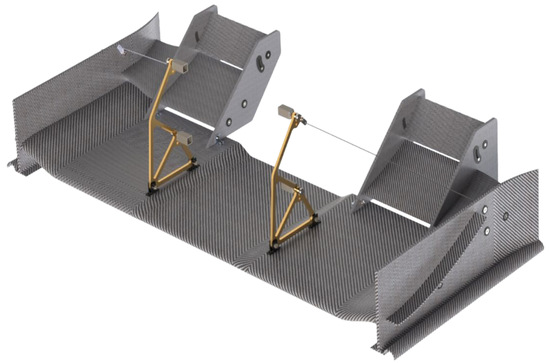
Aerodynamics Overview

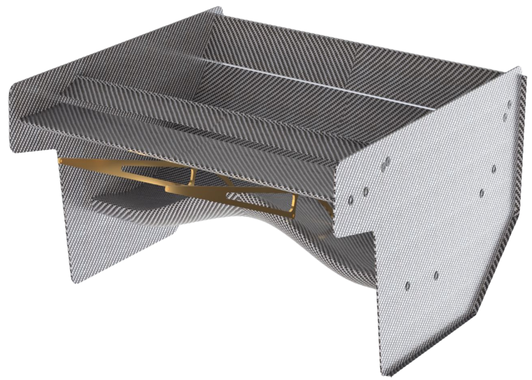
UC Berkeley

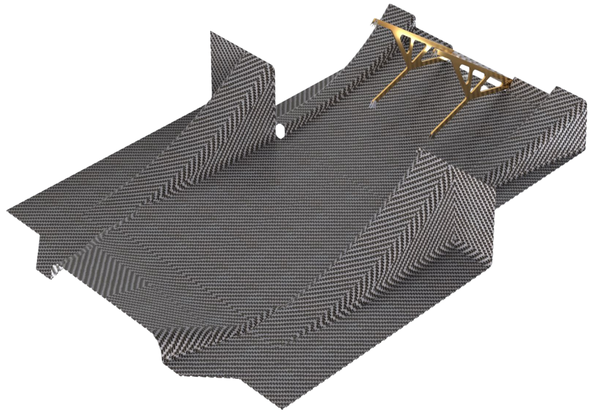
Aerodynamics Overview - Subsystems

UC Berkeley









Bodywork

- Four bodywork panels: nosecone, top clamshell, 2x side panels
- 34% weight reduction by using resin infusion and 3-ply 195 GSM 2x2 twill carbon fiber
- Simplified and improved mounting reliability: removed 6 3D-printed sidepanel mounts, M4 -> M3 bolts, and made slots for alignment

Front Wing

- Front Wing Profiles: NACA 4412, 3405, 6412, 8412
- Less cambered and thinner NACA profile to reduce separation and improve airflow for undertray
- Adjustable height and FE3 to tune CoP and upwash
- Curved endplates and C-channels generate outwash and vortices to suppress tire squirt and wheel wake for improved downstream flow

Rear Wing

- Rear Wing Profile: MSHD, NACA 6412
- Optimized RE1, 2, 3 and added RE4 and RE5 to maximize downforce produced
- New wing construction method on XPS foam core improved trailing edge thickness, and weight (up to -61% mass for comparable element)
- Curved RE1 to prevent flow separation for improved aerodynamic efficiency

Undertray

- 7-channel diffuser to control inwash.
- Side expansion tunnels produce local load at the center of the car
- 4-piece undertray for easy installation
- Undertray inlet & diffuser permanently mounted
- Tow bar integration for diffuser outlet performance
- Laminated copper mesh grounding

Aerodynamics Overview - Structures

UC Berkeley

Structural Optimization:

- Used topology optimization to reduce mass

Load case validation:

- Increased accuracy of FEA through integrating CFD load cases via fluid-structure interaction (downforce surface distribution)
- Mesh convergence studies to improve simulation fidelity at bolt connections

Composite Simulations

- ANSYS OptiSlang used to calibrate material model parameters for various stackups.
- Utilized ANSYS ACP and materials testing data to model carbon-fiber wrapped foam element wings and perform full wing simulations.

Towbar + Undertray Pylons

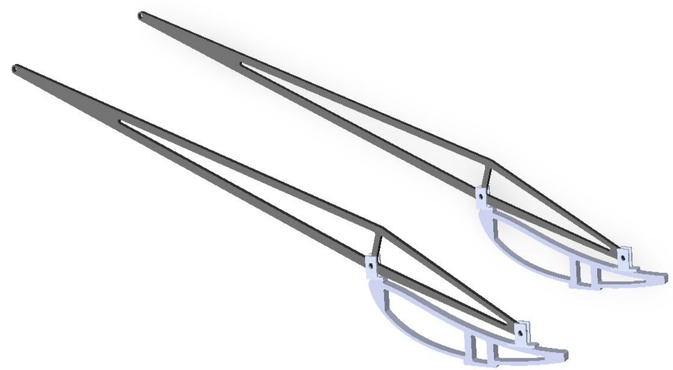
- **0.712 kg.** mass reduction, topology optimized
- **FOS:** 3.026 (Yield), 2.1 (Buckling)

Front Wing Mounting: Mounting to car only through FE1, removing need for mounting ribs in FE2.

- **Pylons: 0.148 kg.** 41%mass reduction (250g to 148g), topology optimized, **FOS:** 3.6
- **Mounting U-Channels : FOS:** 2.4
- **Pylon Mounting Ribs:** Removed internal truss/spar system after optimization. **FOS:** 3.2

Rear Wing Mounting: Mounted only to RE1, removing the need for mounting ribs in RE2.

- **Swanneck:**
  - **FOS:** 2.4
- **Mounting Ribs:**
  - **FOS:** 6.5



SN5 Guiding Principles - L1 → L2 Goals

UC Berkeley

1. **Safe and reliable**
   - Aerostructures *shall have* FOS ~1.5 at part and assembly level, 2.5 for composites at part level
2. **Risk mitigation**
   - SN5 *shall* have only 4 aerodynamic subsystems
   - SN5 *shall* not have active aerodynamic components
   - Aerodynamics package *shall be* mountable and removable within 5 minutes
3. **Inarguably rule compliant**
   - SN5 front and rear wing *shall* have 25-50 mm tolerance to clear bounding box rules check without issue
4. **First principles + rigorous substantiation**
   - Composite simulations and material testing *shall* support our design robustness
5. **Tested and validated**
   - SN5 CFD *shall* be validated with wind tunnel testing and dynamic testing plan

L2 Requirements

UC Berkeley

SN5's aerodynamics package shall:

- **Increase dynamics performance:** Generate >950N overall downforce: increase effective normal force of 240kg car by ~40%, resulting in simulated **-1.353s** autocross time, **-.197s** skidpad time, **-0.127s** acceleration time, and **-2.089s** endurance lap time.
- **Improve efficiency**: >2.4 Cl/Cd and 4.5 CIA, where lap time gained from downforce outweighs endurance energy consumption penalty (~7kW @ 341 N drag, 20 m/s).
- **CoP**: Between 45-60% of wheelbase, avoid excessive oversteer/understeer bias.
- **Structural: >=**1.5 FOS at part and assembly level, >=10% mass down per component. Safe & reliable, rule-compliant.
- Develop SOP for CF panels/molds/R&D for future manufacturing & mass reduction.

Key Highlights

UC Berkeley

- Started design phase early; prioritized **full-car sims** rather than isolated surface analysis
- Implemented new foam-core airfoil construction method to reduce weight up to **61%**, maintain stiffness, and cut **6+ hrs** of machining time per element
- Front wing and undertray designed to generate outwash for improved downstream effects
- **2.72 Cl/Cd ratio achieved**
- **CoP: 45-60%** COP aim for 50% as a neutral starting point avoiding aggressive over/understeer and include adjustability range for driver preference and selecting dynamics ranges
- Focusing on **overall team performance, reliability** and subteam integration

Why Aerodynamics

UC Berkeley

1. **Improves dynamic event performance** by **increasing tire grip** — reduces lap times through **enhanced traction, braking and cornering**.
2. Supports **long-term mass reduction** goals through **continued composites R&D** across the vehicle.
3. Enables **future aerodynamic development** once vehicle reliability is established and additional performance gains are needed for **top 10 dynamics event placement**.
4. **Expands team knowledge** and technical skills through hands-on experience in composites, aerodynamics, design, and testing.

Aerodynamic Values @ 20m/s

UC Berkeley

- Reference Area: 1.08 m^2
- ClA: -3.97 (973N)
- CdA: 1.46 (359N)
- ClA/CdA: 2.72
- Aero Balance (CoP): Front: 55%, Rear: 45% in current configuration

CFD Values Overview

UC Berkeley

	SN4 Mass (kg)	SN5 Mass (kg)	% Change
Bodywork	3.169	2.370	-25.2
Front Wing	4.600	3.902	-15.2
Rear Wing	6.900	5.861	-15.1
Undertray	11.4	9.200	-19.3

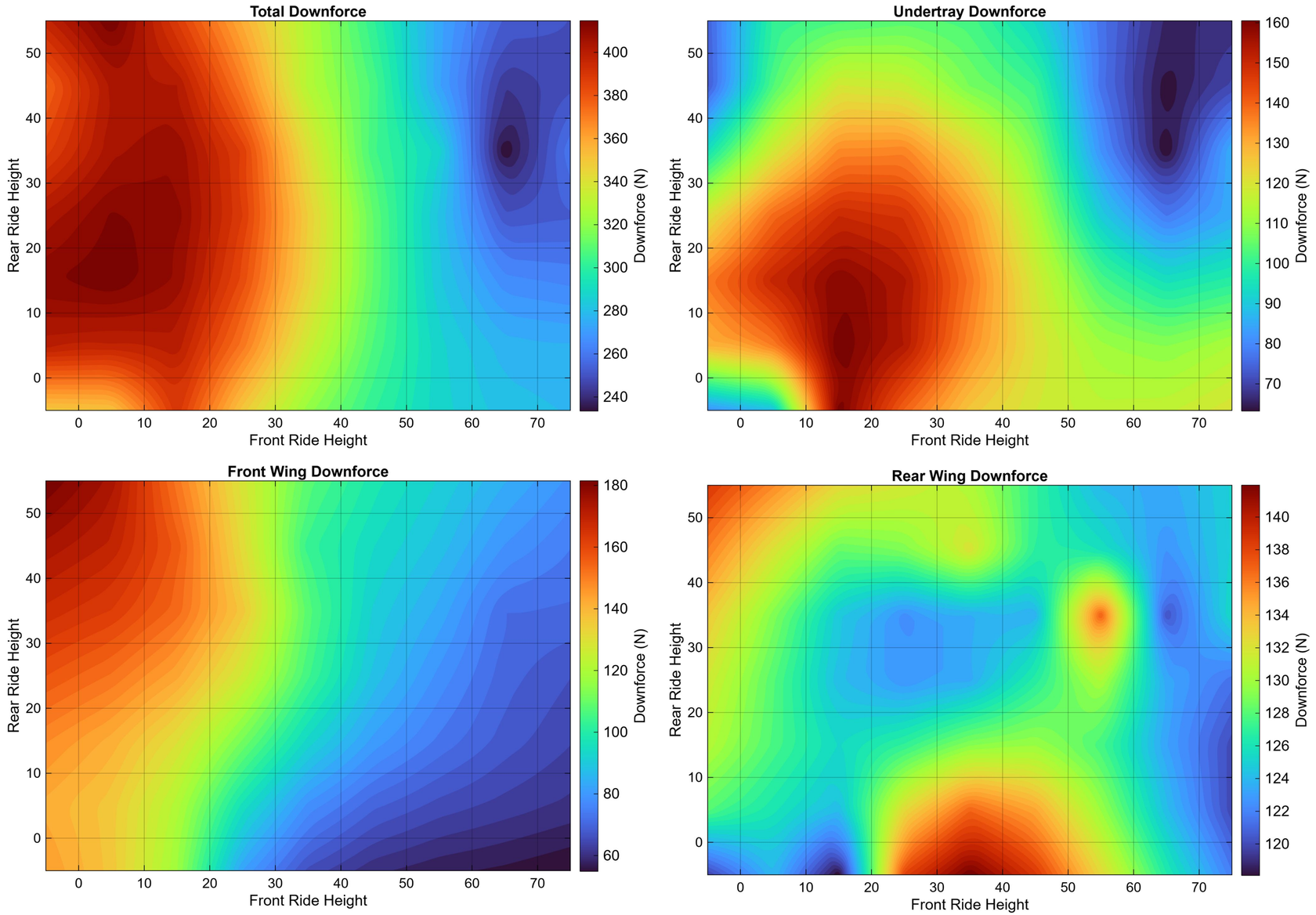
	SN4		SN5		% Change	
Assembly	**Downforce (N)**	**Drag (N)**	**Downforce (N)**	**Drag (N)**	**Downforce**	**Drag**
Bodywork	-38	93	5.6	143.4	+114.4%	+53.3%
Front Wing	345	24	313.8	39.8	-9.00%	+65.8%
Rear Wing	347	149	282	125.4	-18.7%	-15.8%
Undertray	330	48	383	50.6	+16.1%	-5.4%
Total	984 4.16 ClA	314 1.55 CdA	973.2 3.97 ClA	359.2 1.46 CdA	-1.1%	+14.39%

Aero Map

UC Berkeley

Lift vs Front vs Rear Ride Height

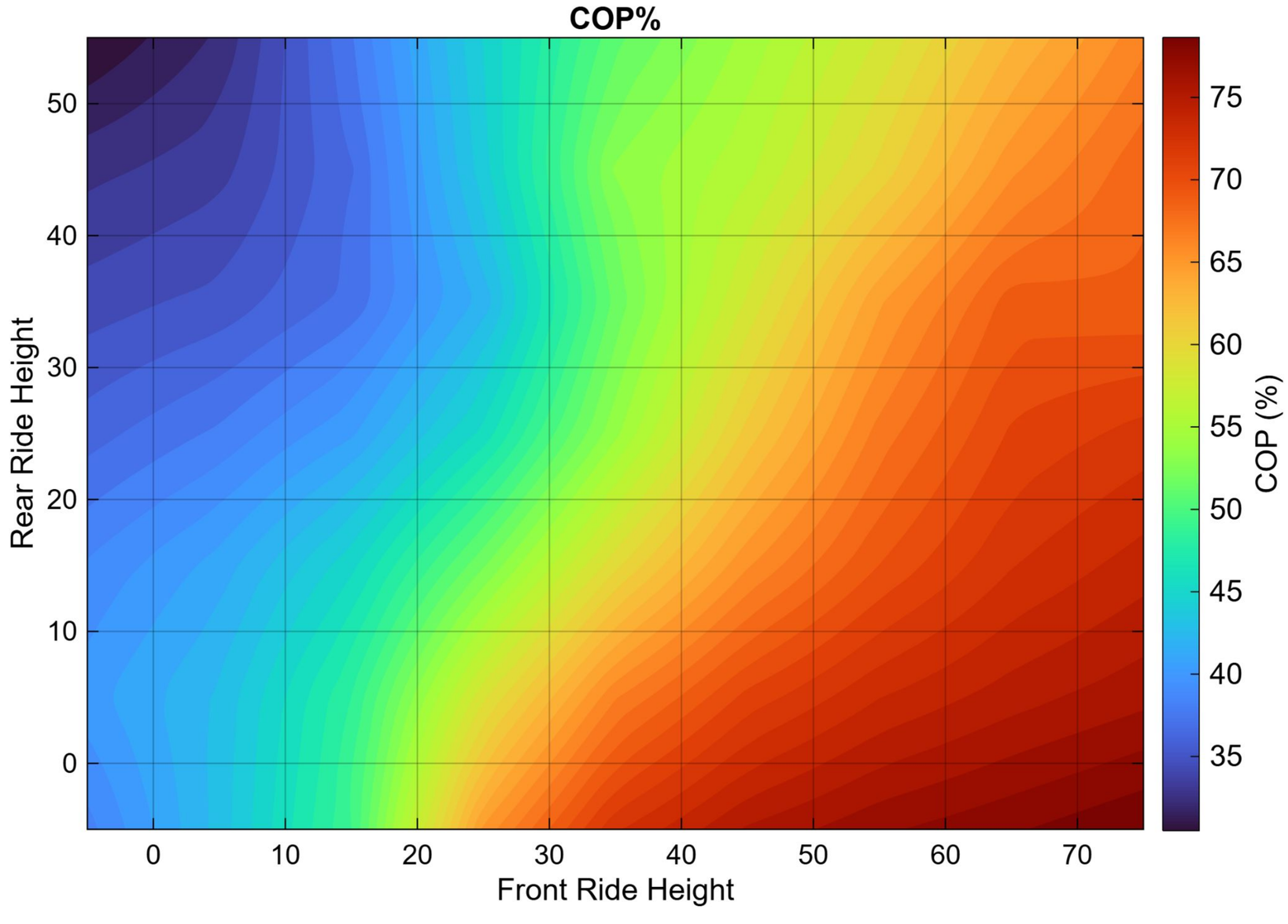
Total, Undertray, Front Wing, Rear Wing



Aero Map - COP

UC Berkeley

COP% vs Front vs Rear Ride Height



Development Philosophy & Challenges

UC Berkeley

SN5 Development Philosophy

- Prioritize overall vehicle performance over maximizing aerodynamic load.
- Focus on manufacturing quality, structural reliability, and stiffness-to-weight improvements.
- Emphasize mass reduction, which provides greater overall vehicle performance gains than increasing raw downforce.
- Focus on undertray development; SN5 is the team's second undertray and third complete aero package.

Challenges

- RTML and driver headrest disrupt rear wing flow.
- Reduced ground clearance and significantly different chassis geometry compared to previous vehicles.
- Limited aero data due to significant design changes.

Subteam Interaction

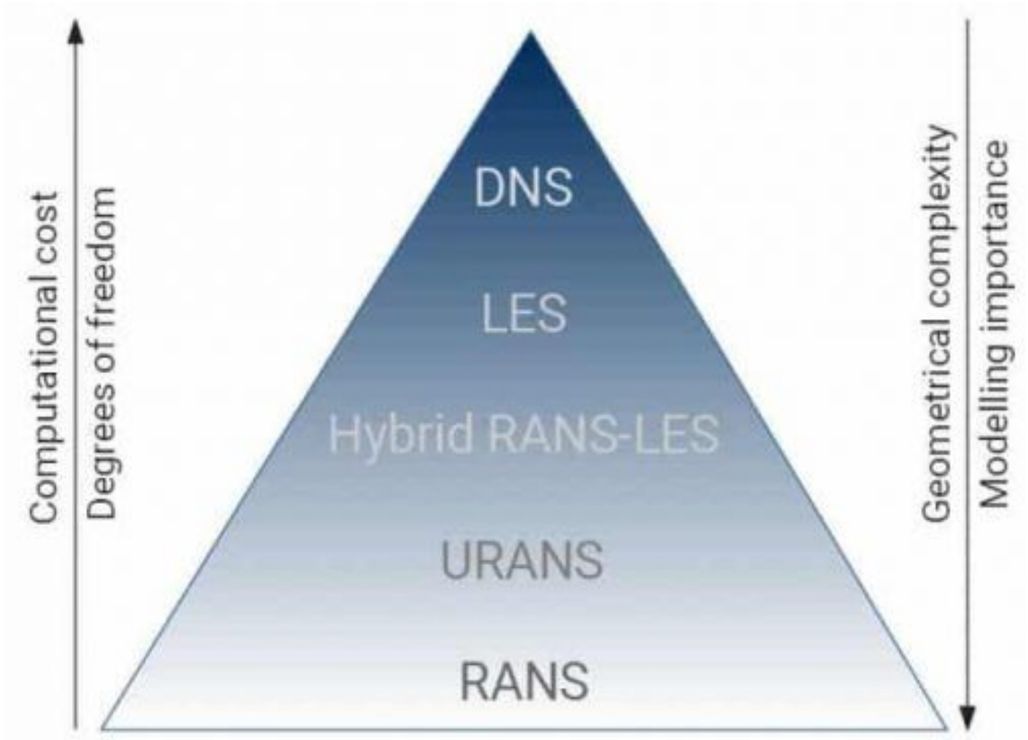
UC Berkeley

- Improves dynamic event performance through increased tire grip and reduced lap times.
- CFD targets set from vehicle-level simulations:
  - Determine maximum allowable drag from endurance energy budget and accumulator sizing.
  - Maximize **Cl/Cd** within the drag constraint.
- Works with Composites to develop lightweight, stiff aerodynamic structures.
- Collaborated with Driver Interface on a new ergonomic jig that lowered the driver headrest position, reducing an airflow constraint on rear wing development.

CFD - Solver & Model

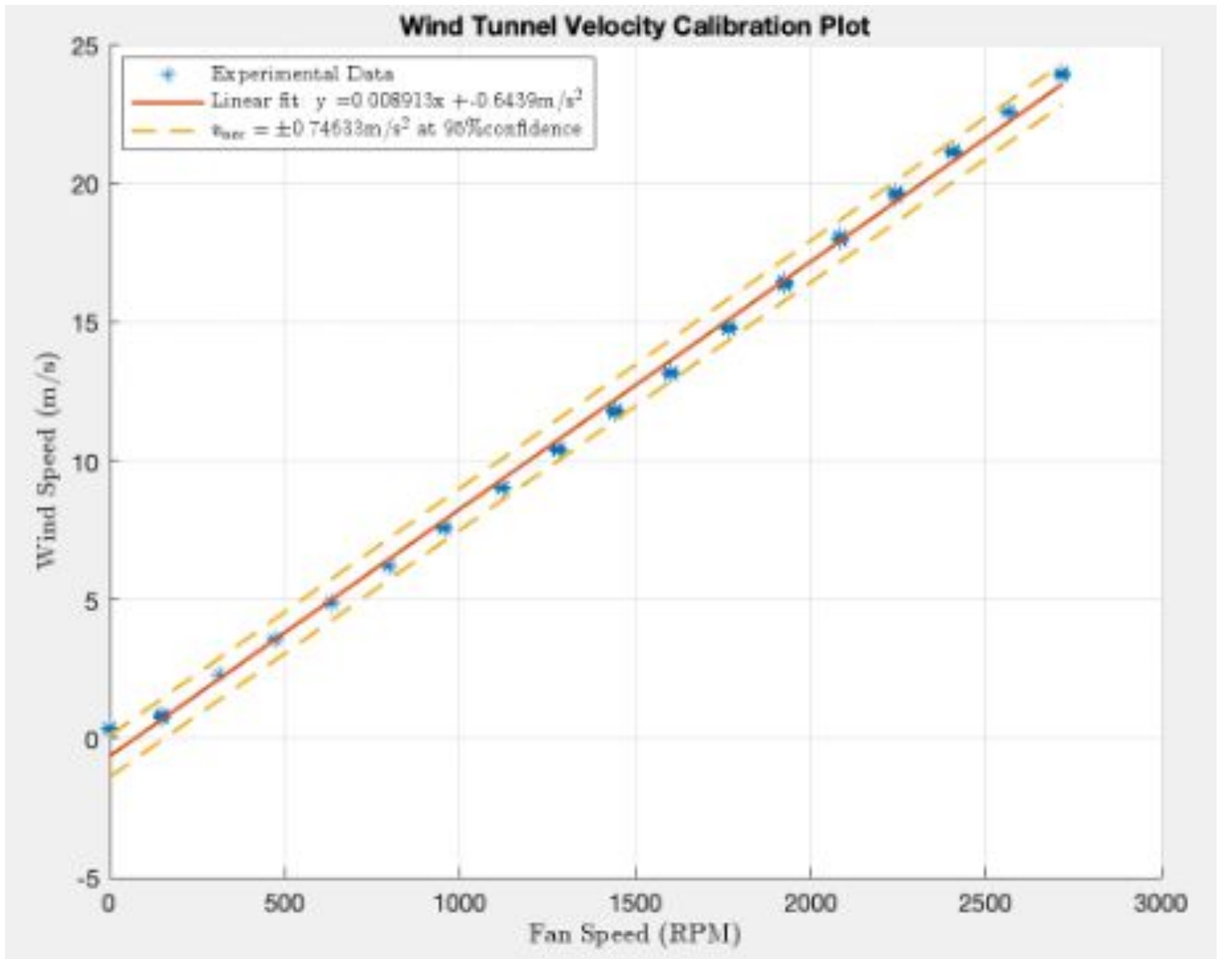
UC Berkeley

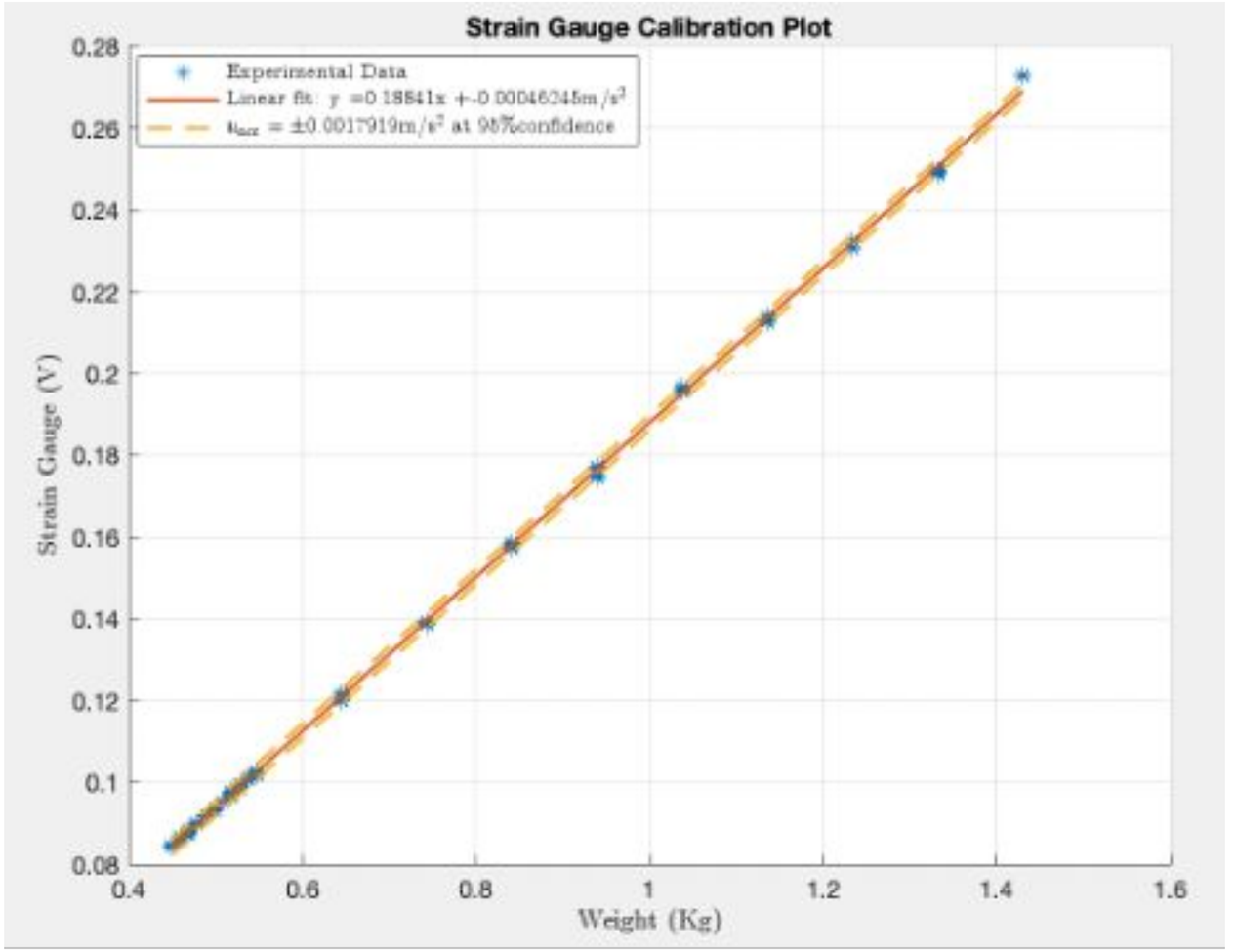
- Simulations run at 20 m/s
- RANS for better computational efficiency
- ANSYS Fluent
- k-ω SST turbulence model
  - Curvature correction enabled



Wind Tunnel Validation - 1 Element

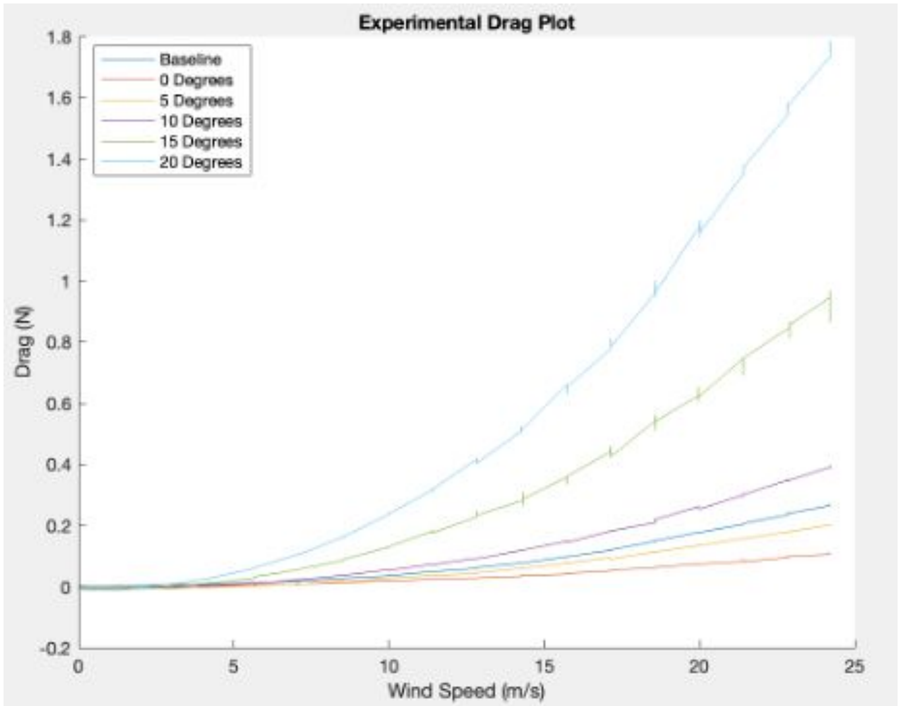
UC Berkeley

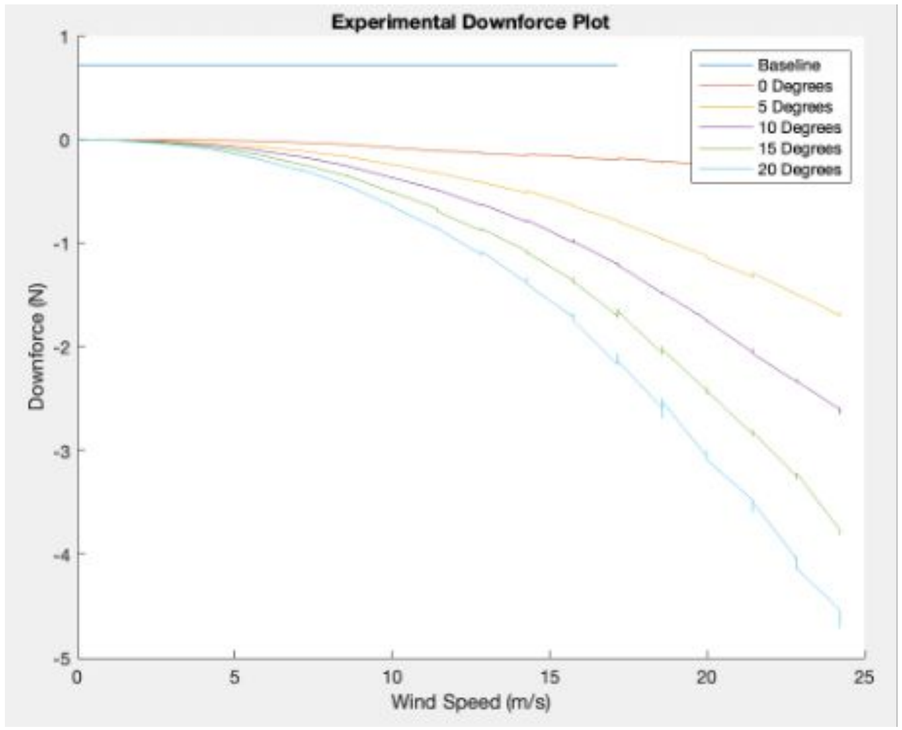


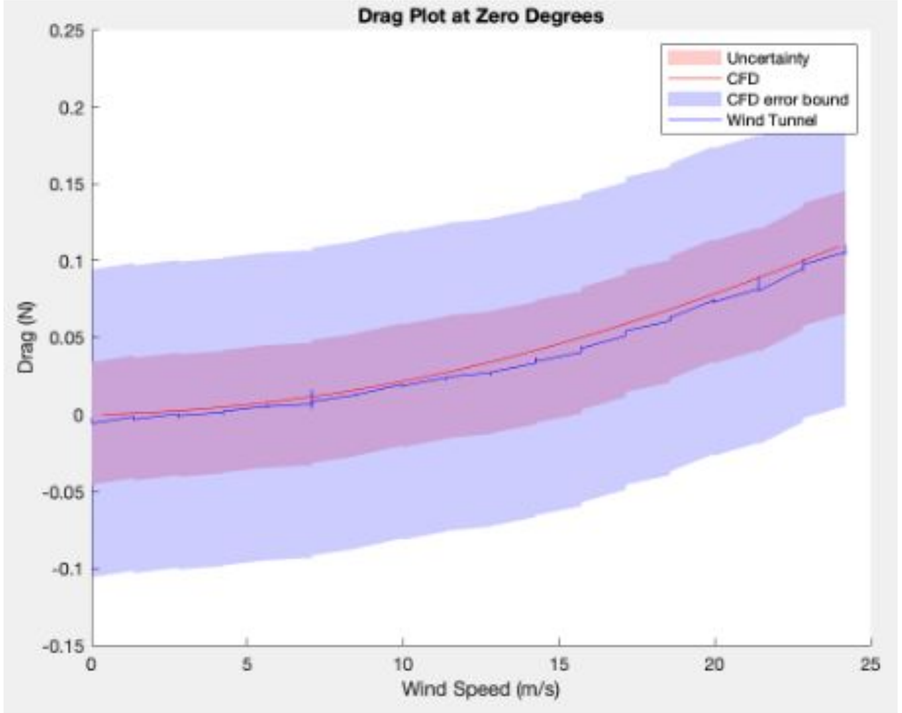


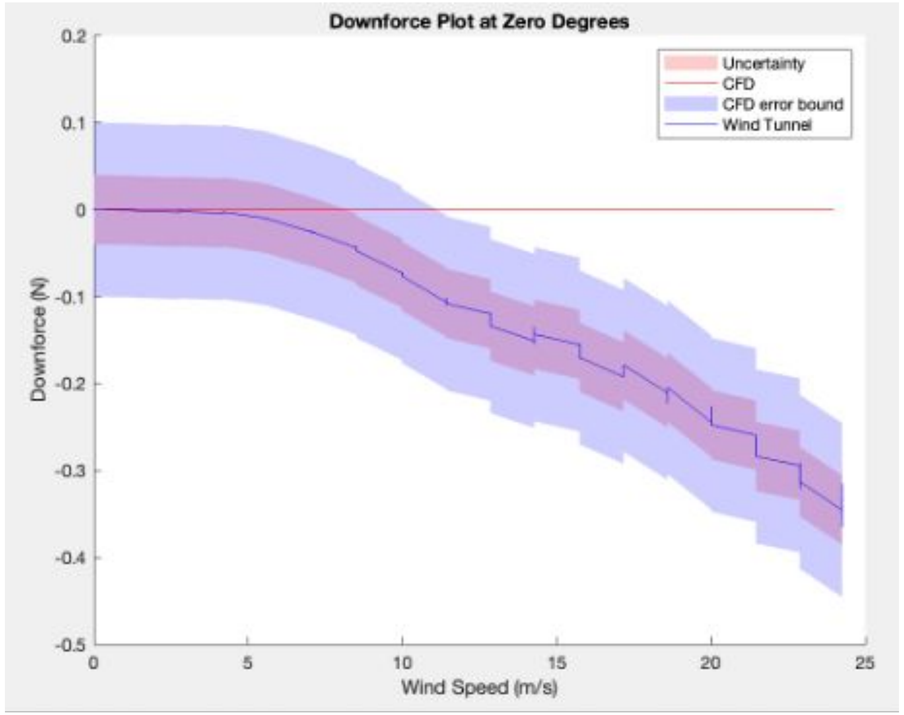
Wind Tunnel Validation - 1 Element

UC Berkeley



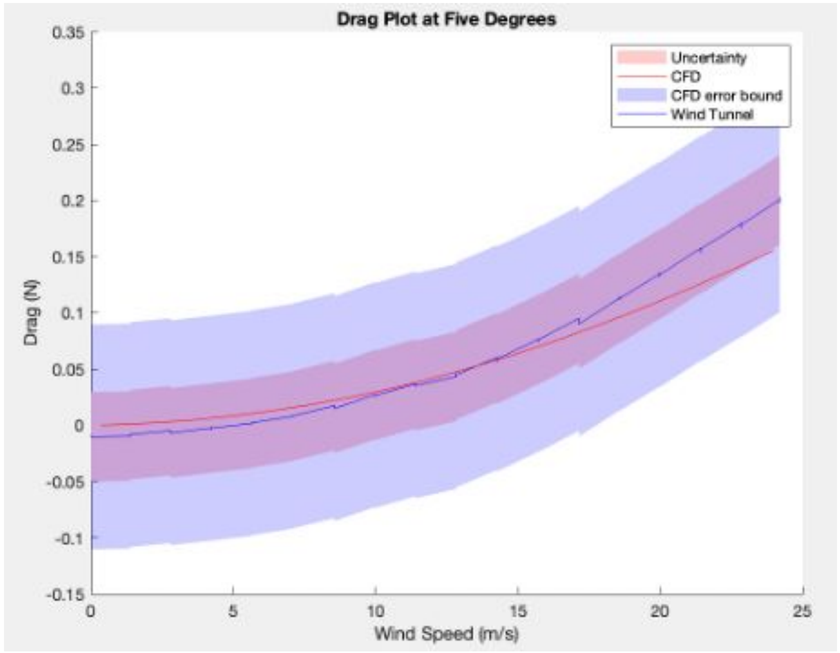


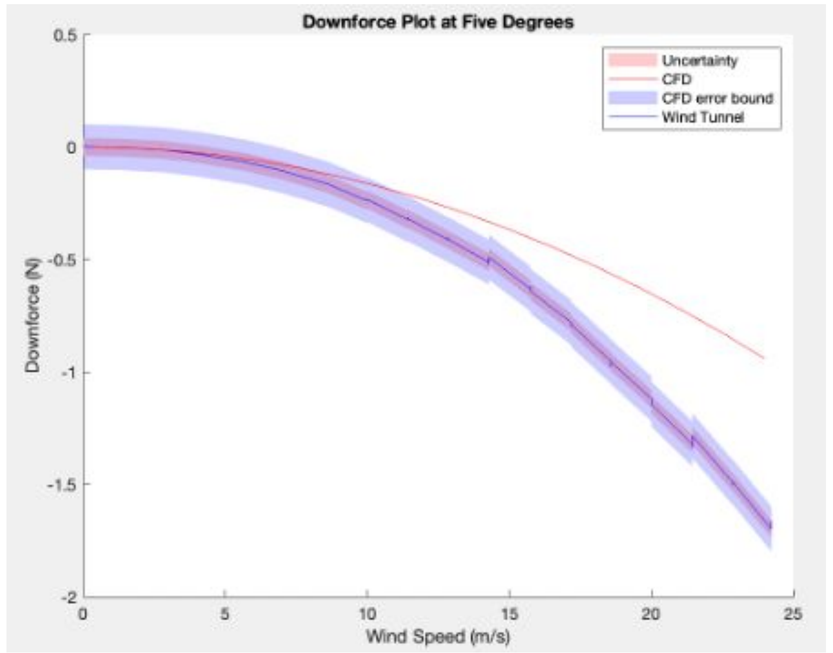


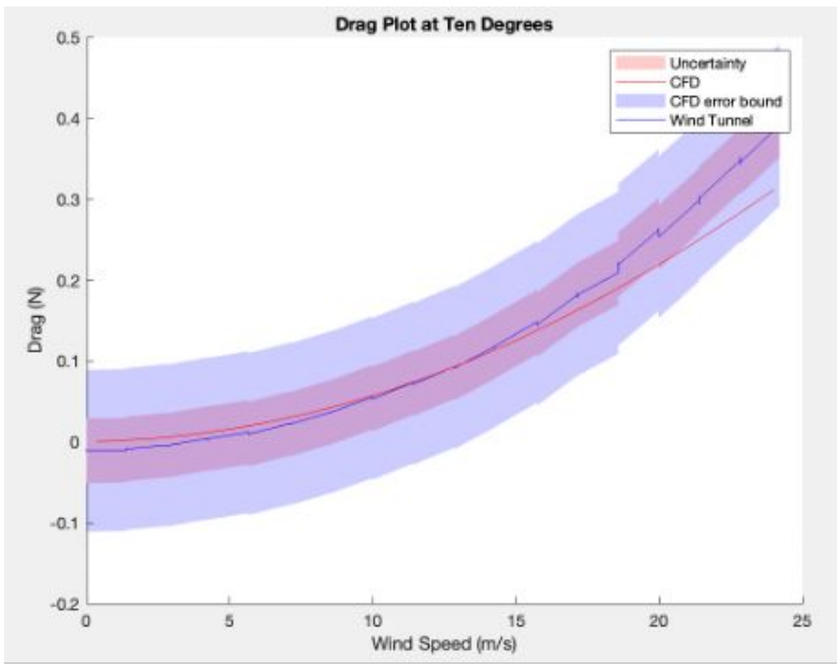


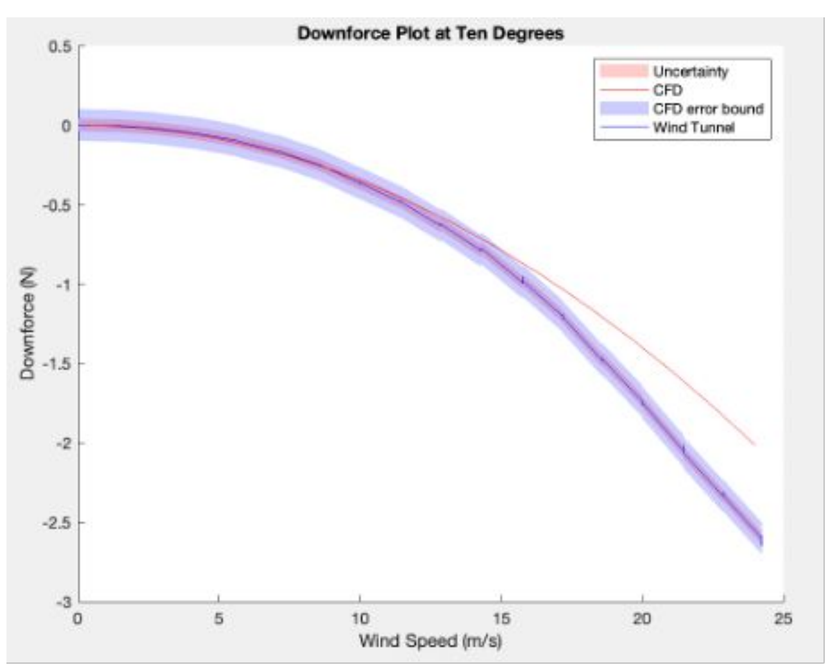
Wind Tunnel Validation - 1 Element

UC Berkeley



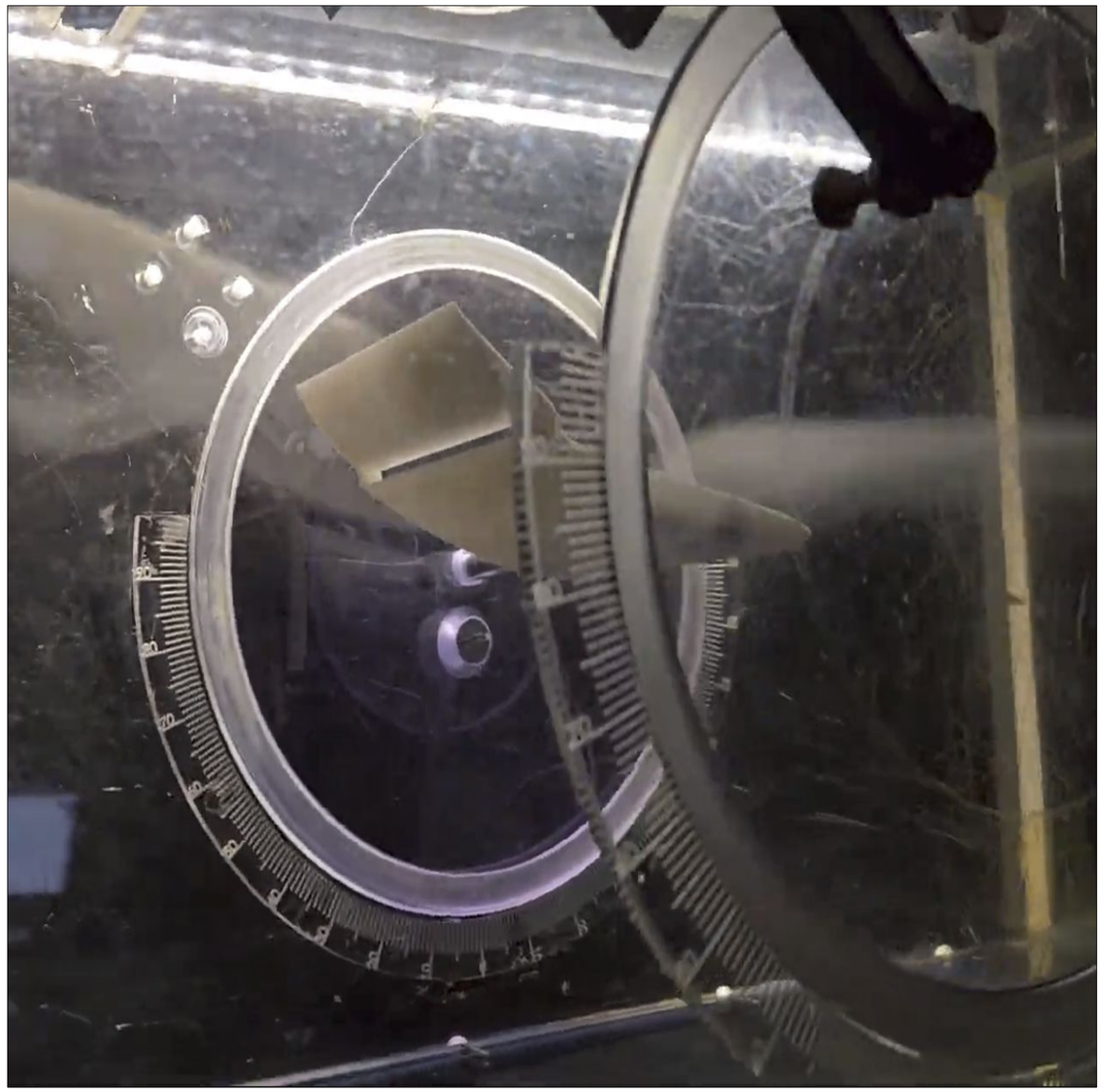


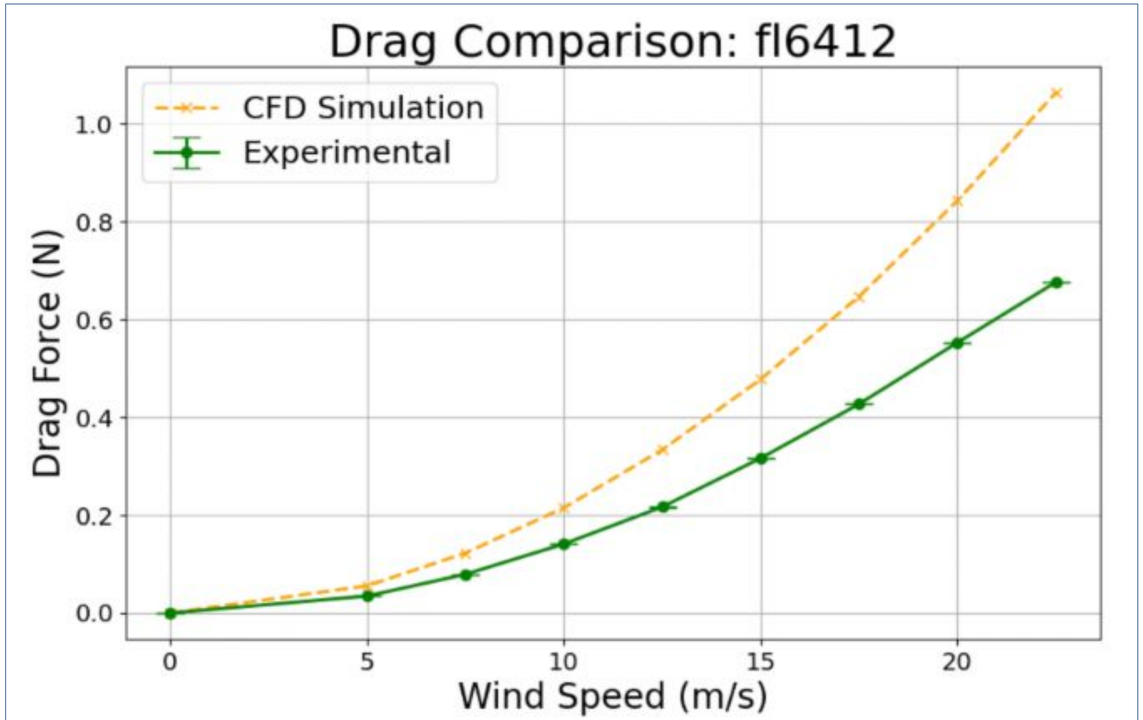




Wind Tunnel Validation - 2 Element

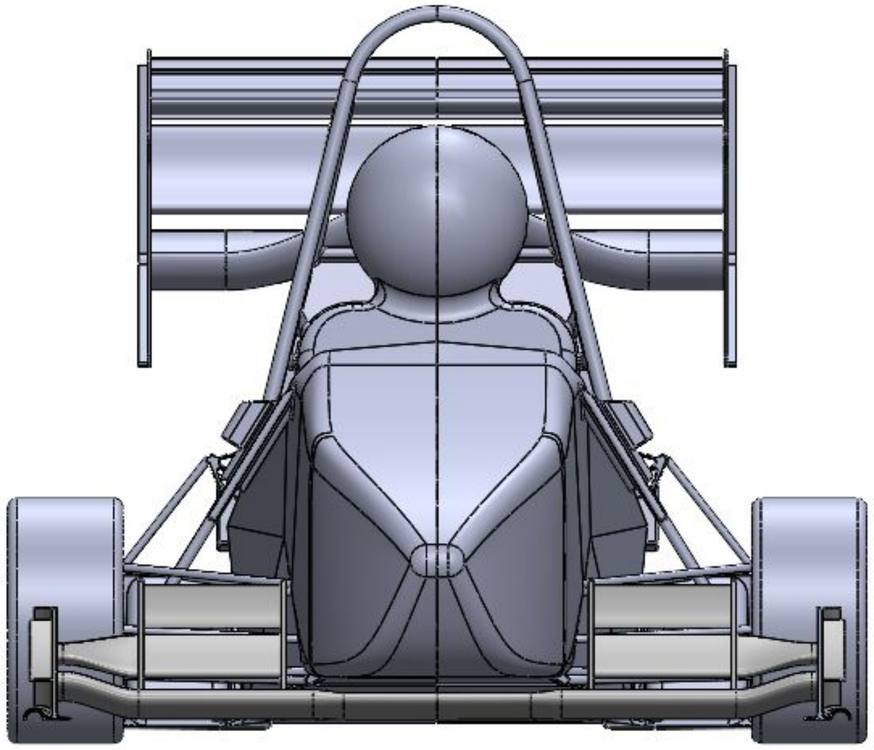
UC Berkeley

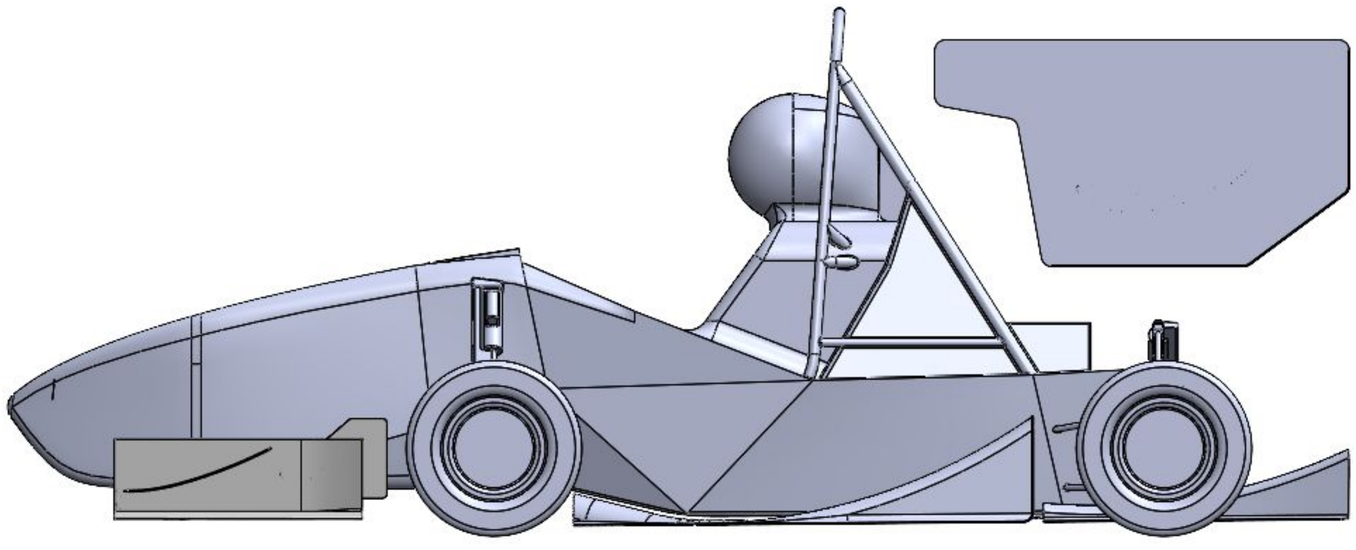


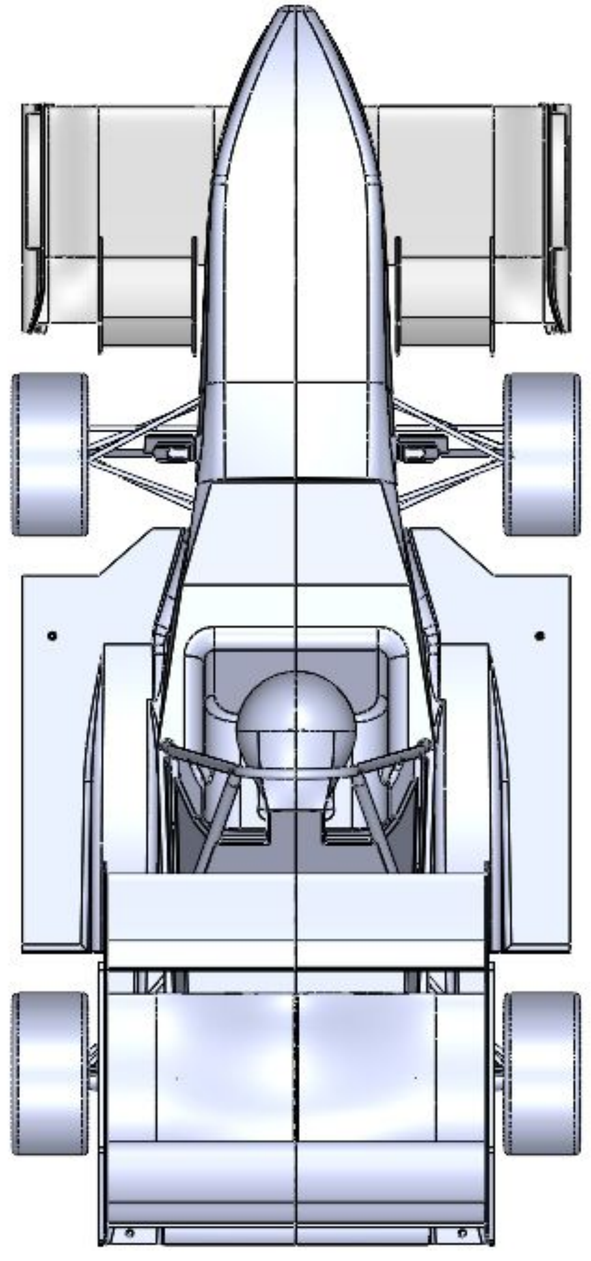


CFD - Simulation rig

UC Berkeley

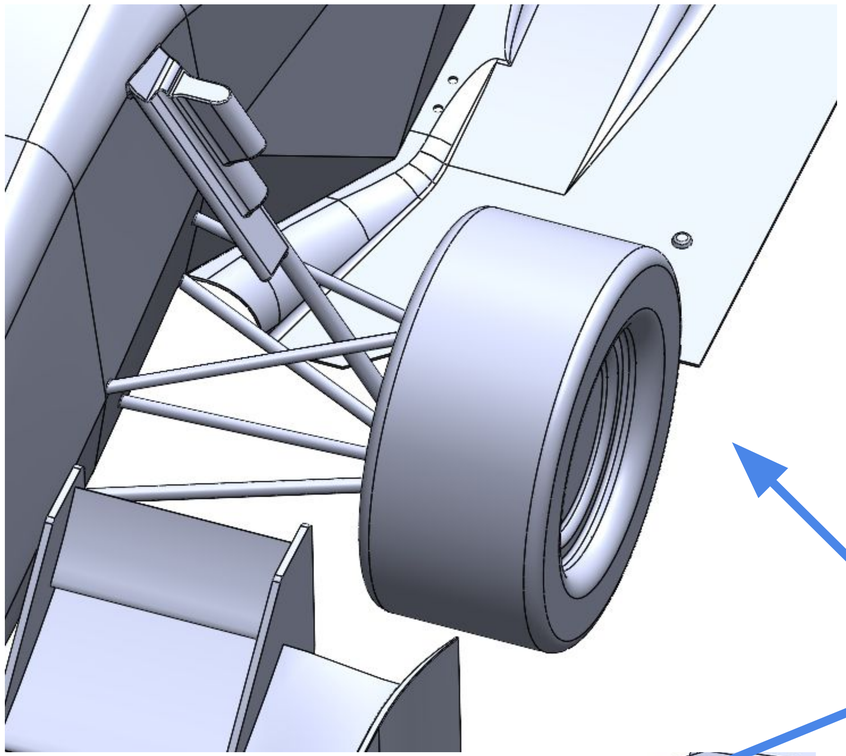




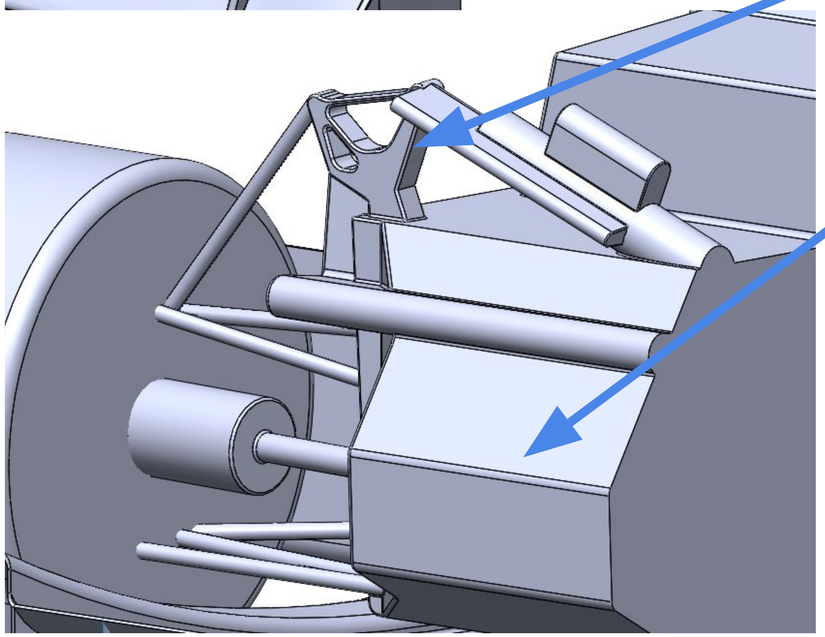


CFD - Simulation Rig

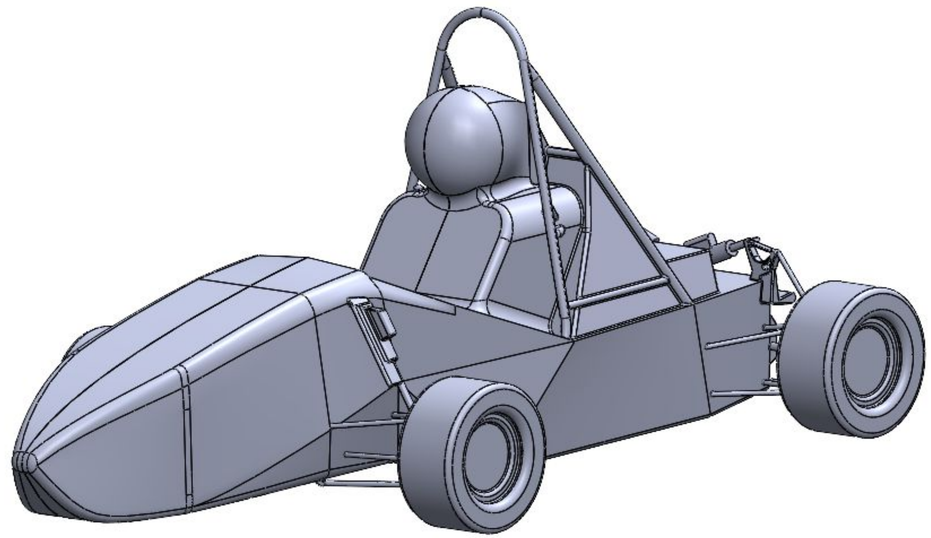
UC Berkeley



Simplified suspension

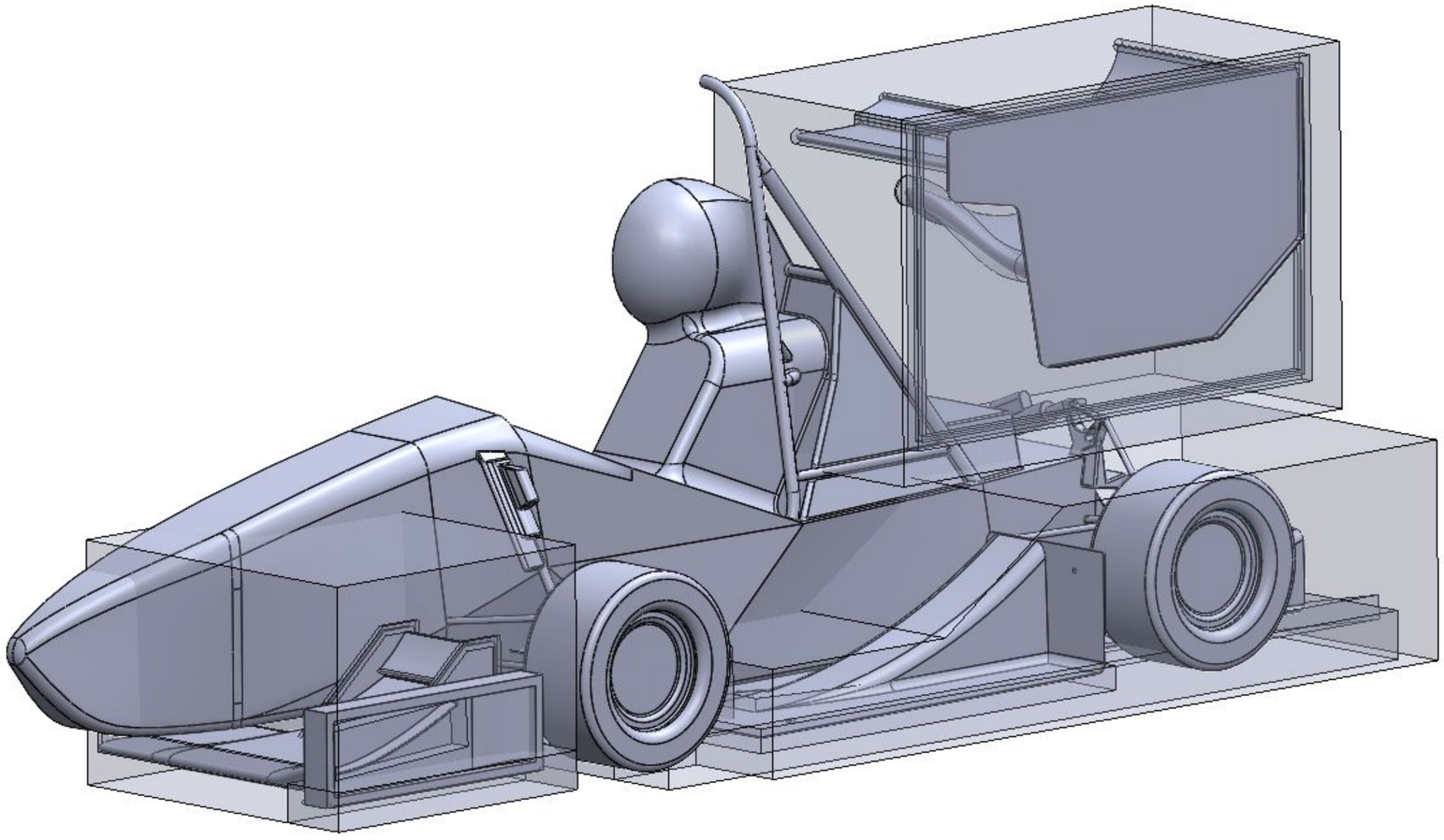


Simplified chassis, bodywork, and driver model



CFD - Simulation Rig

UC Berkeley



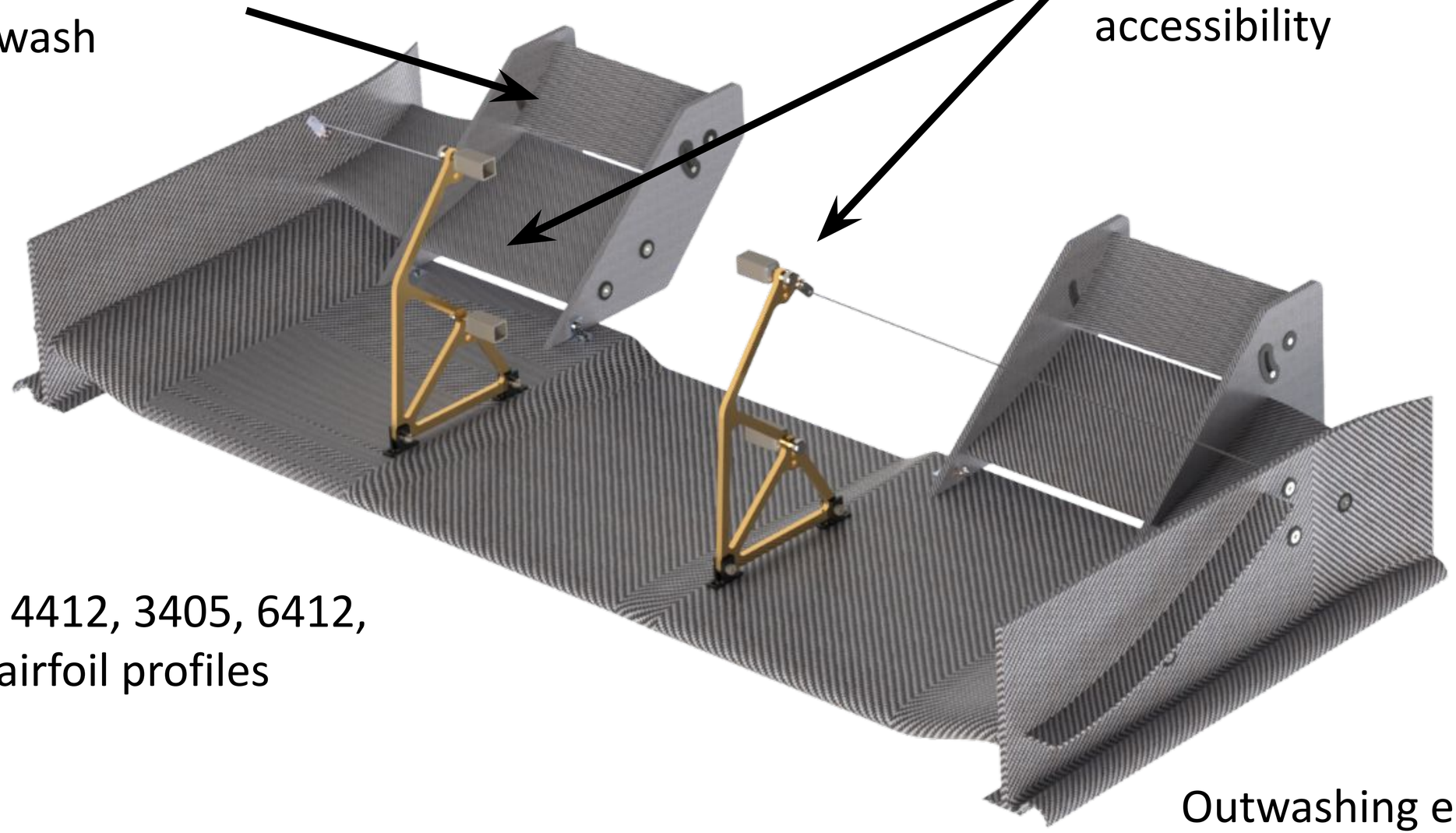
- Bodies of influence on leading and trailing edges to capture complex curvature and fine geometry

Front Wing

UC Berkeley

Adjustable FE3 AoA for CoP and outwash tuning

Pylons mounted outside of bodywork for easy accessibility

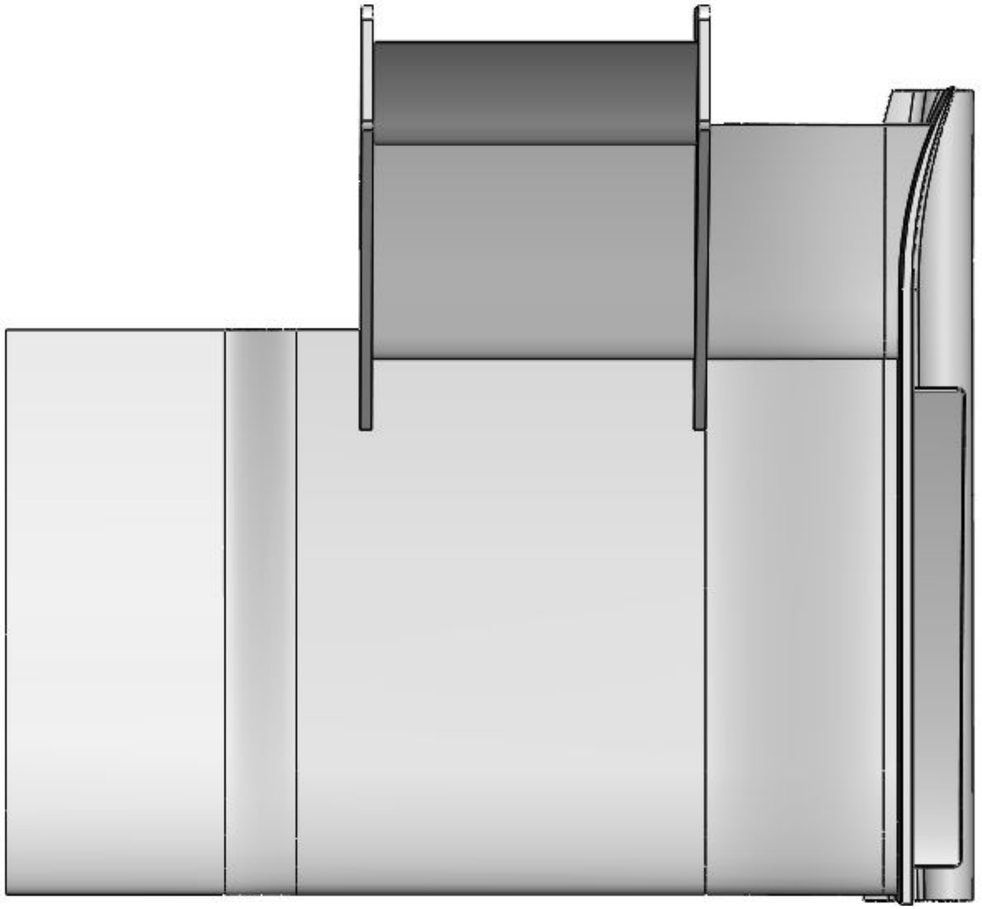


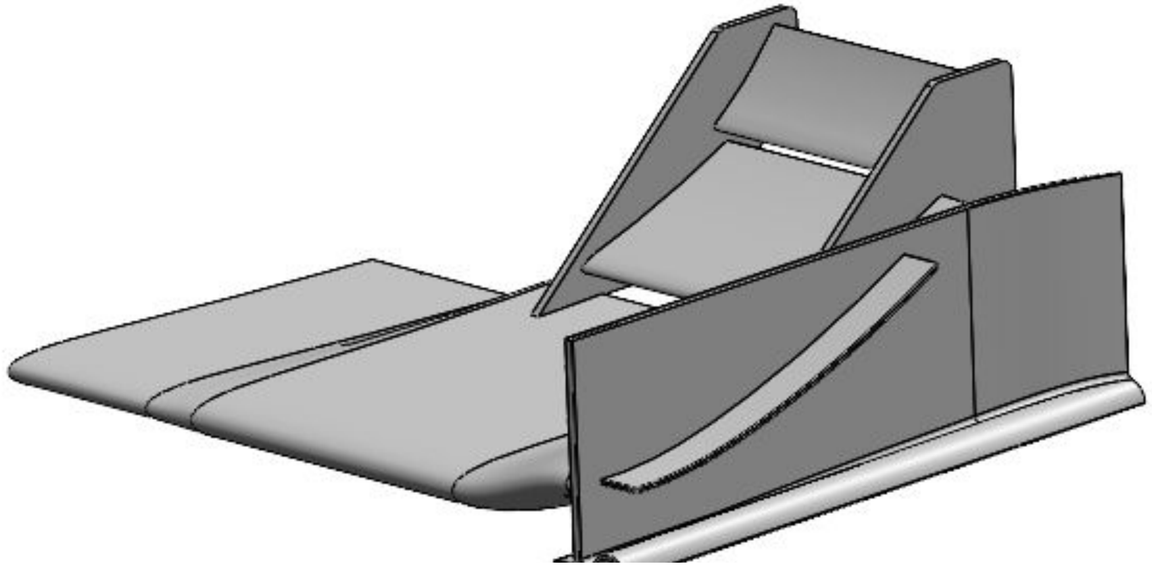
NACA 4412, 3405, 6412, 8412 airfoil profiles

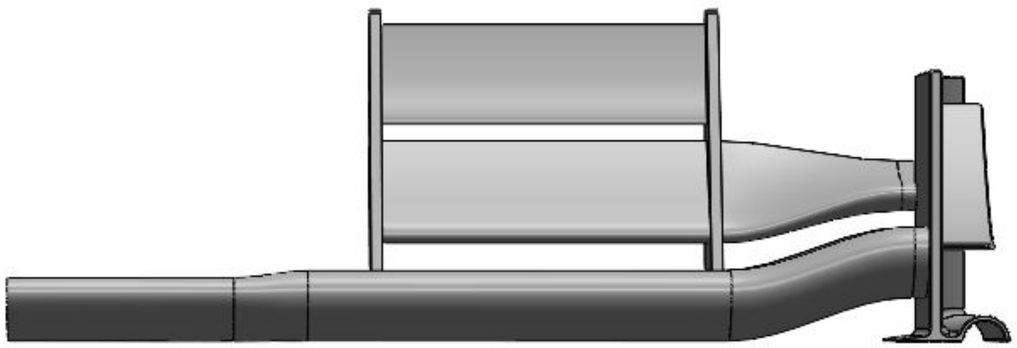
Outwashing endplate to drive wheel wake away from body

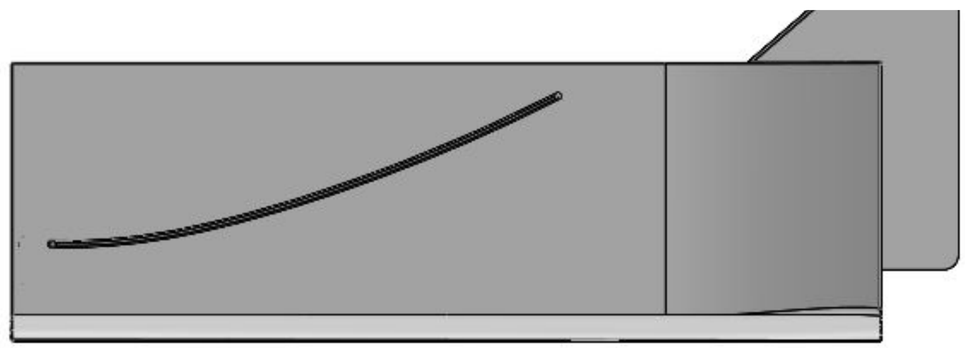
Front Wing - CAD Overview

UC Berkeley





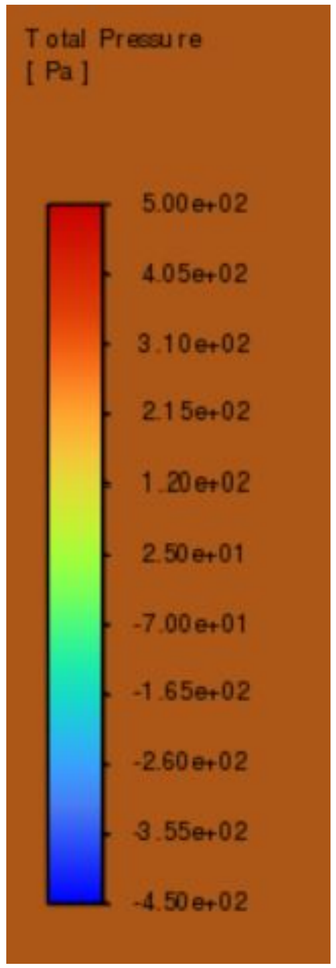


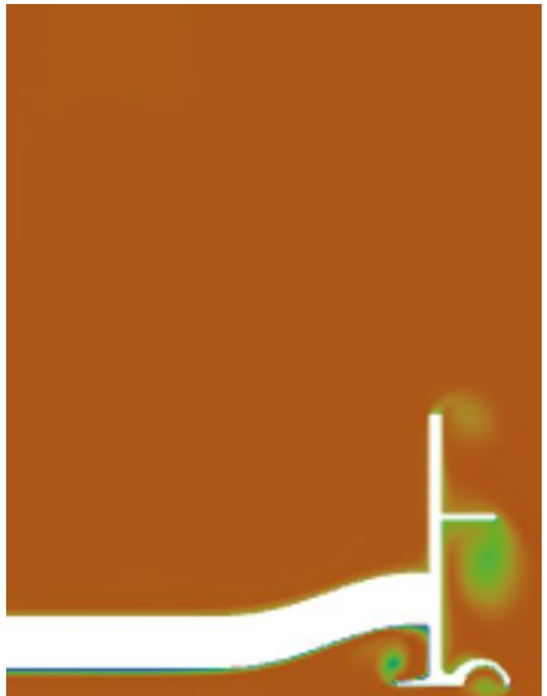


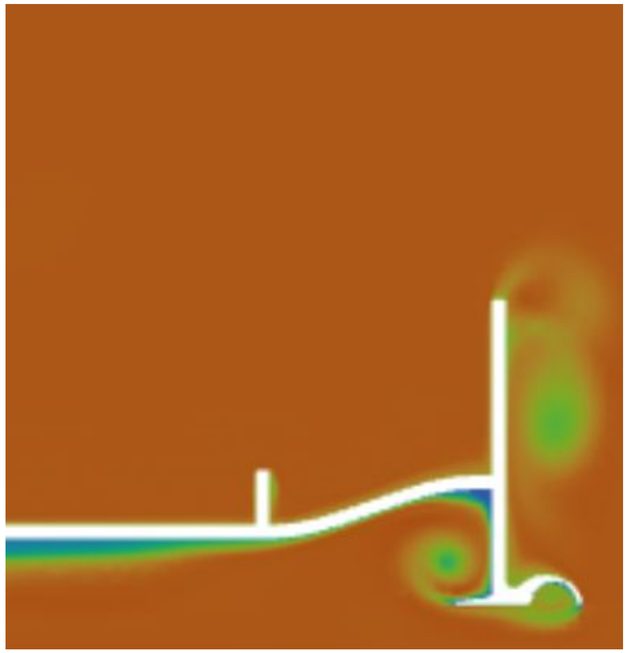
Vortex Blow-Up

UC Berkeley

- Vortex generated on front wing benefits undertray airflow

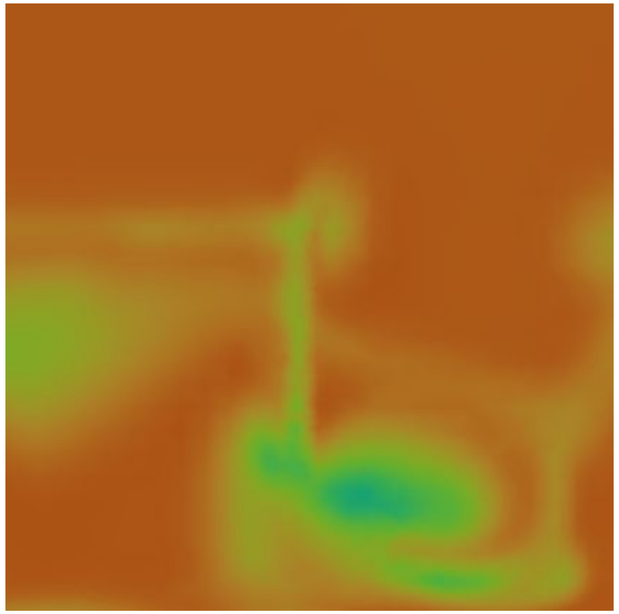


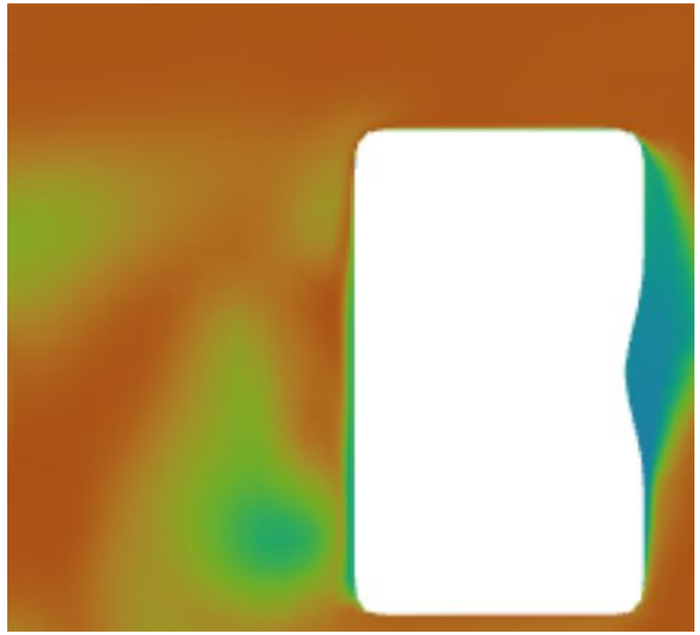


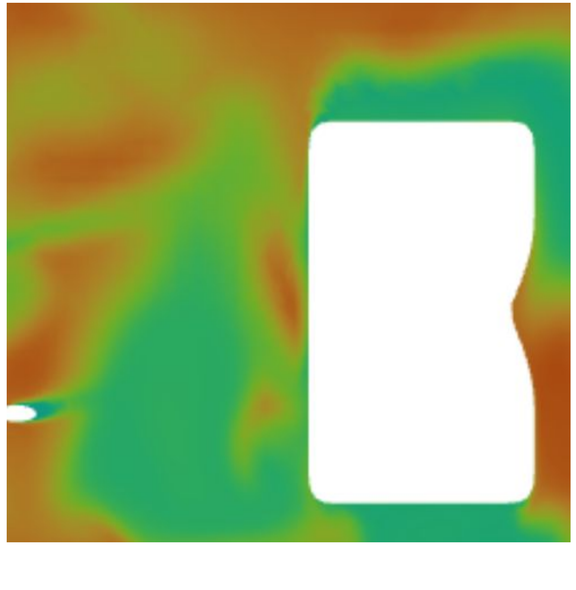


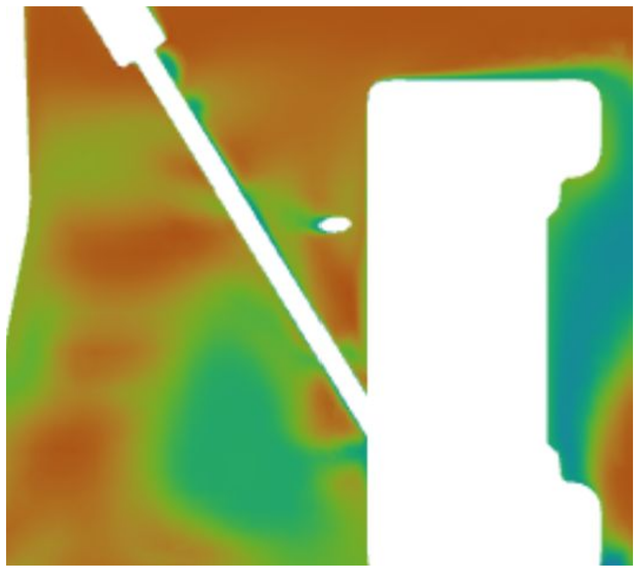
Tire Squirt & Blow Up Control

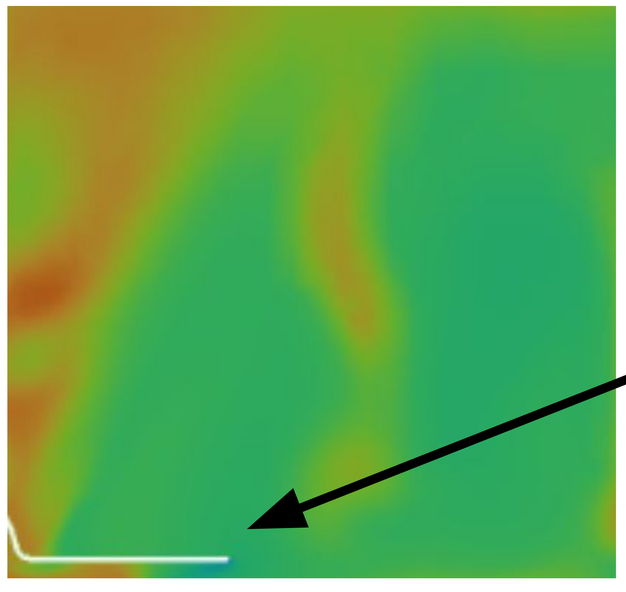
UC Berkeley











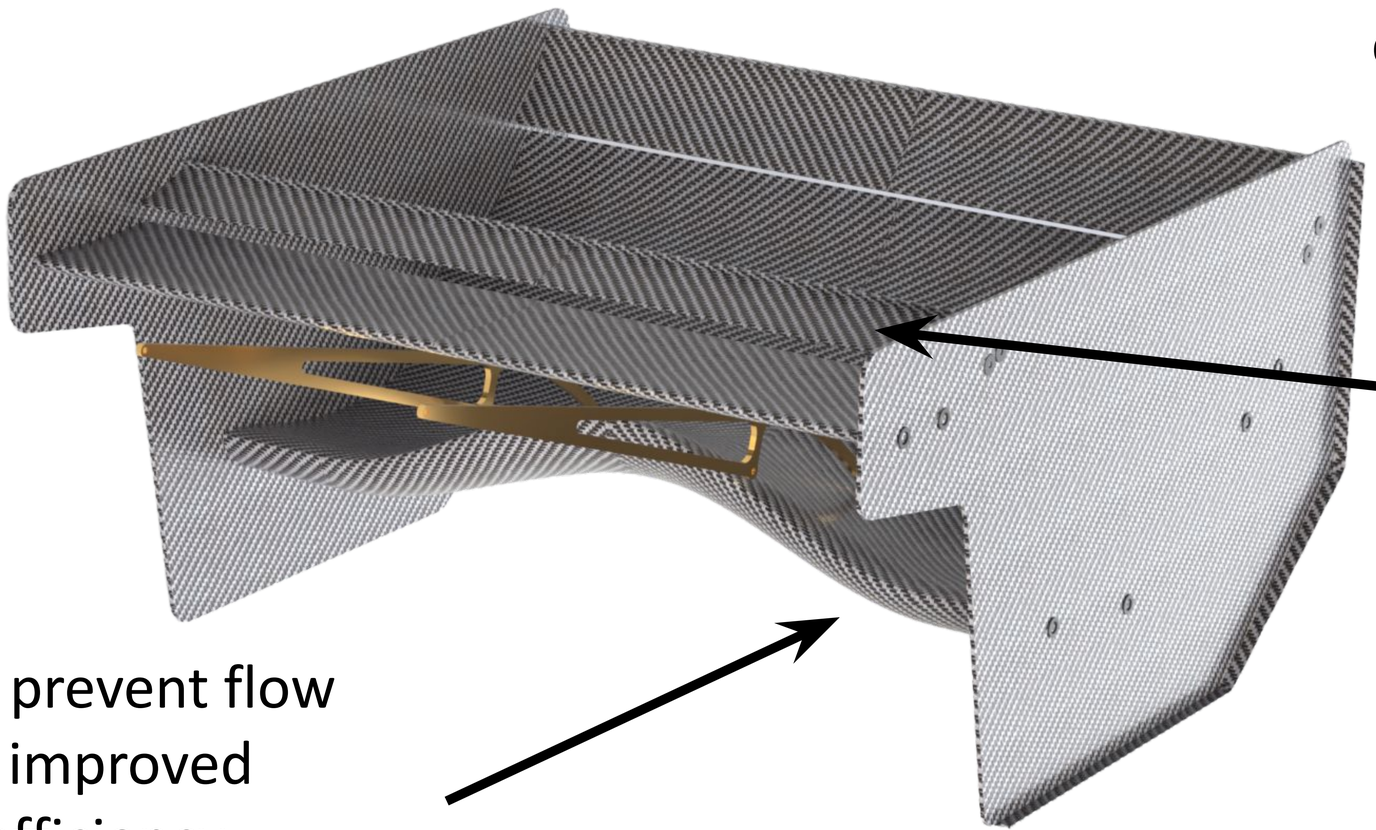
- Tire squirt brought up over inlet

Rear Wing

UC Berkeley

MSHD and NACA 6412 airfoil profiles

XPS foam-core wing construction

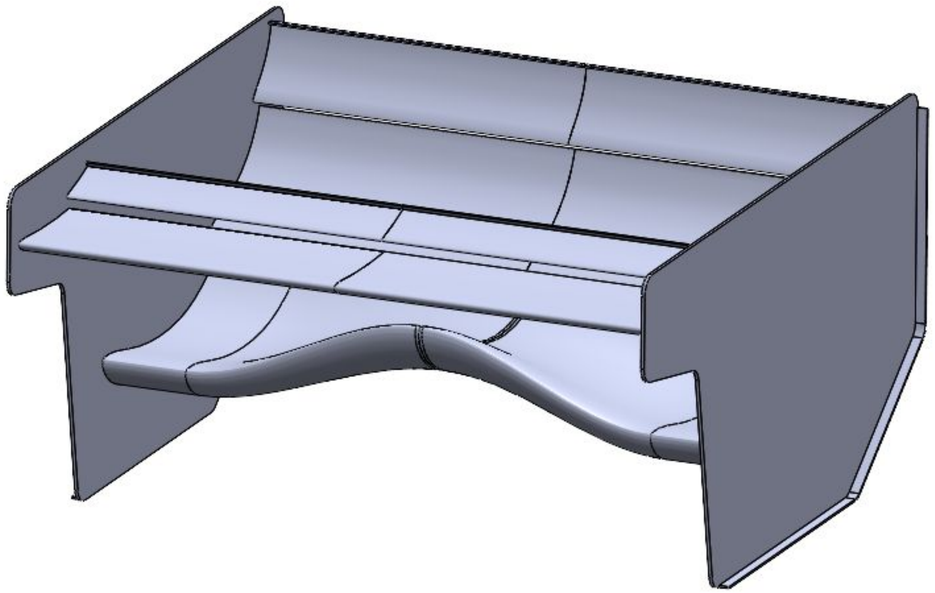


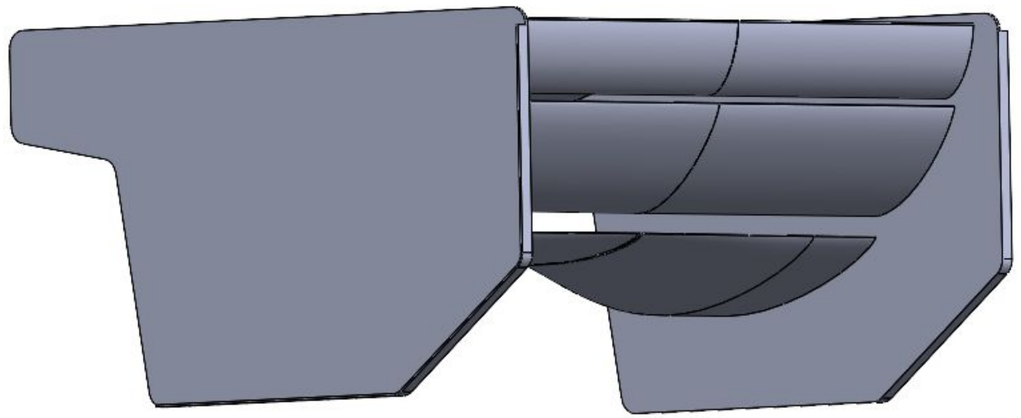
Added RE4 and RE5 to maximize downforce produced

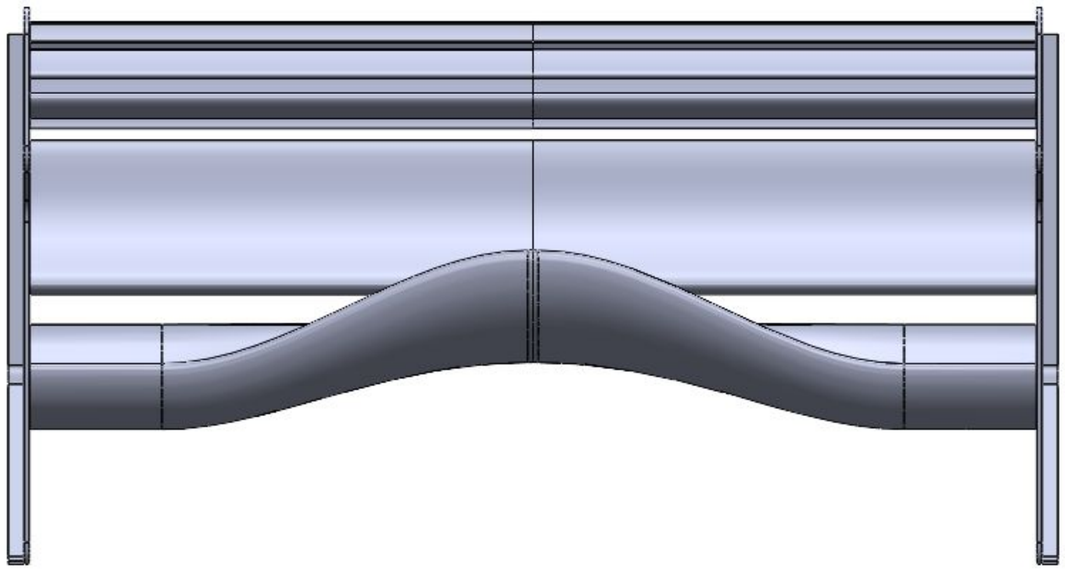
Curved RE1 to prevent flow separation for improved aerodynamic efficiency

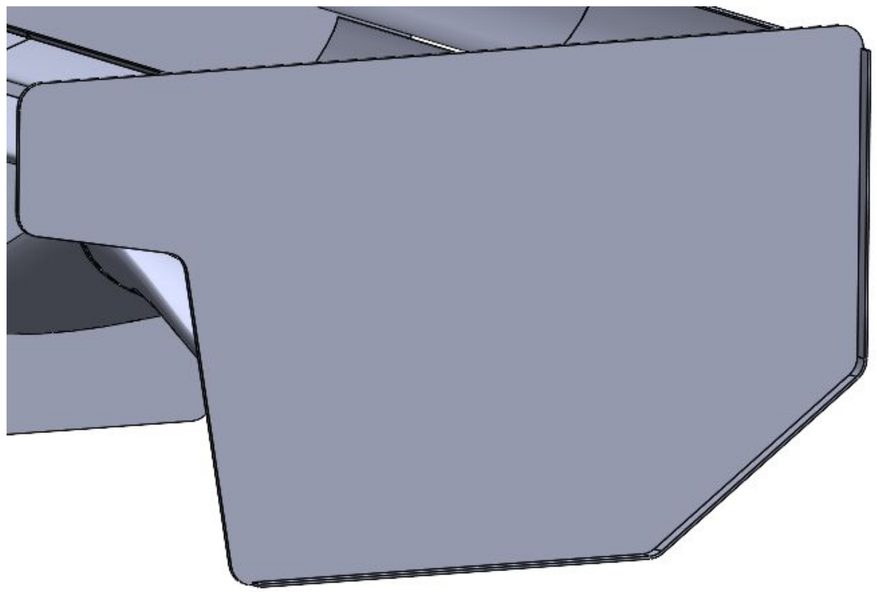
CFD - Rear Wing

UC Berkeley







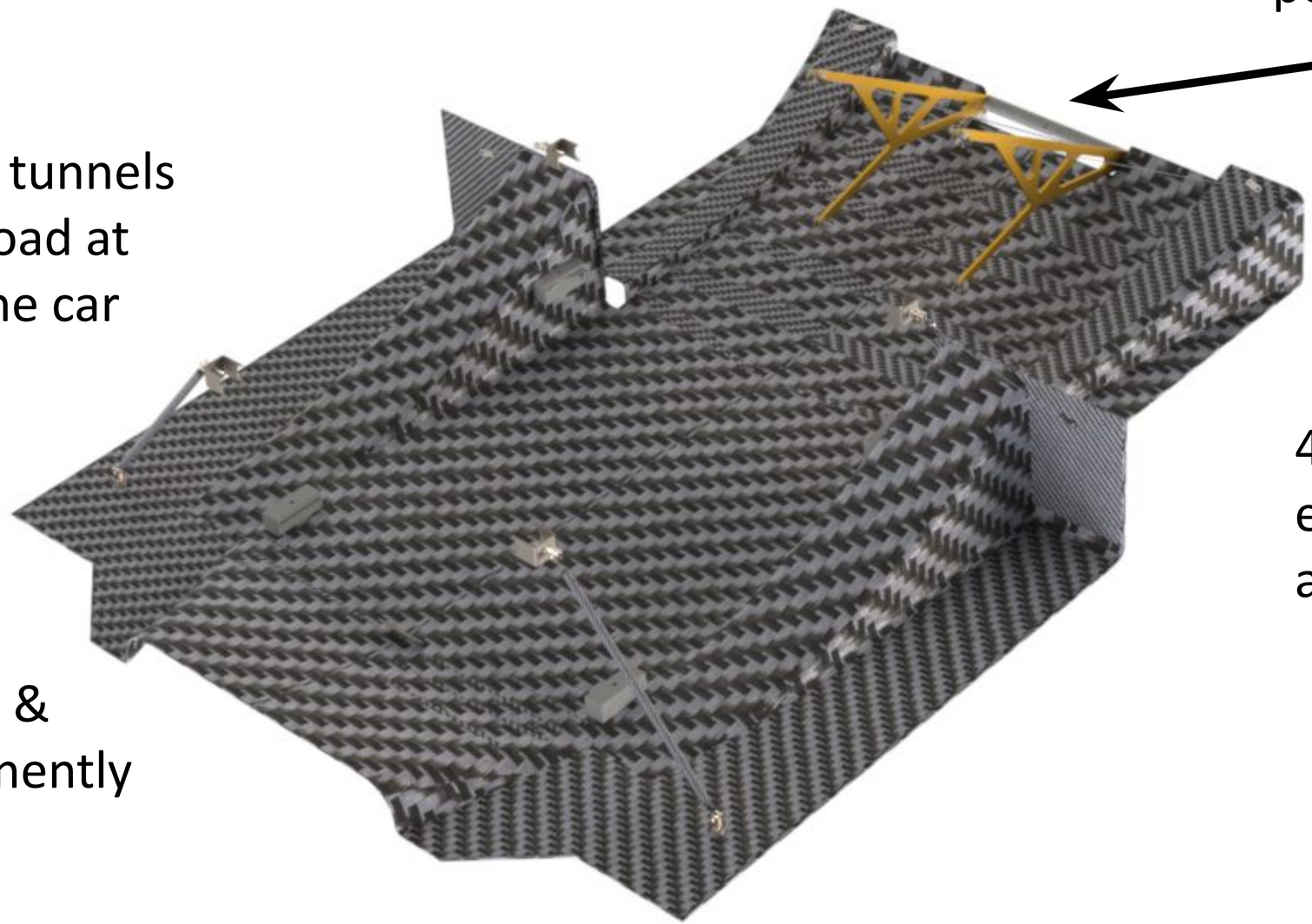


Undertray

UC Berkeley

7-channel diffuser to control inwash

Tow bar integration for diffuser outlet performance



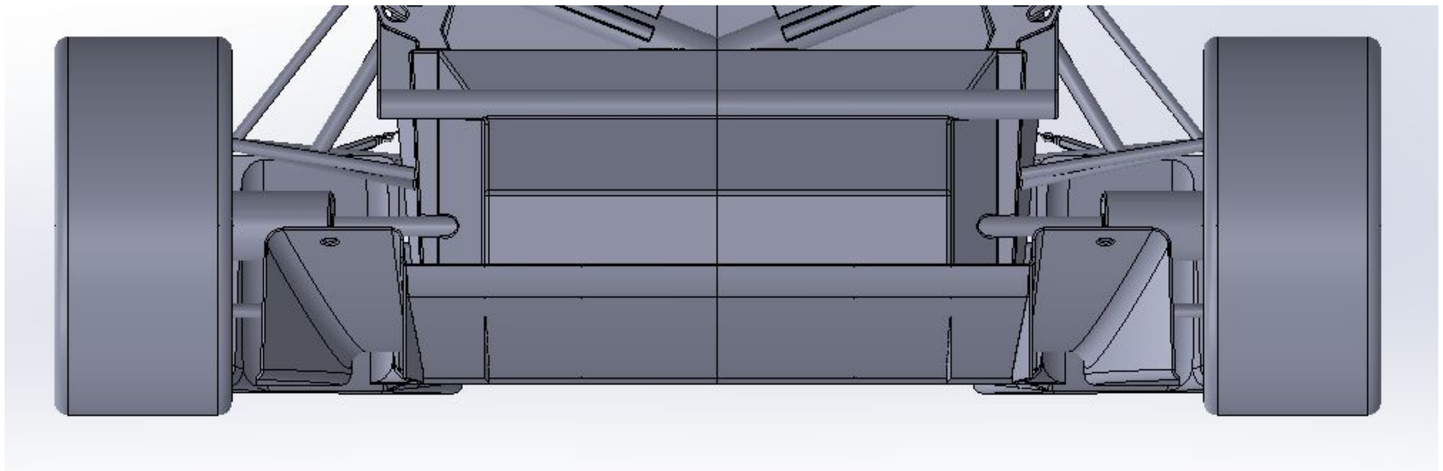
Side expansion tunnels produce local load at the center of the car

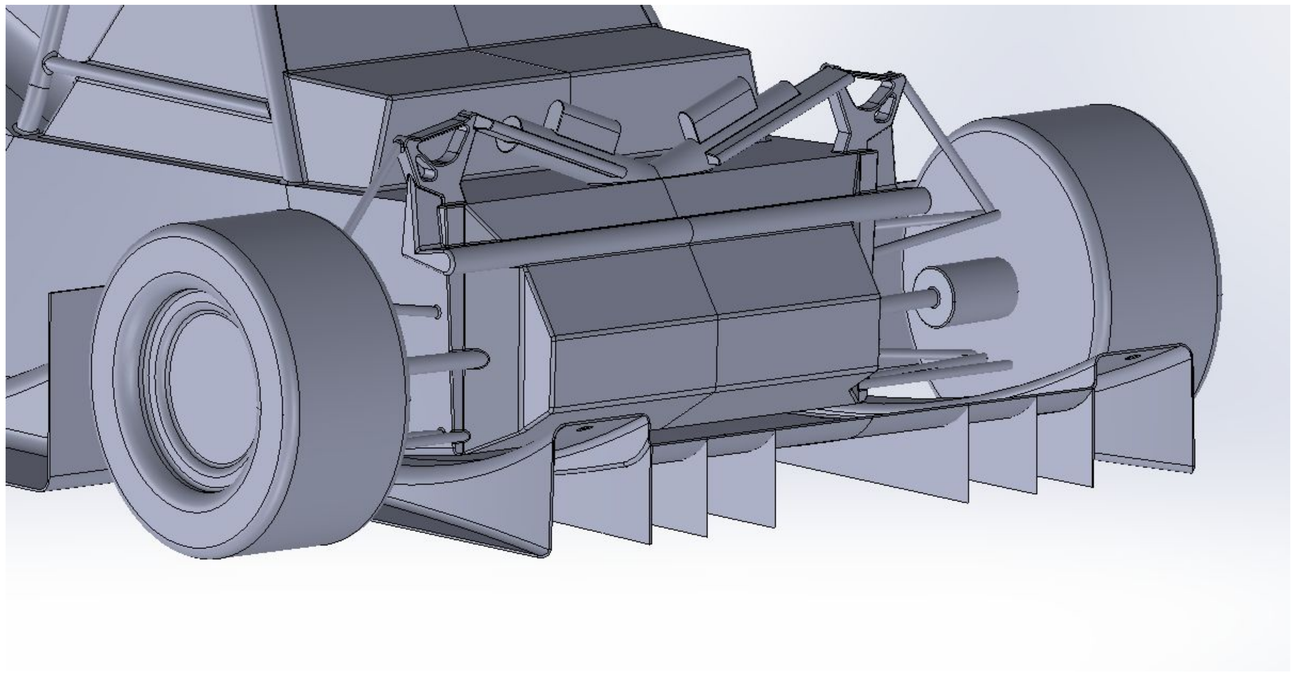
4-piece undertray for ease of installation and manufacturing

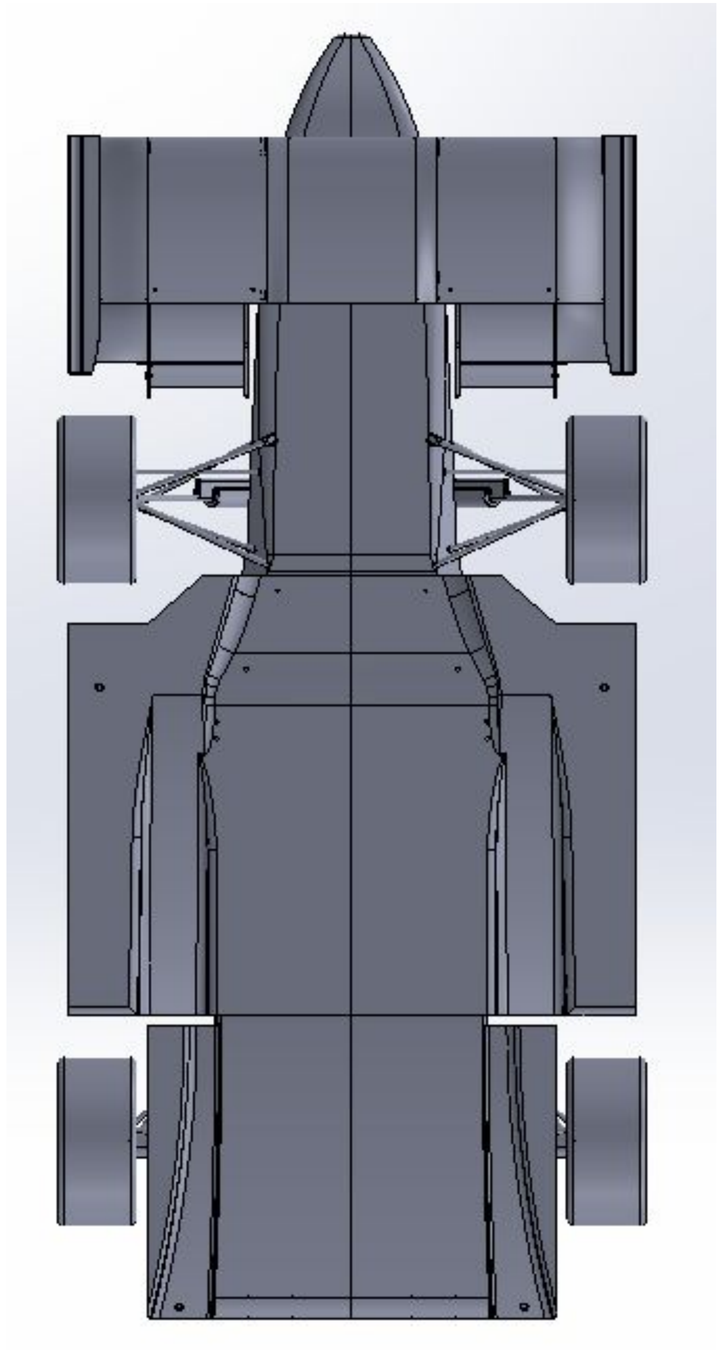
Undertray inlet & diffuser permanently mounted

CFD - Undertray

UC Berkeley

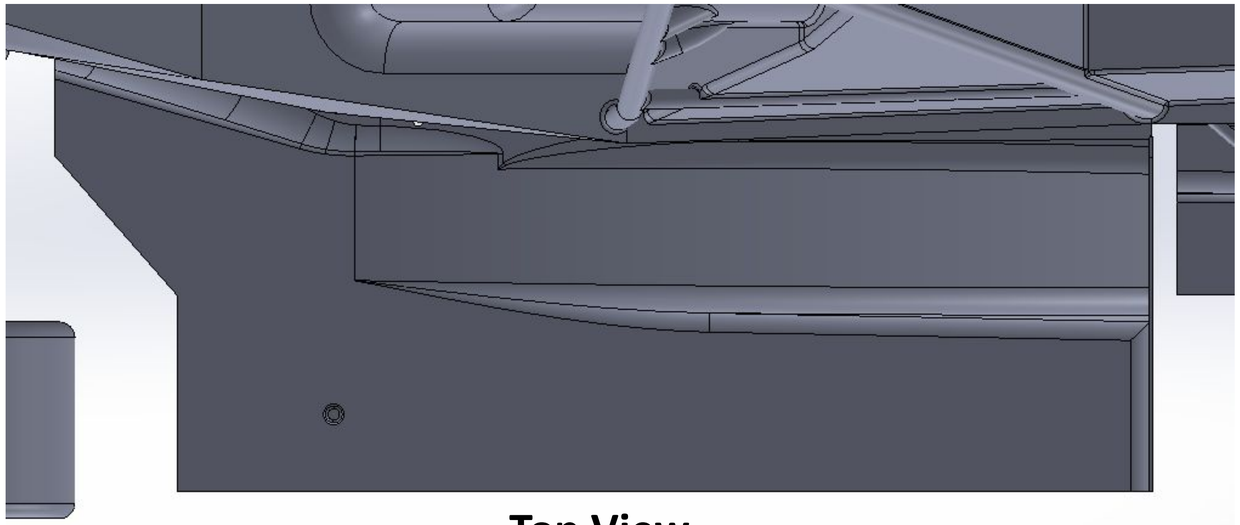




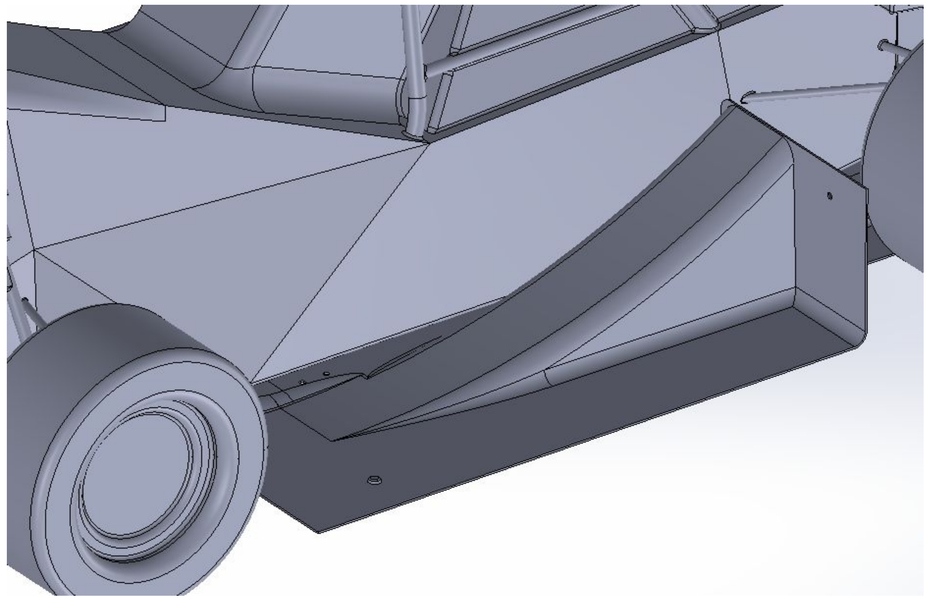


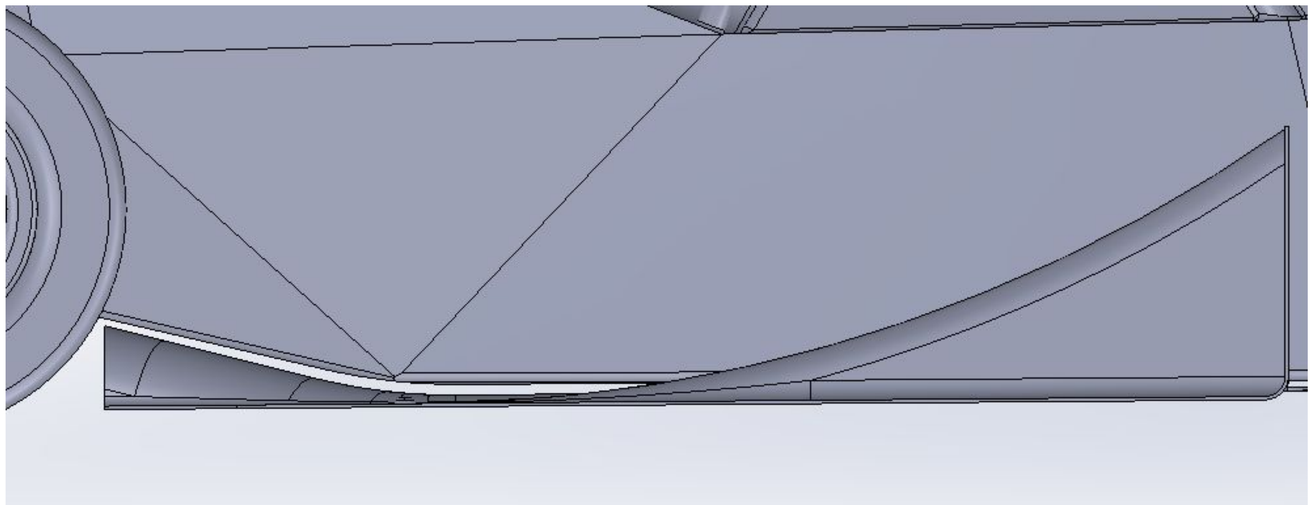
CFD - Undertray

UC Berkeley



Top View



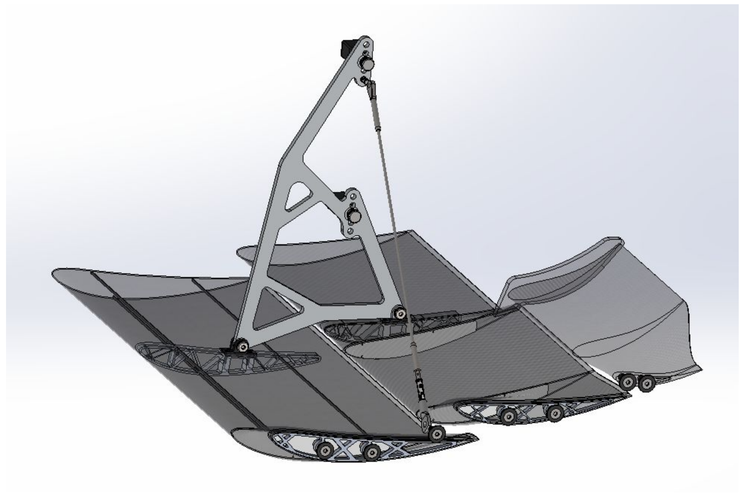


Side View

Mechanical Design

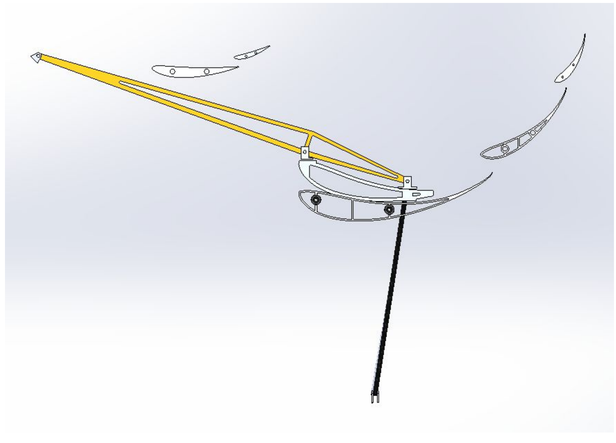
UC Berkeley

- Standardized M3/4/6 bolts
- Simplified structures
  - Removed RE2 mounting rib
  - Removed carbon spars along the span

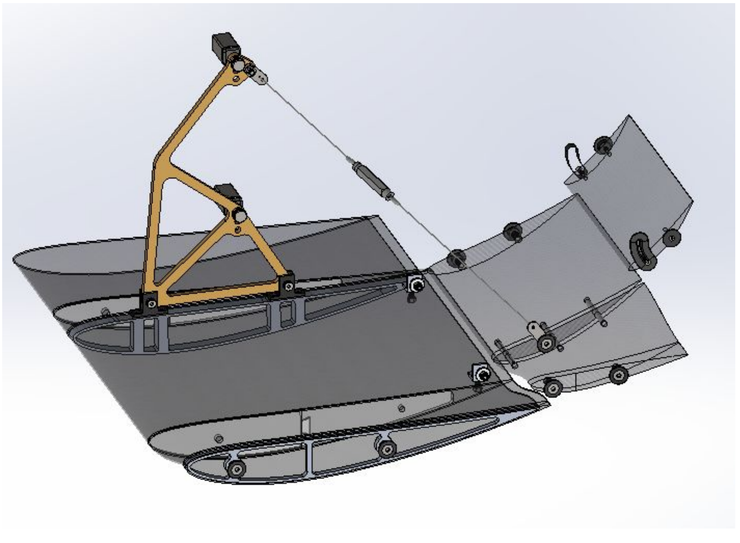


SN4 FW Structure

SN4 RW Structure



SN5 RW Structure



SN5 FW Structure

Structure Optimization & Validation

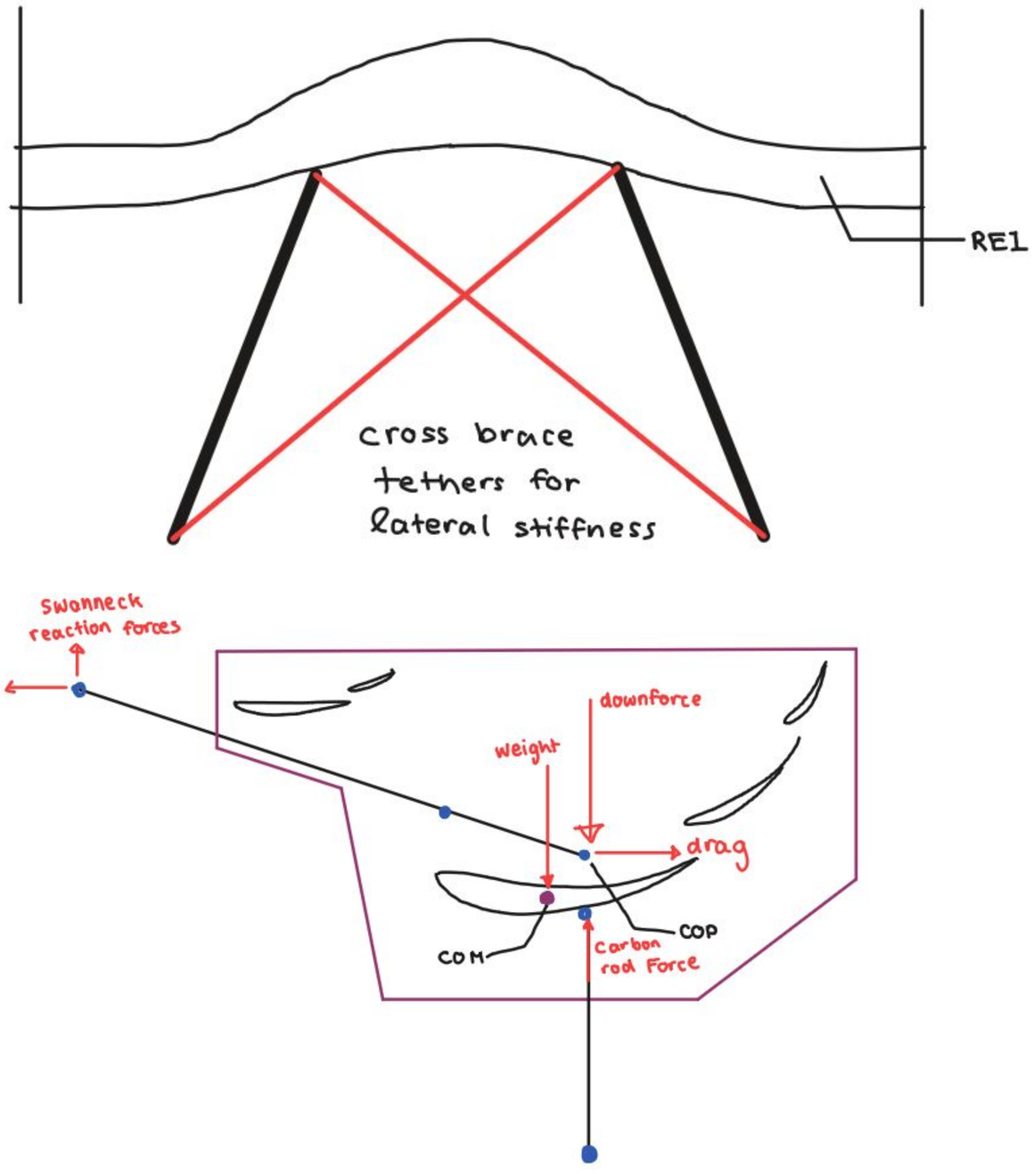
UC Berkeley

- Used topology optimization to reduce mass
- Increased accuracy of FEA through integrating CFD load cases via fluid-structure interaction (downforce surface distribution)
- Mesh convergence studies to improve simulation fidelity at bolt connections
- ANSYS OptiSlang used to calibrate material model parameters for various stackups
- Utilized ANSYS ACP and materials testing data to model carbon-fiber wrapped foam element wings and perform full wing simulations

Rear Wing Design - FBD

UC Berkeley

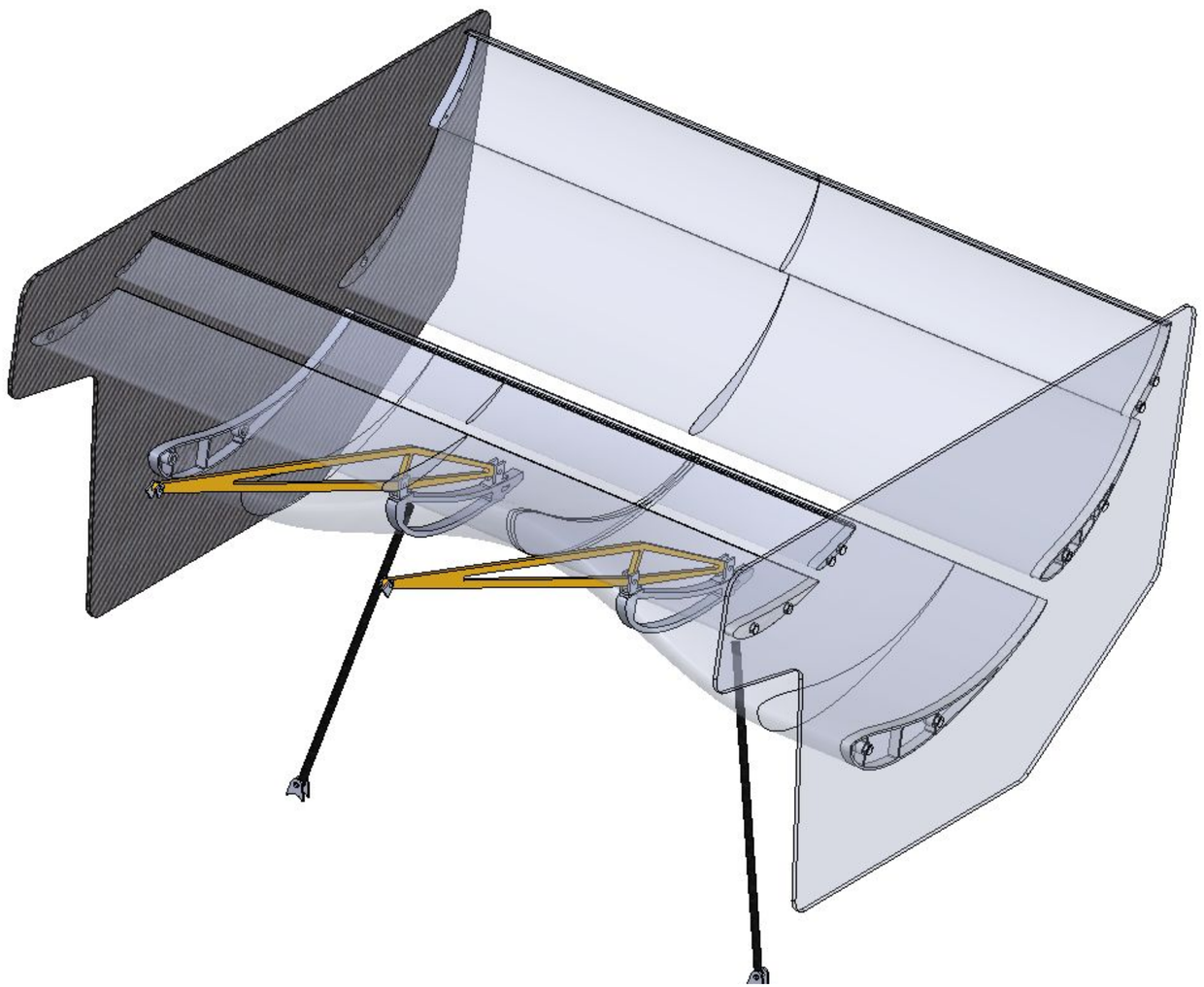
- Swanneck constrains Drag and Forward Acceleration
- Carbon rods constrain downforce
- Tethers constrain lateral forces

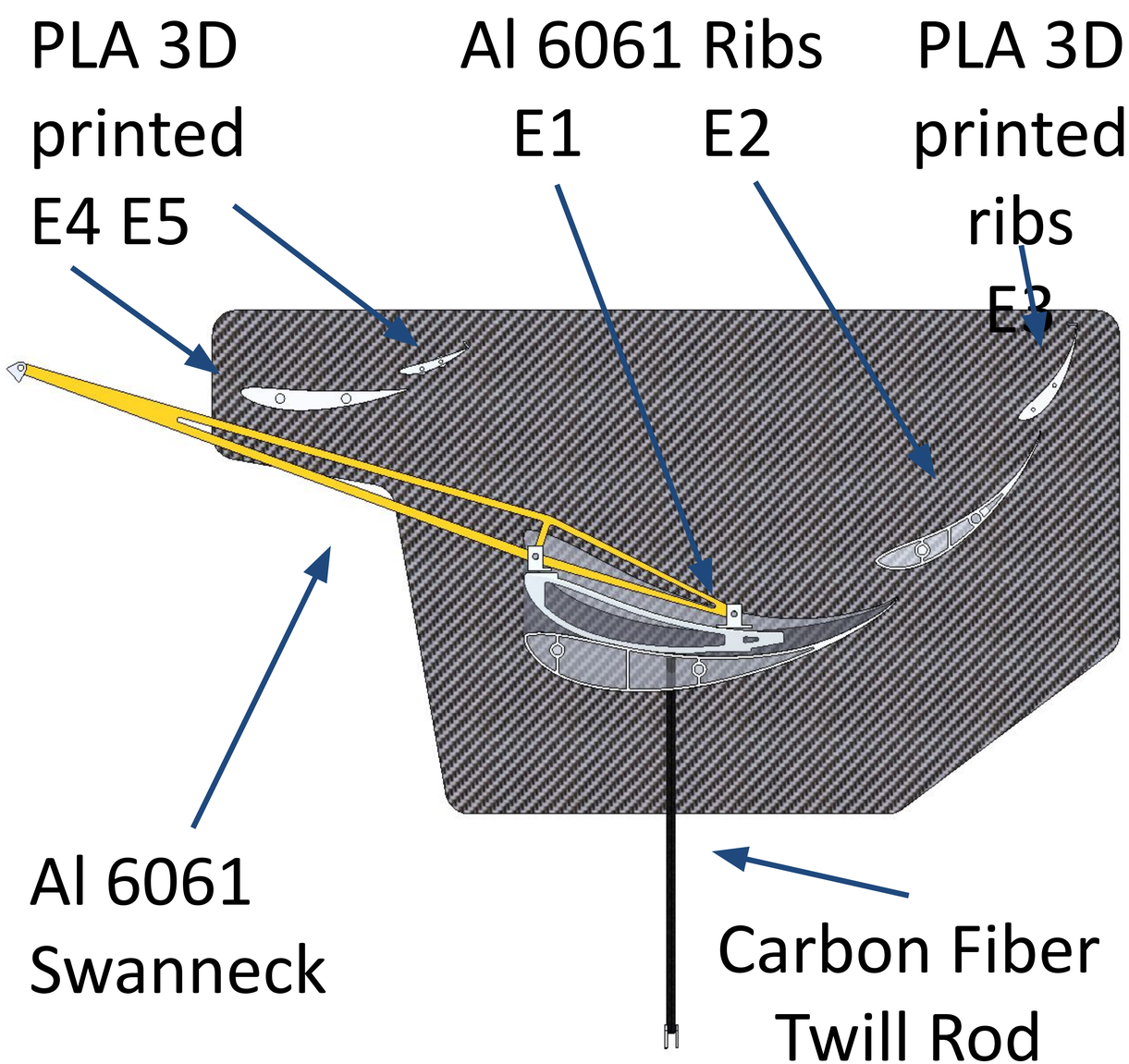


Rear Wing Design - Main Structures

UC Berkeley

Rear Wing Assembly

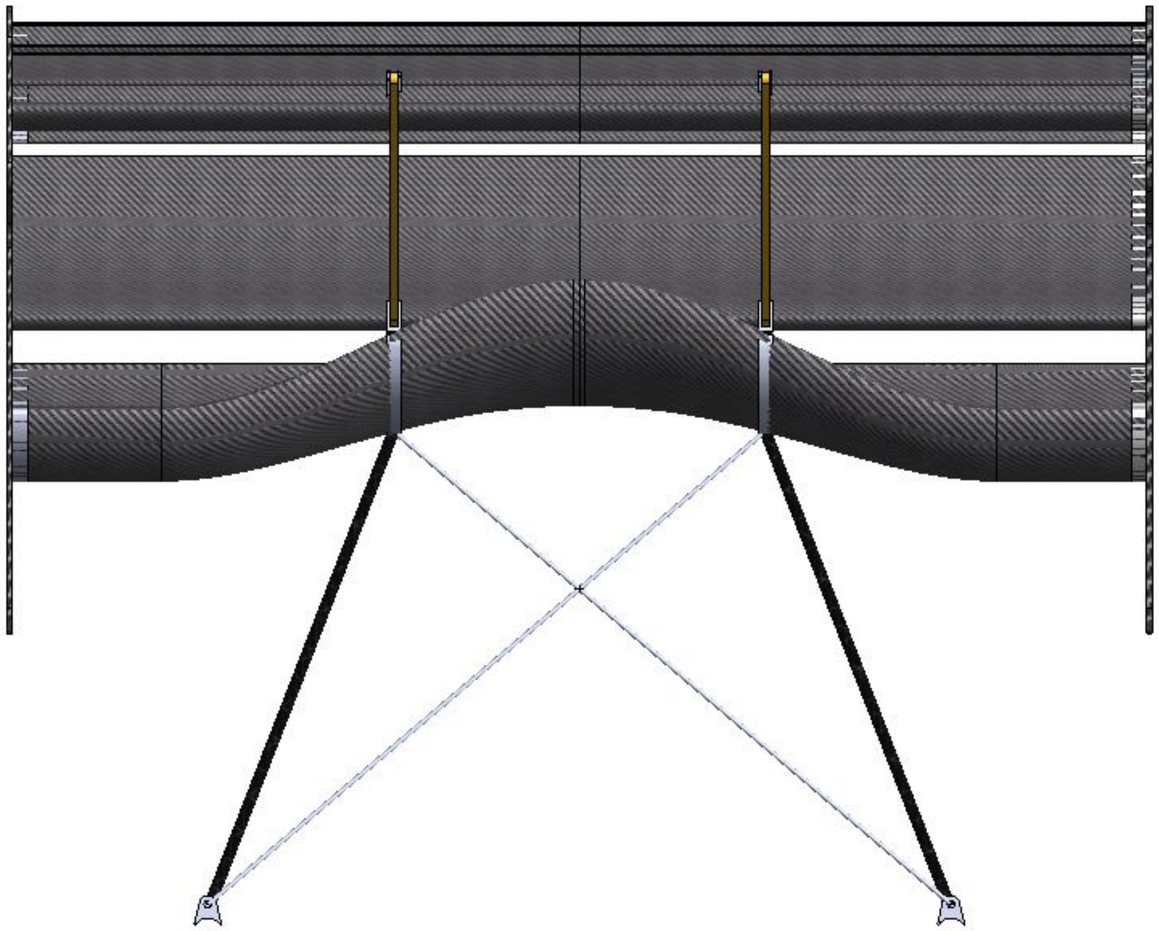




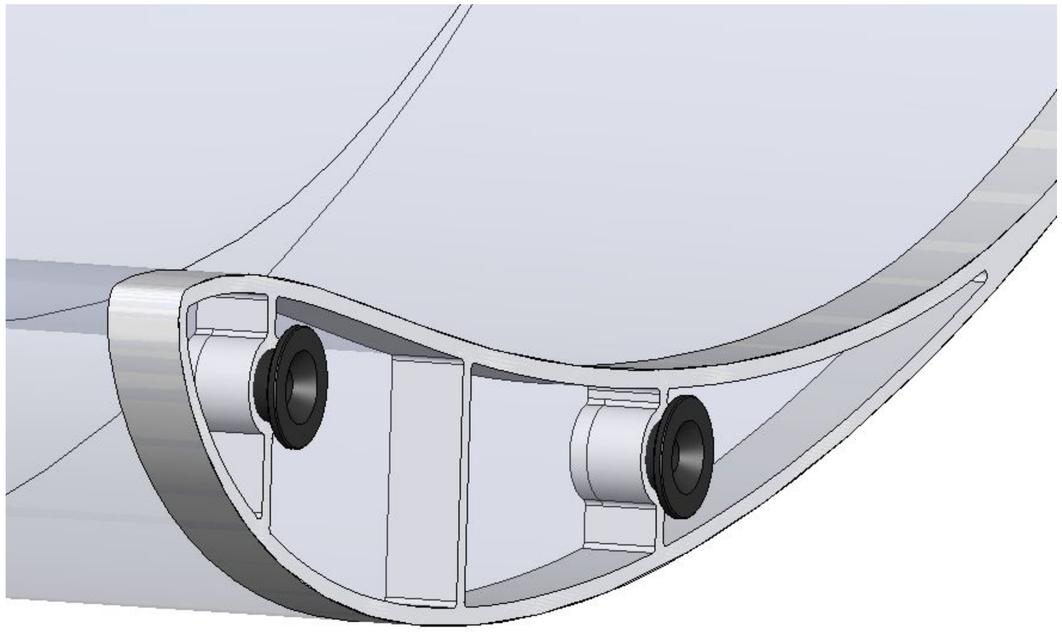
Rear Wing Mounting

UC Berkeley

Tether Layout

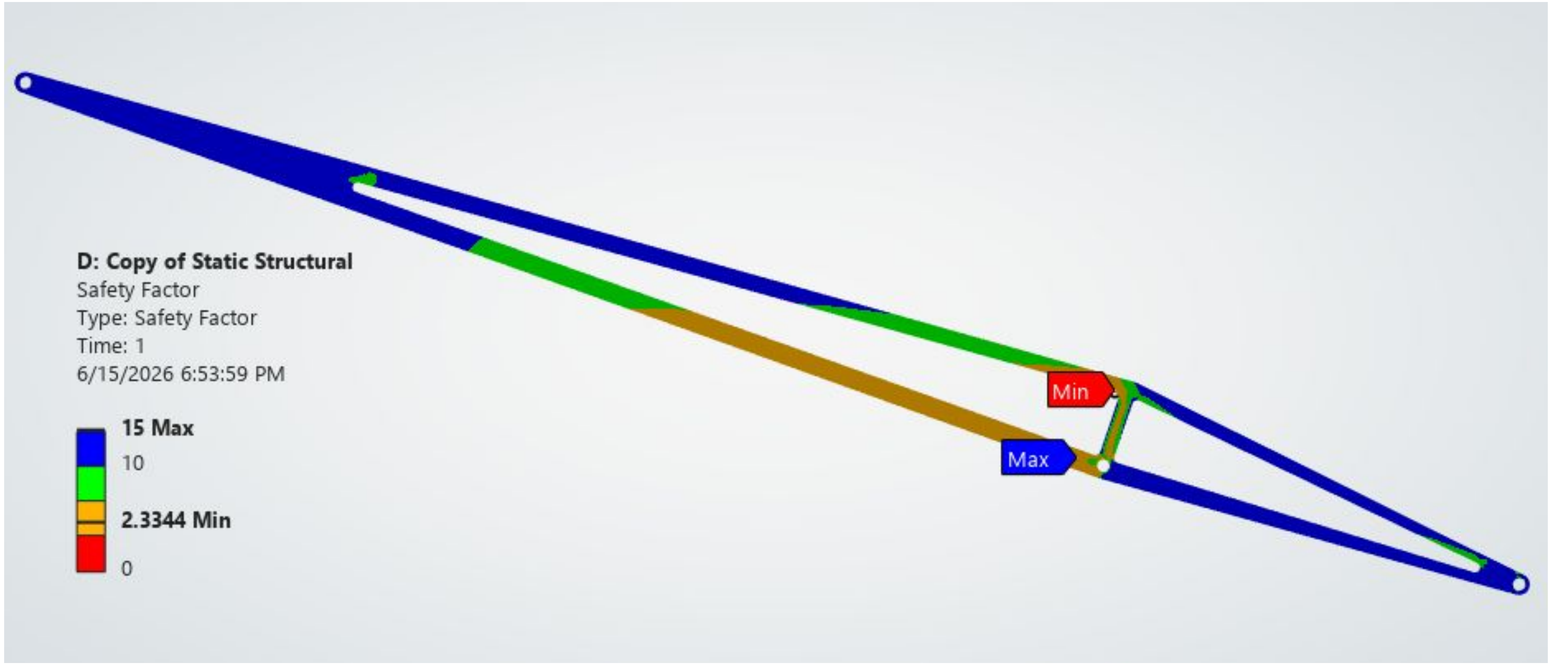


PLA T-Hat Compression Limiter

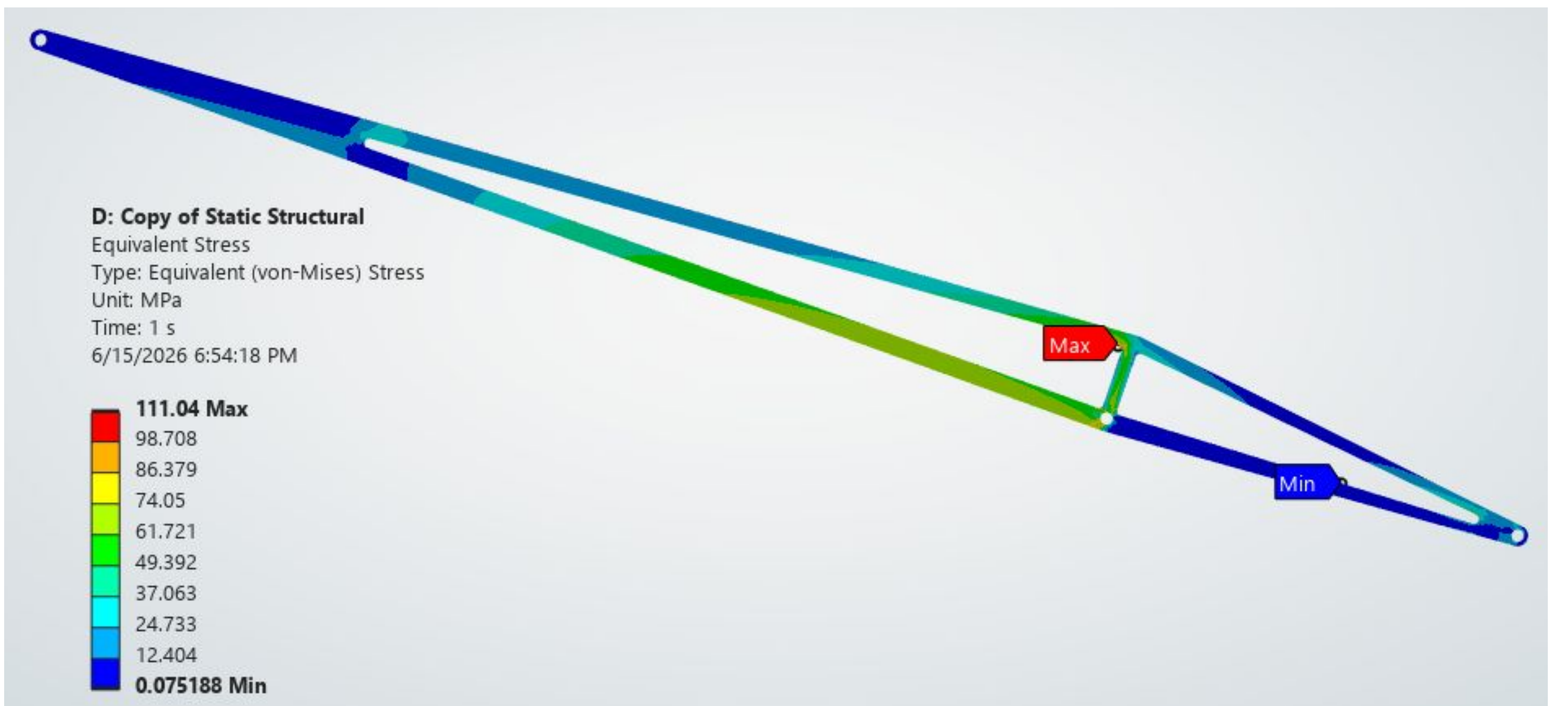


Rear Wing Design FEA

UC Berkeley



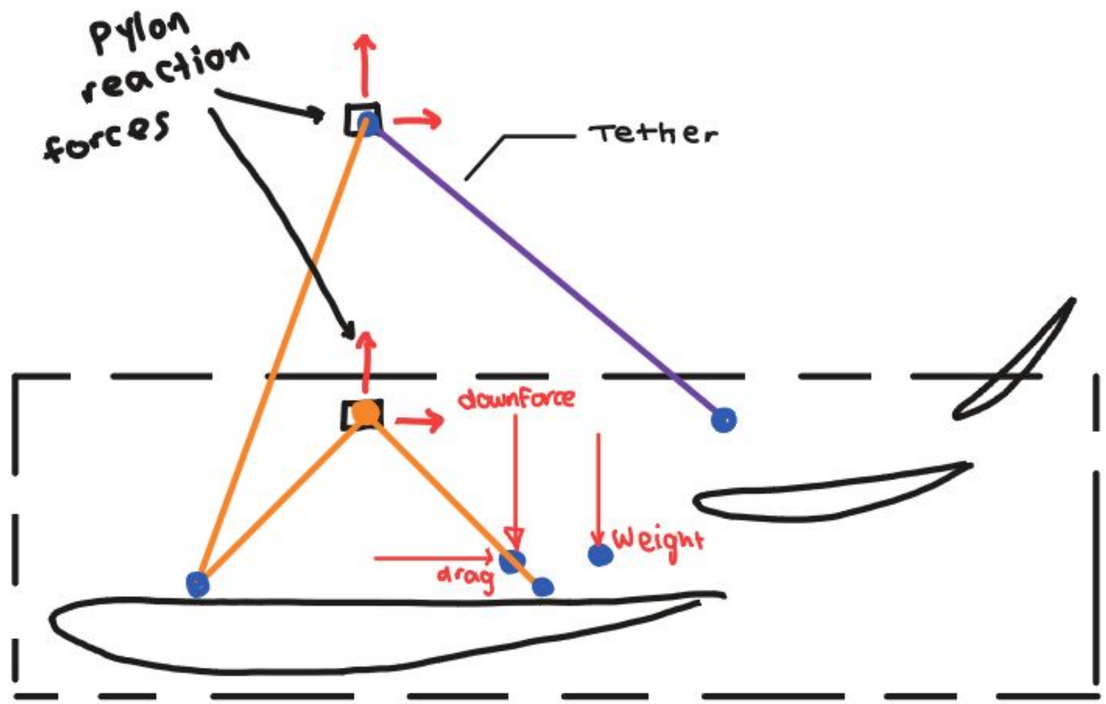
- FOS 2.33
- 62.5N Drag (per half)
- 1.4G Braking Force

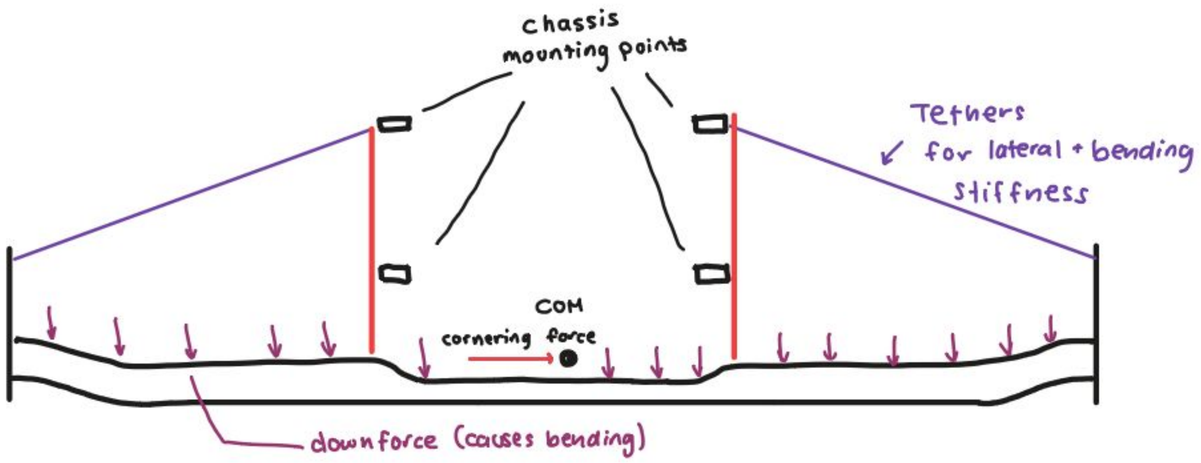


Front Wing Design FBD

UC Berkeley

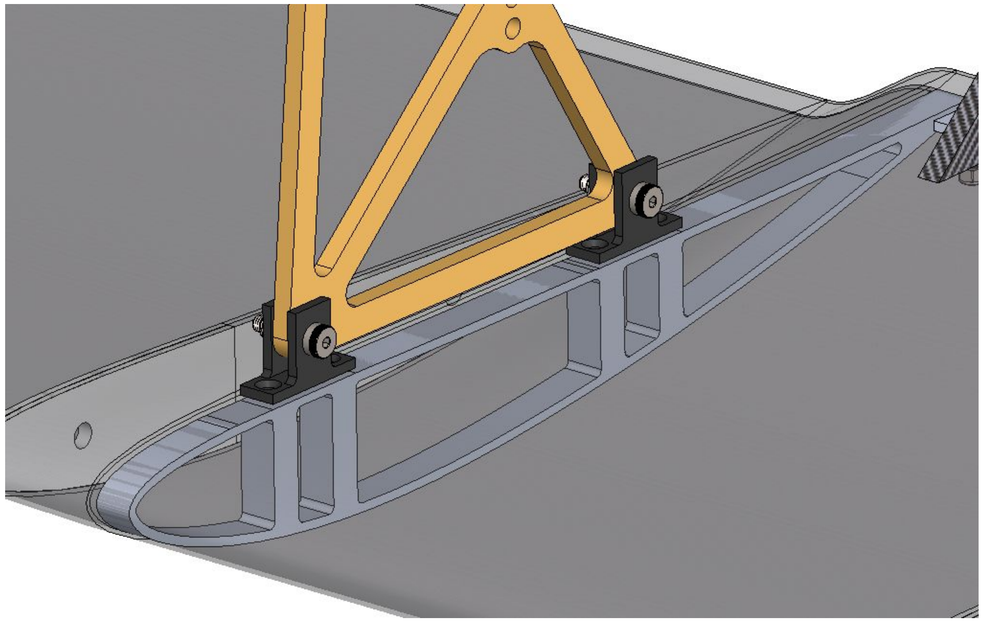
- Pylons constrain downforce and drag
- Tethers constrain lateral forces and bending

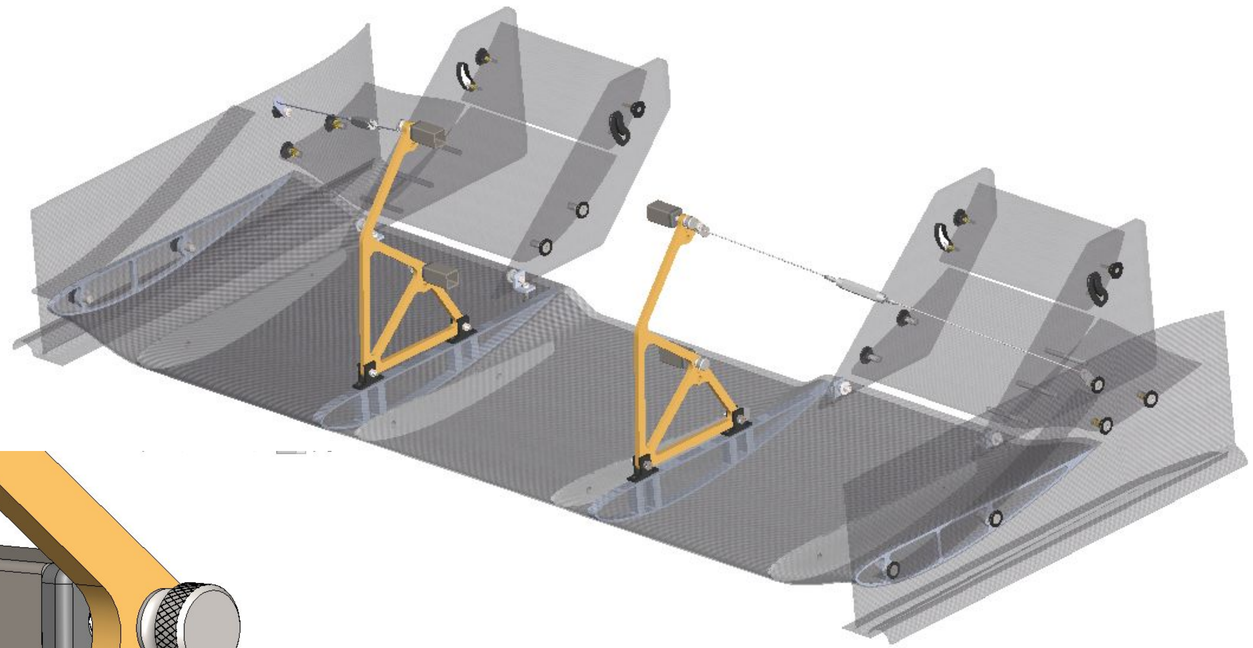


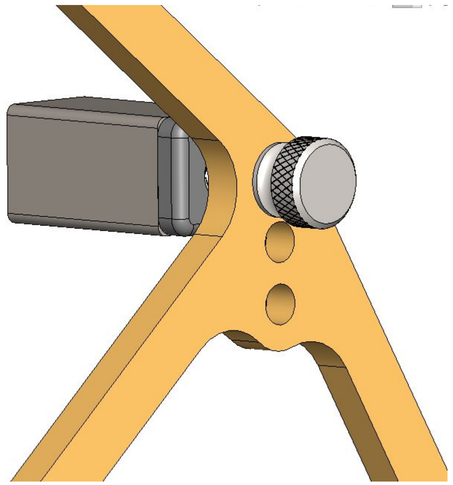


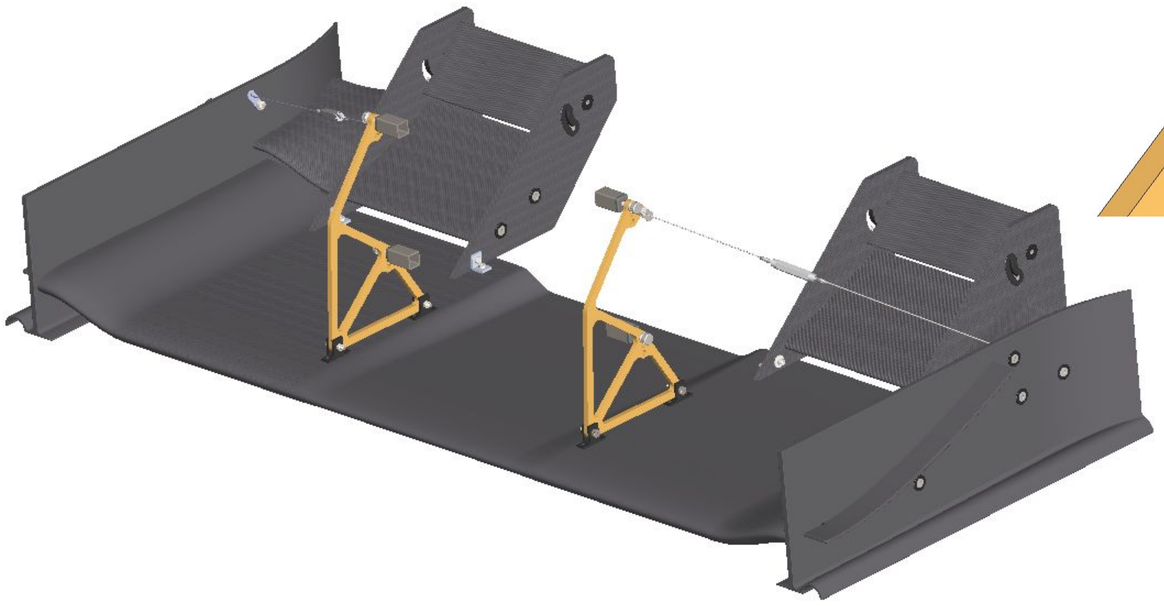
Front Wing Mounting

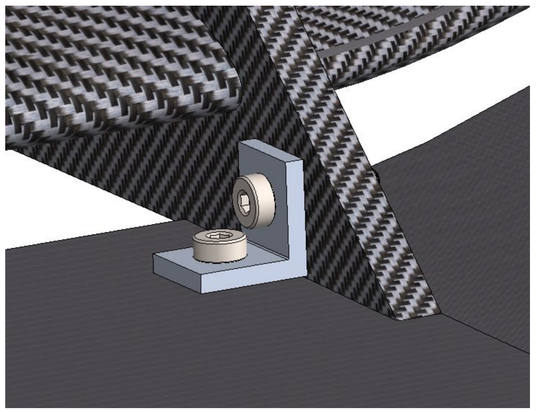
UC Berkeley

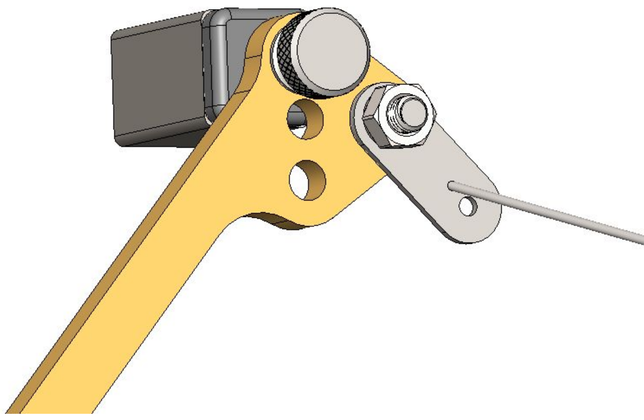






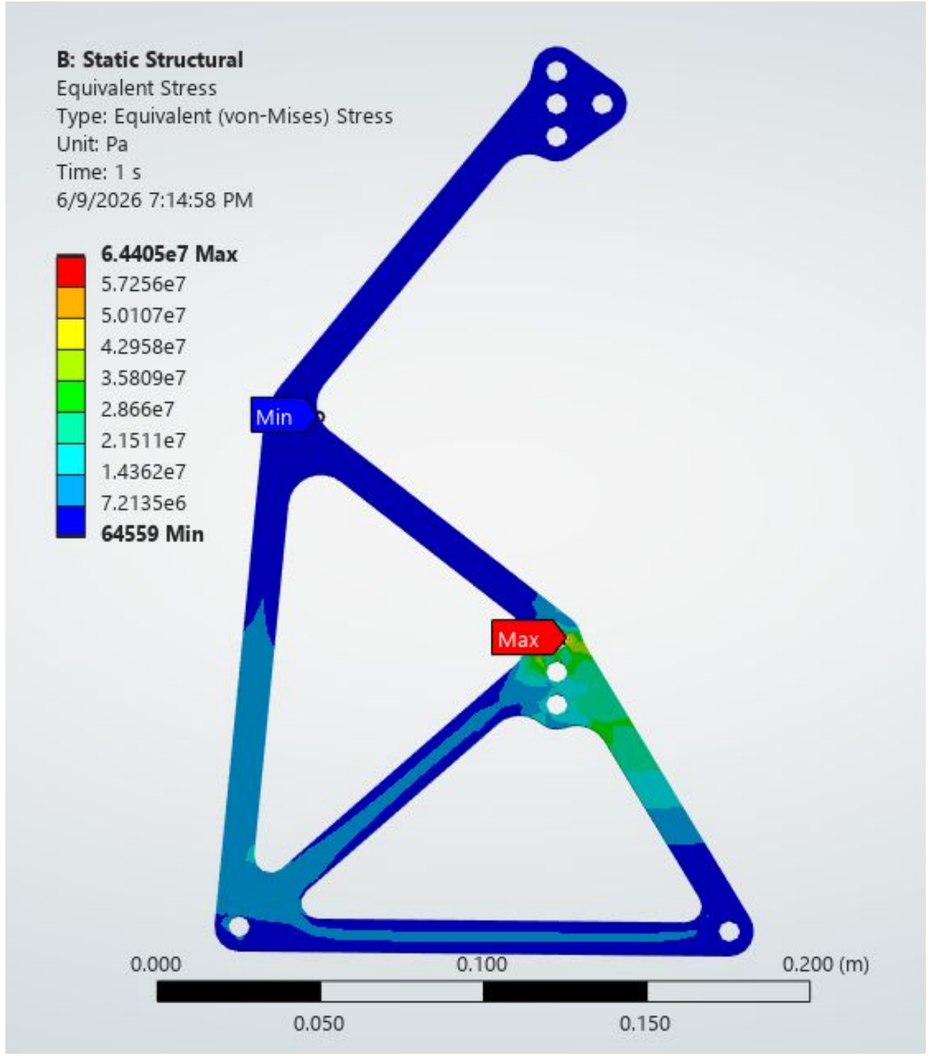


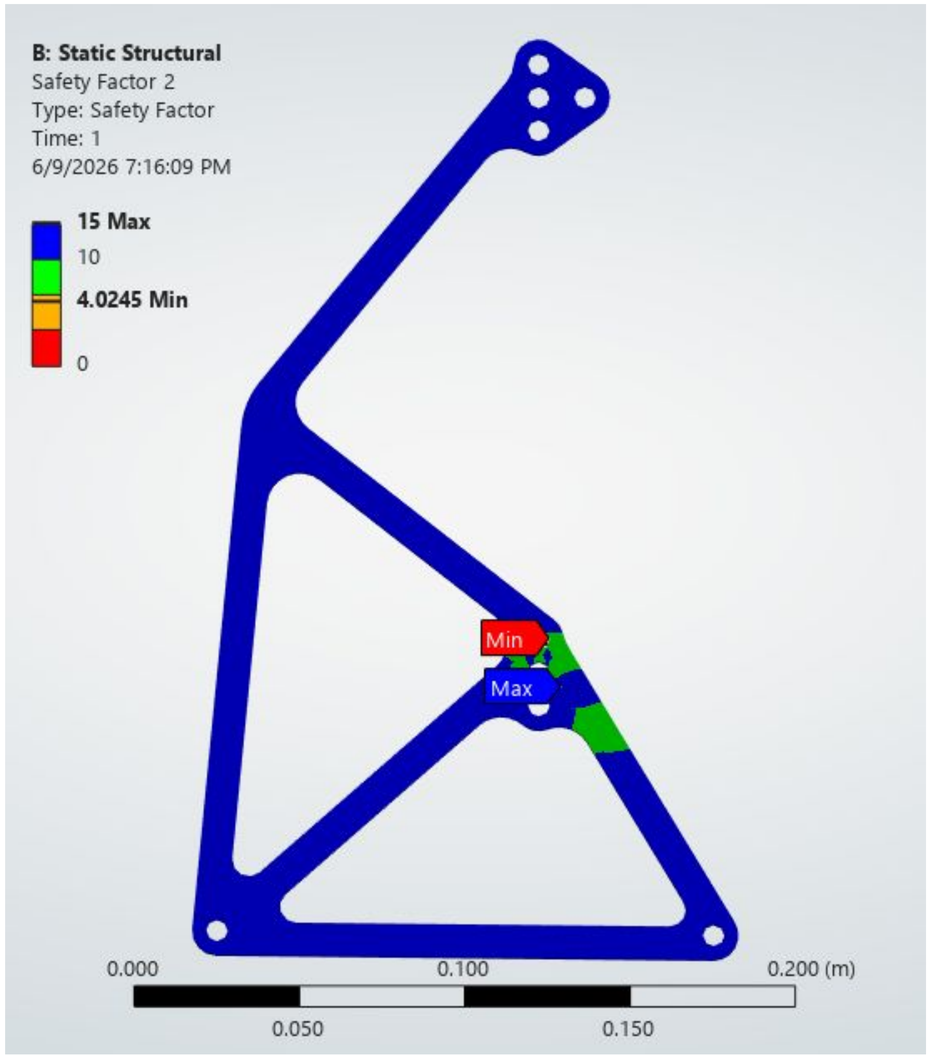




Front Wing Pylon FEA

UC Berkeley



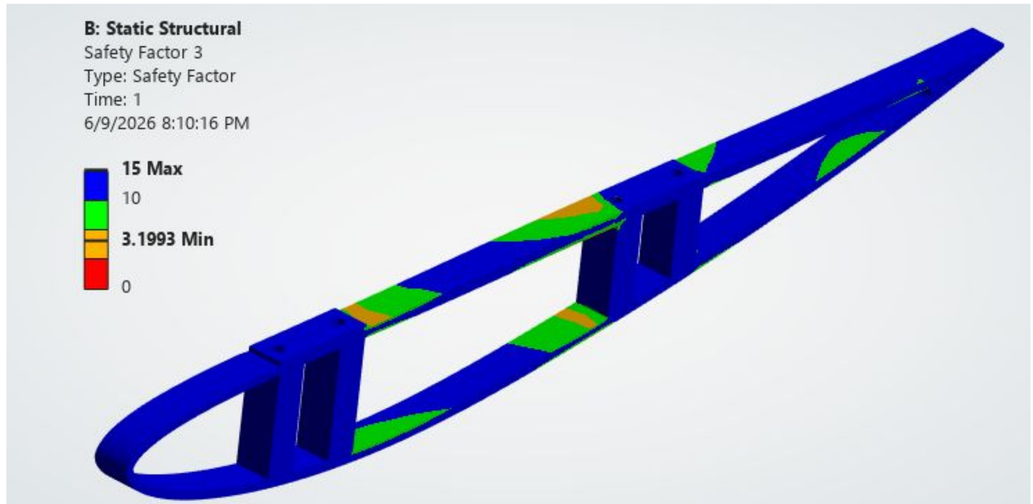


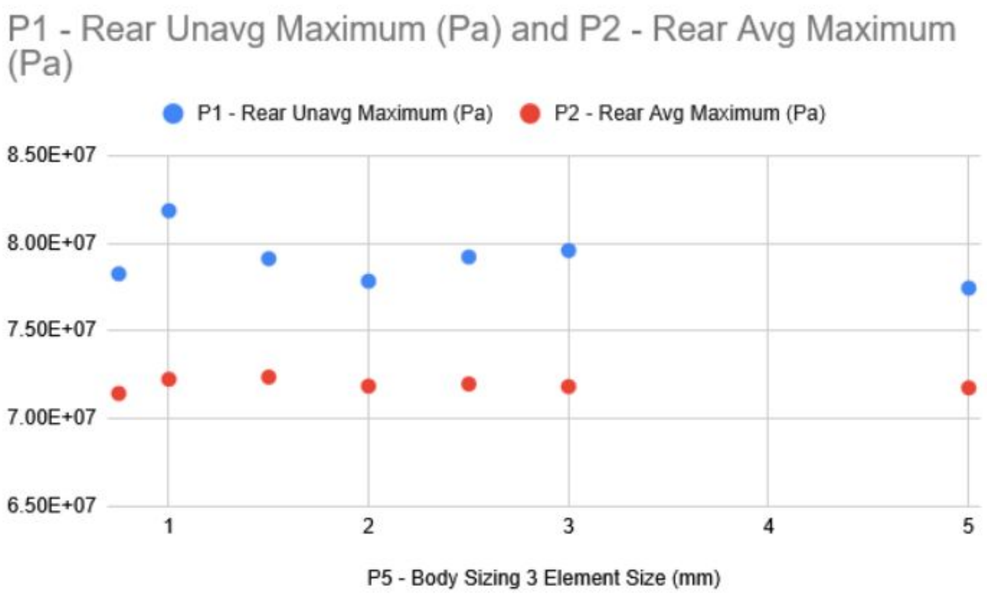
- FOS: 4.0
- Downforce (160 N)
- Drag (20 N)
- Lateral Force (40 N)
- Mass (25N)

Front Wing Mounting

UC Berkeley

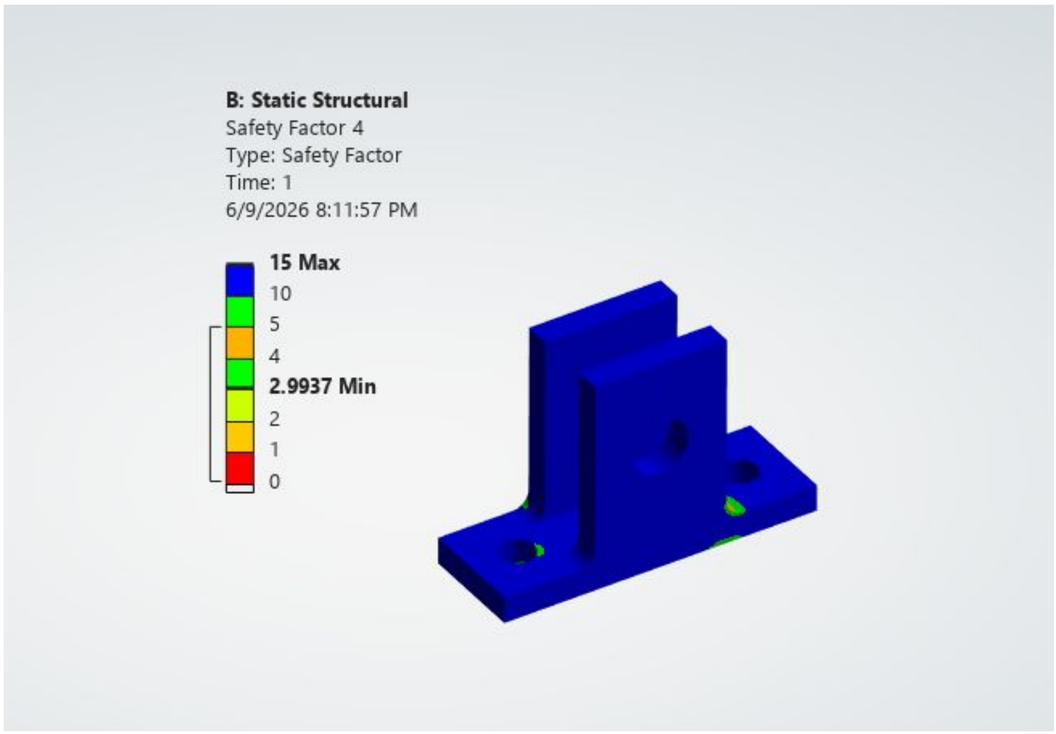
E1 Pylon Rib FOS: 3.2





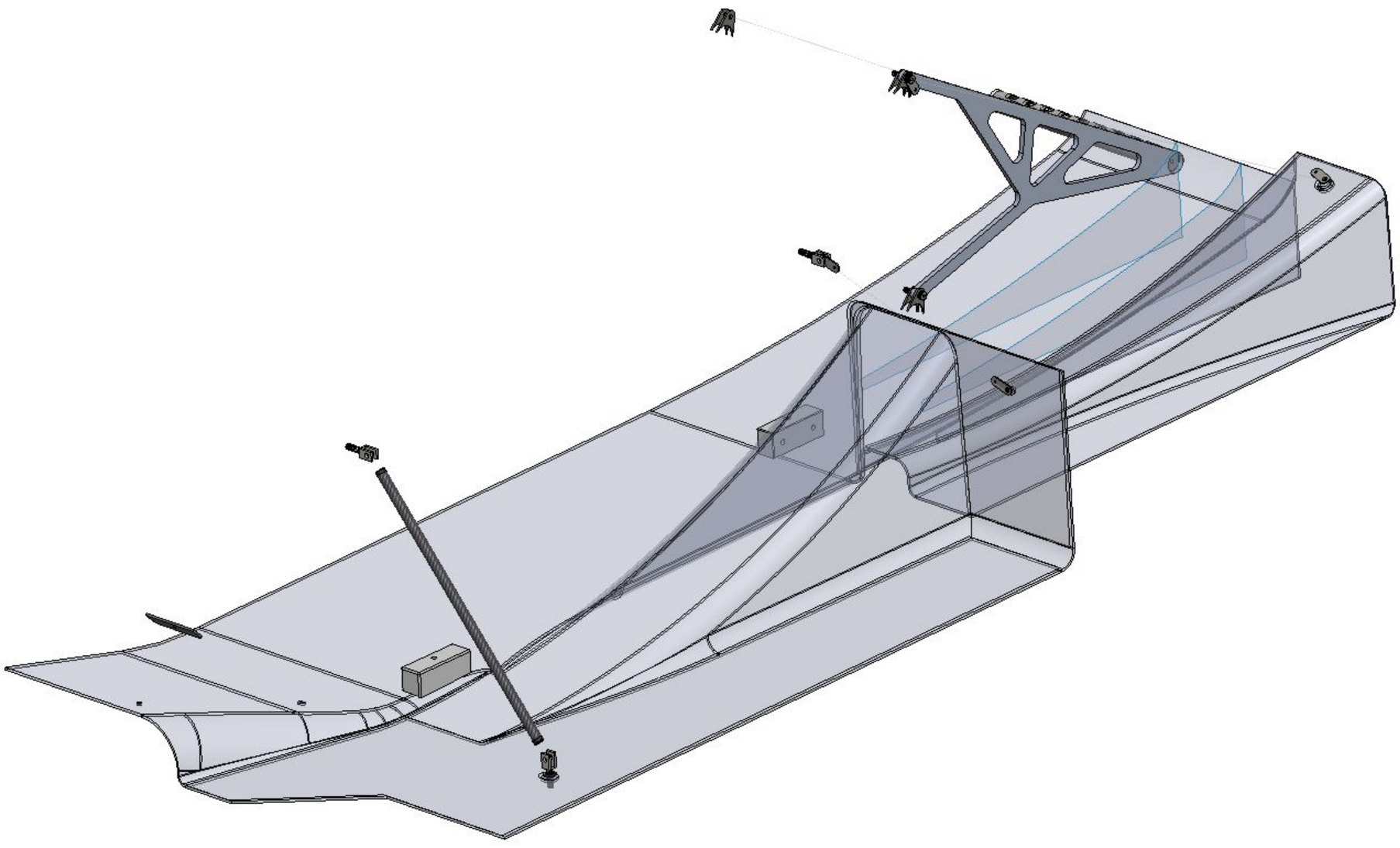
Mesh Independence Study

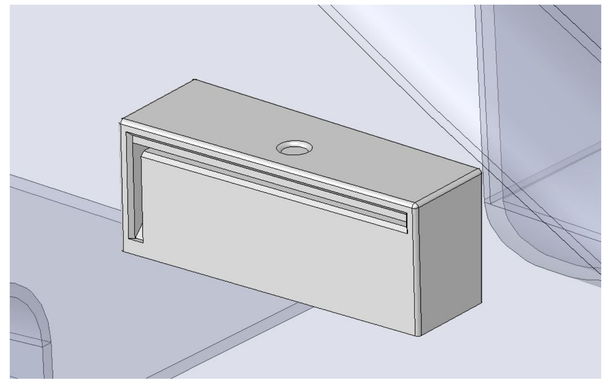
E1 U-Channels FOS (minimum on rear U-channel): 3.0

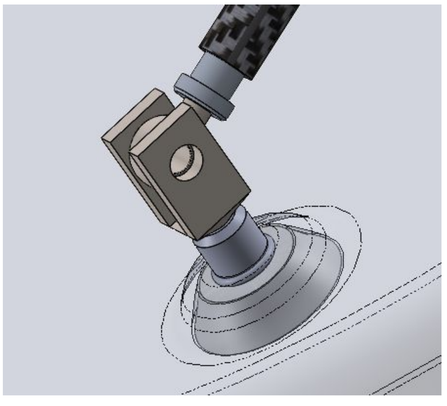


Undertray Mounting

UC Berkeley

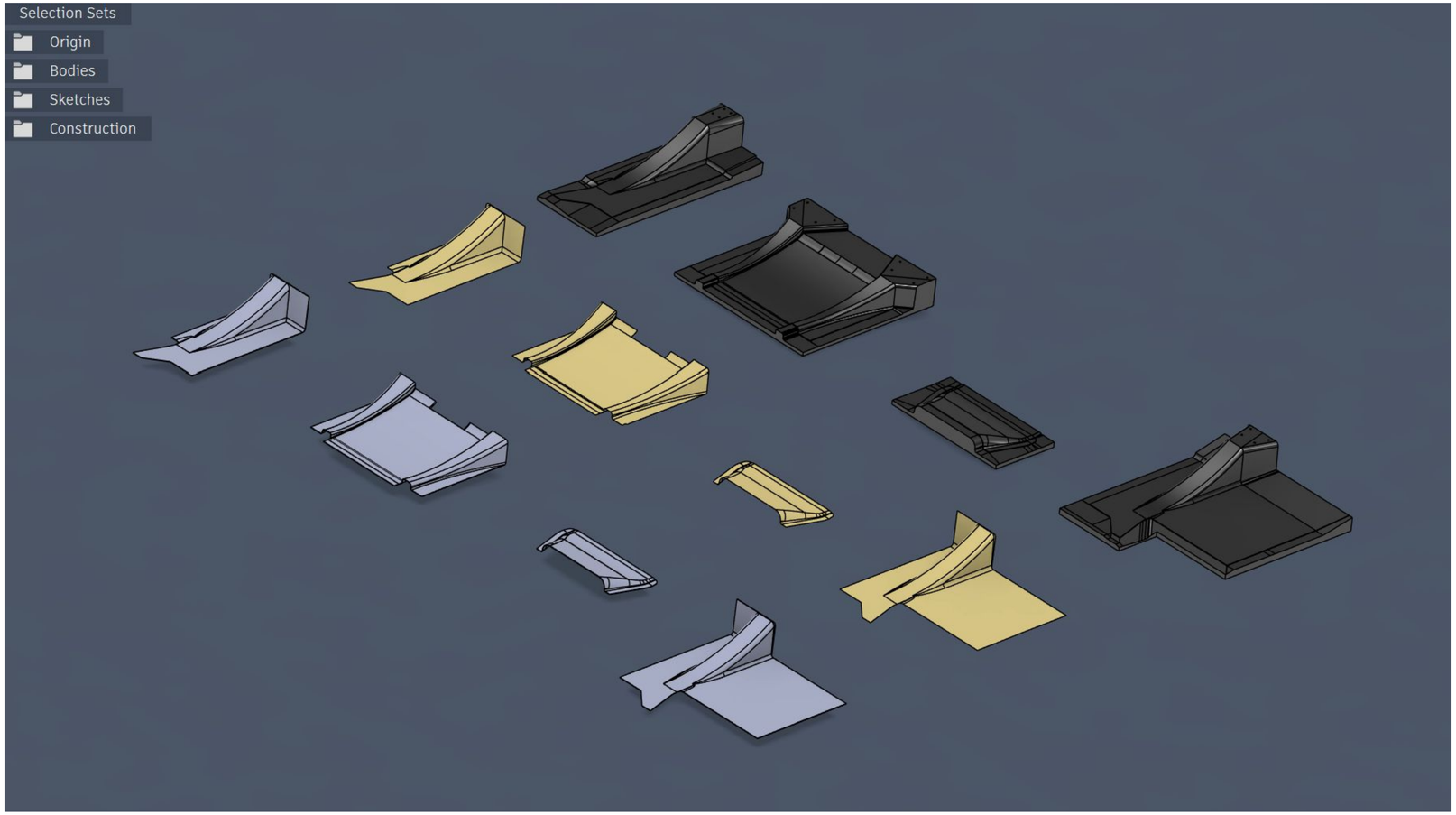






Undertray Split

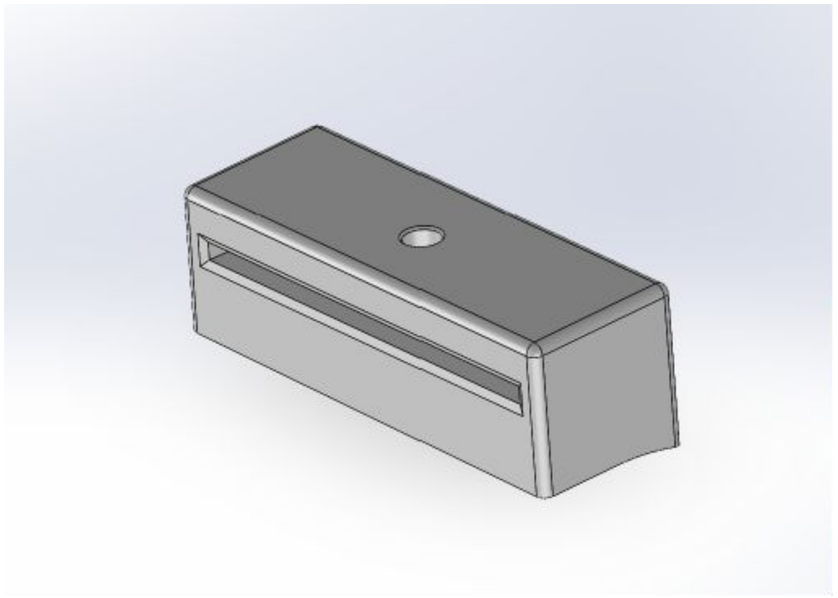
UC Berkeley

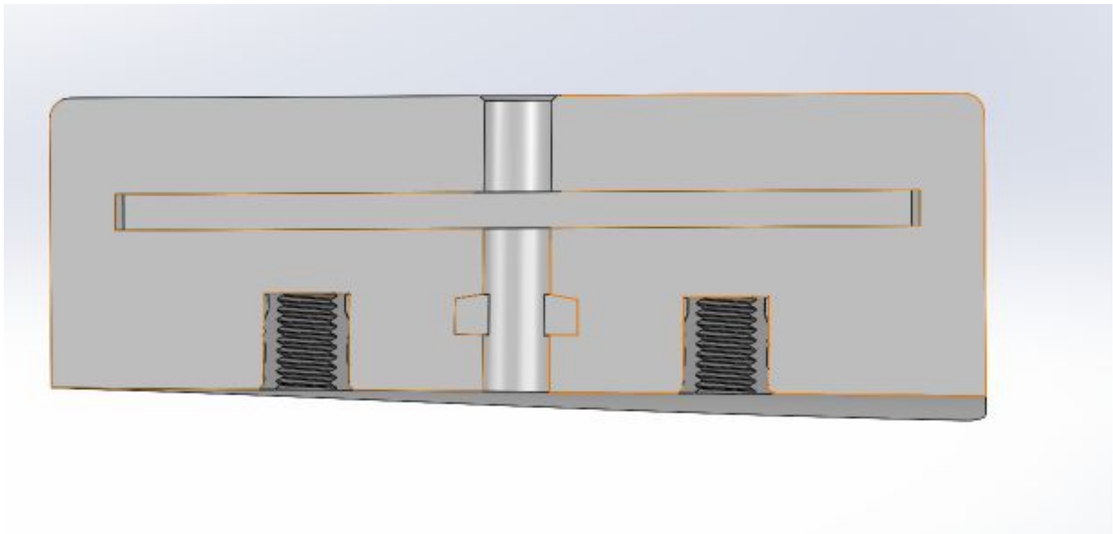


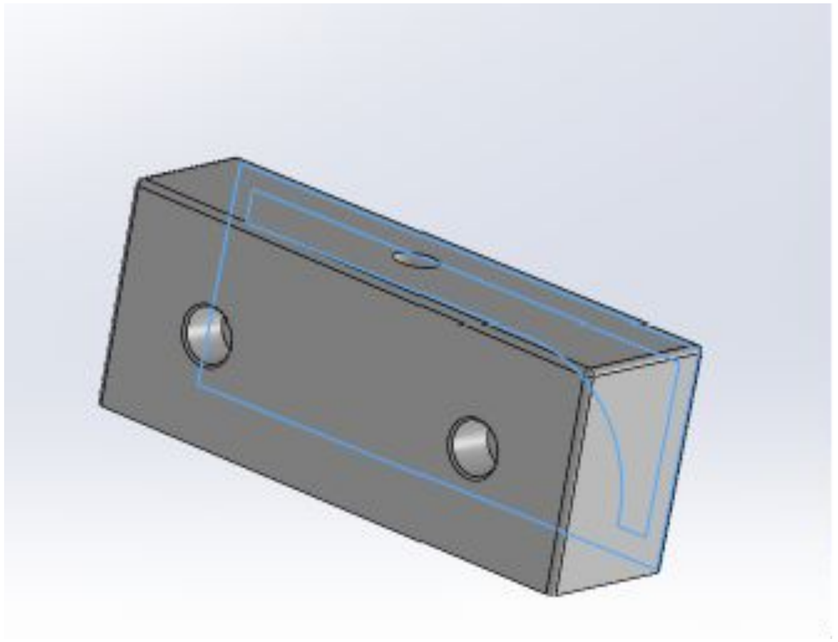
- The Undertray designed as a modular four piece assembly consisting of: the inlet, two side dams, and the diffuser to improve serviceability around the accumulator.
- The inlet and diffuser are permanently mounted to the car, while the side dams are designed for rapid removal to allow easier accumulator installation, removal, and maintenance.

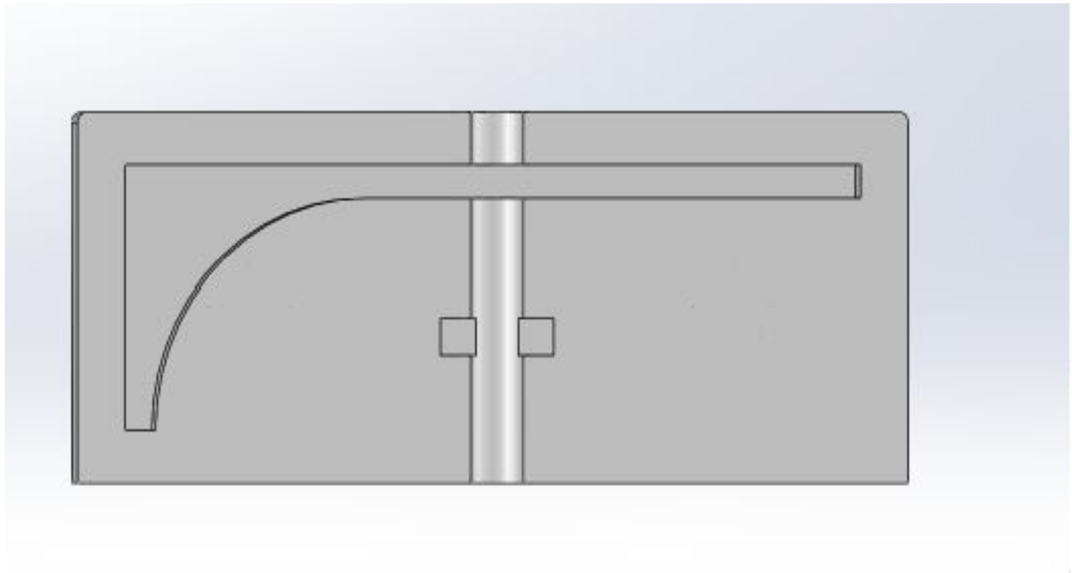
Undertray Mounting -

UC Berkeley









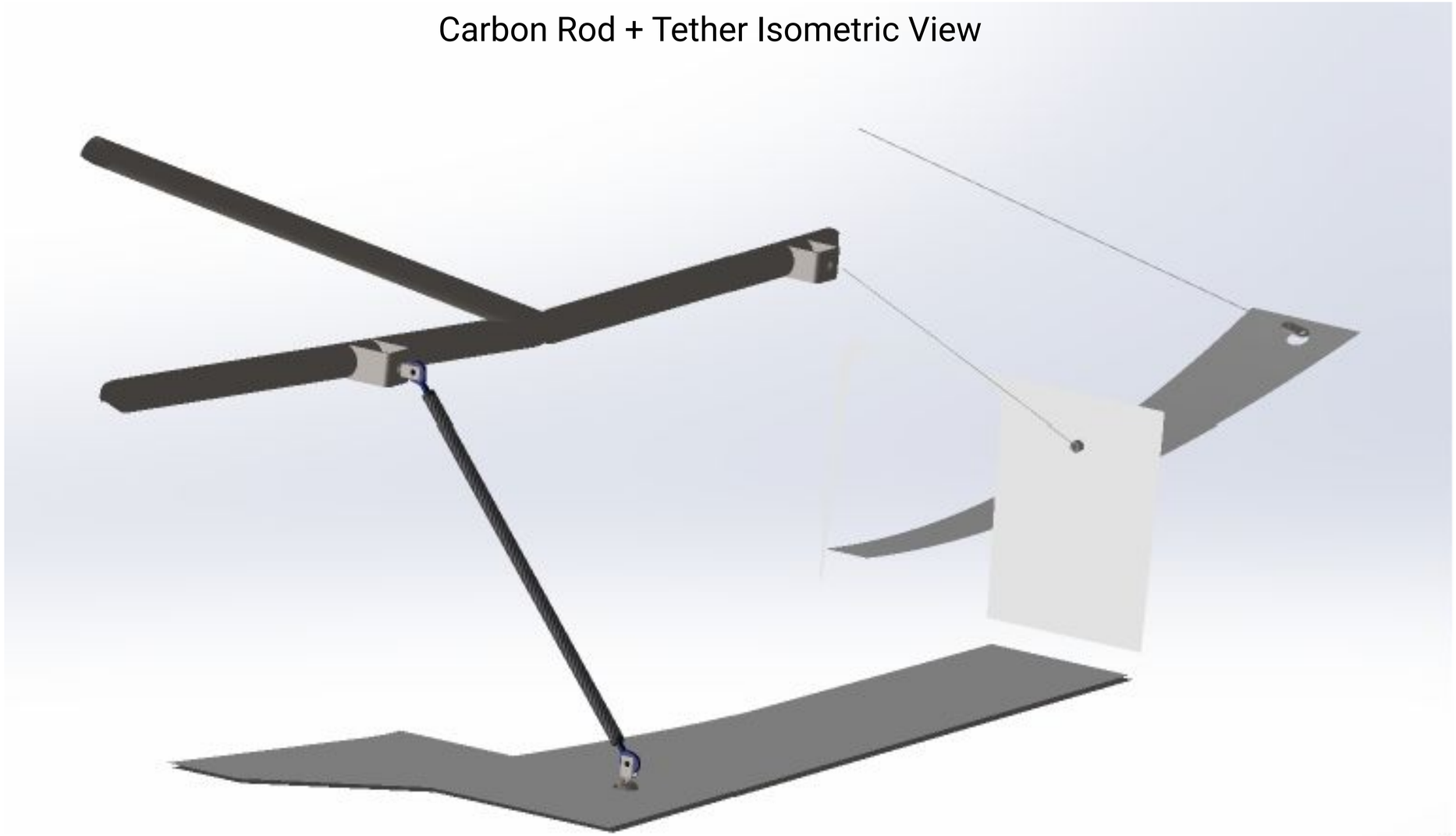
UT Box Mounts:

- UT box mounts 3D printed with guiding integrated slots
- Mounts constrained using locking pins, enabling fast assembly and disassembly during service.

Undertray Mounting -

UC Berkeley

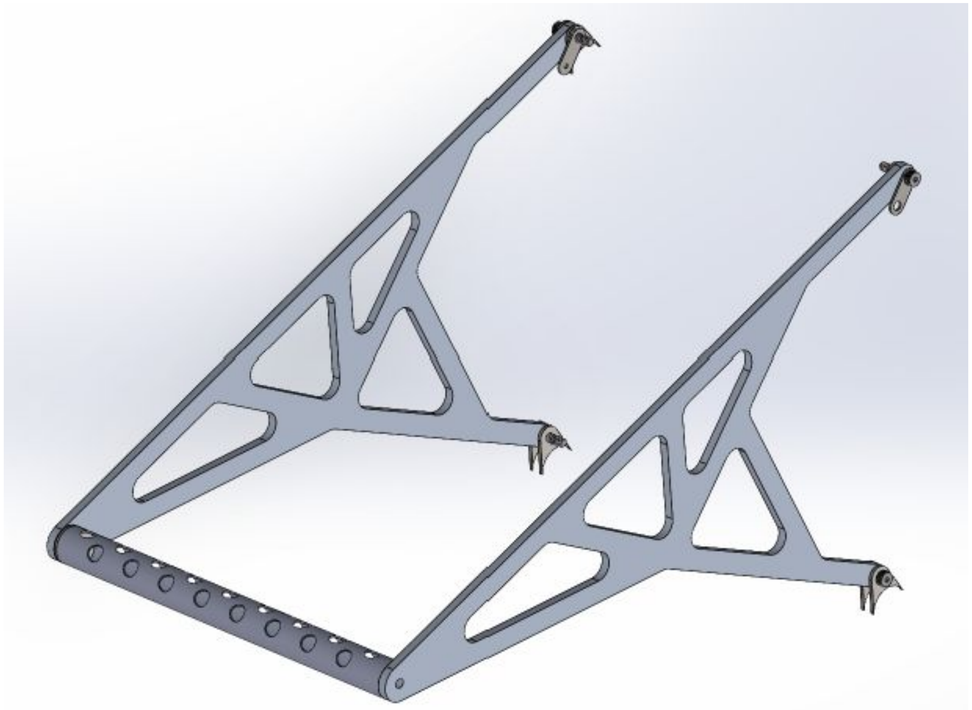
Carbon Rod + Tether Isometric View

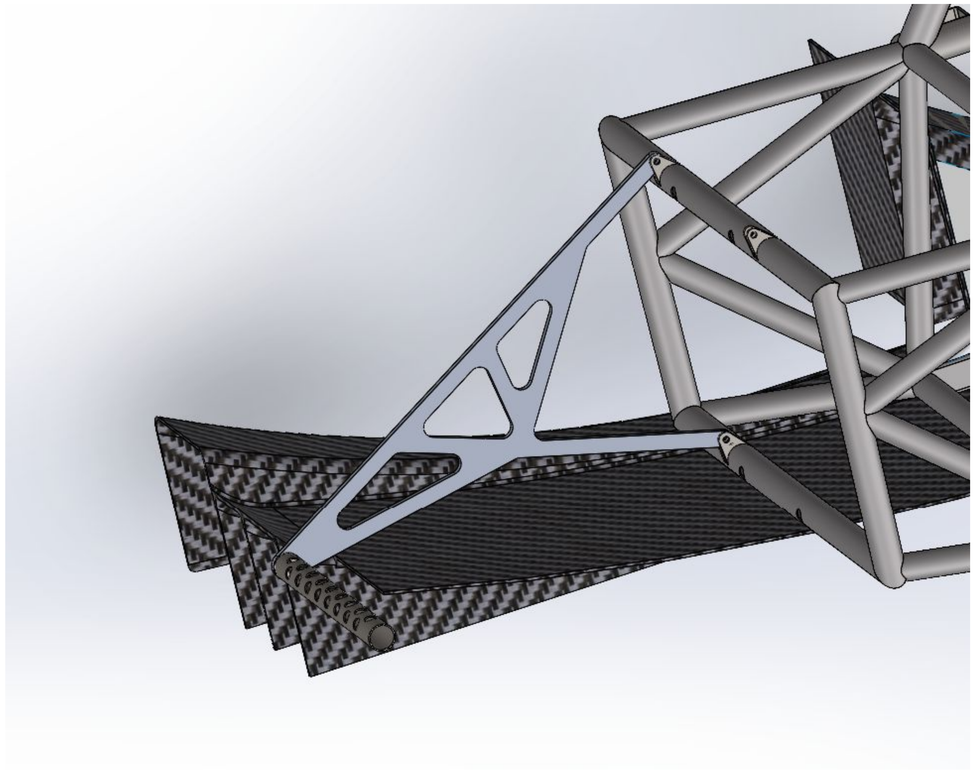


- Carbon rods were integrated into the undertray support system to reduce deflection and improve stiffness under aerodynamic loading.
- Tethers limit undertray droop and provide secondary support, helping the carbon rods carry and transfer downforce loads into the chassis.

Jacking Bar

UC Berkeley

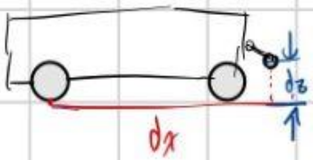




- 25.4mm, 0.095 ID, 4340 Steel
- Integrated with Diffuser for aerodynamic performance and structural purposes.
- During Jacking at 200mm above the floor:
  - Tire Height = 13.6mm with maximum heave and droop.
- Spotted Upper Half for reduced mass

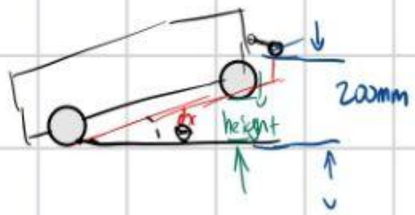
Jacking Bar - Heave / Droop Check

UC Berkeley



$dx = 196.5\,mm$

$dz = 134.7\,mm$

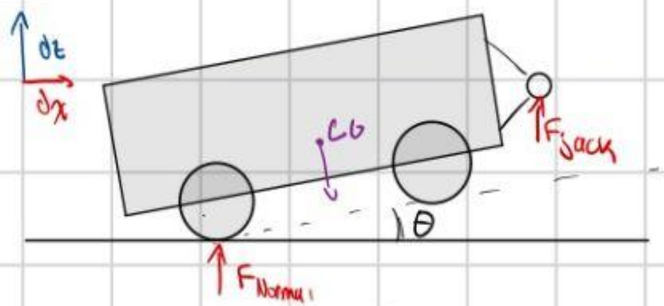


$\theta = \arcsin\left(\frac{200mm - 134.7mm}{196.5mm}\right) = 1.9°$

Wheelbase dist: 1550mm

Wheel distance Travel height: $1550mm \cdot \sin(\theta) = 51.46_{mm}$ > Wheel Droop ✓

FBD



Assume similar to SN4

CG height: 273mm

-Not using SW, not accurate, use PSS

-Assume about 50% CG, SN4 mass: 237kg, use 250kg to Overspec,

Try 46.51, Wheelbase

↳ i) 25-30 mm OD

ii) Minimum Clearance -> 75mm from ground -> Bottom of Tube

iii) When raised 200mm above floor, Wheels do not touch ground in full rebound.

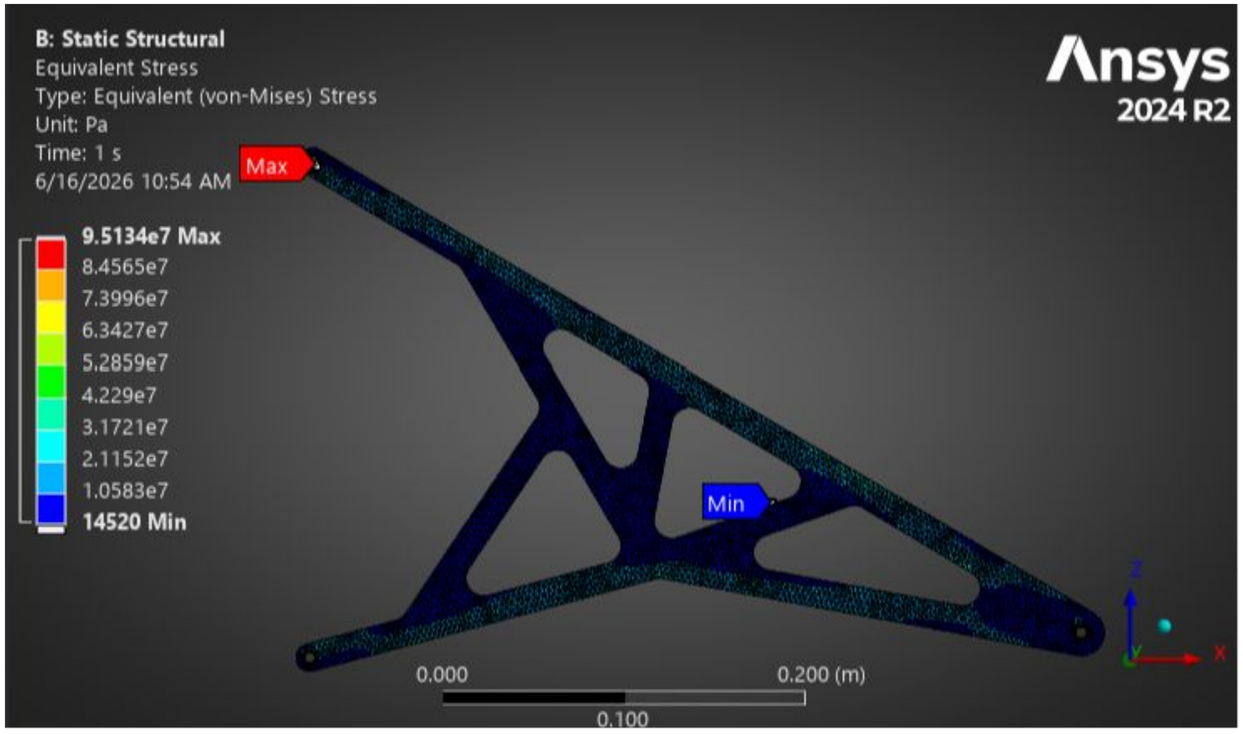
↳ Max Droop: 27.9mm

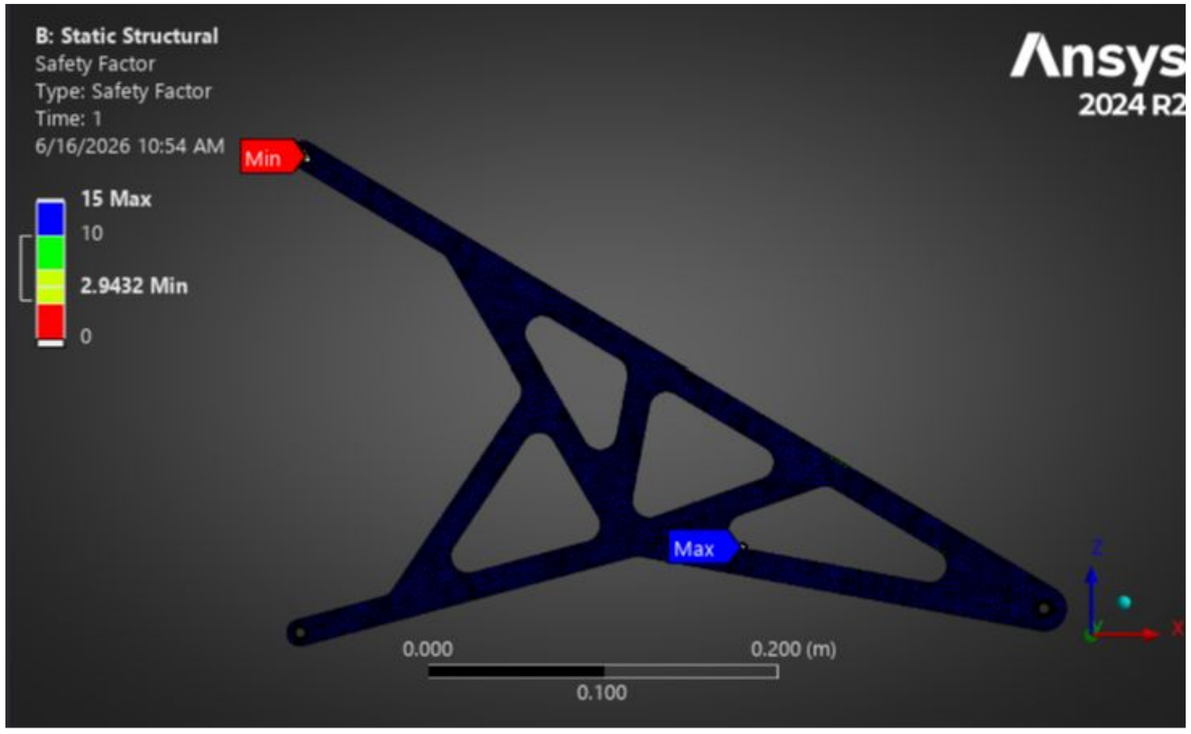
↳ Max Heave: 38.1mm

↳ Front Ride Rate (Per Wheel) - 16.99 $\frac{N}{mm}$

Jacking Bar Pylon FEA

UC Berkeley





- 0.25’’ 6061 Aluminum
- BC: Cylindrical Supports + Bearing Load
- Jacking Load: 650N
- Bearing Load : 15.5MPa
- Stress Singularity at Hole, estimated FOS at hole using 276 Mpa yield strength = 17.7

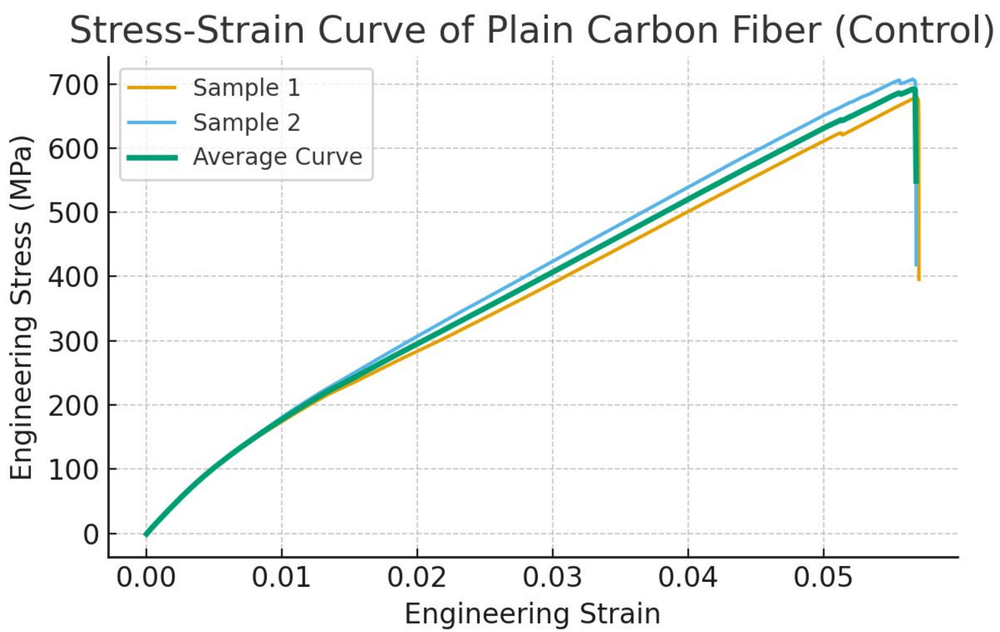
Composites - Tensile Testing

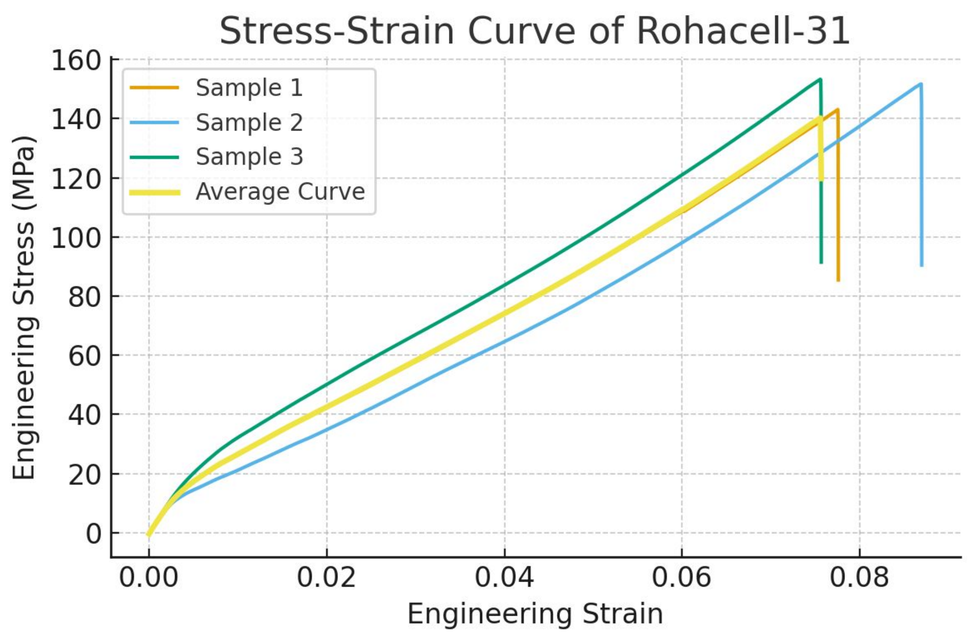
UC Berkeley

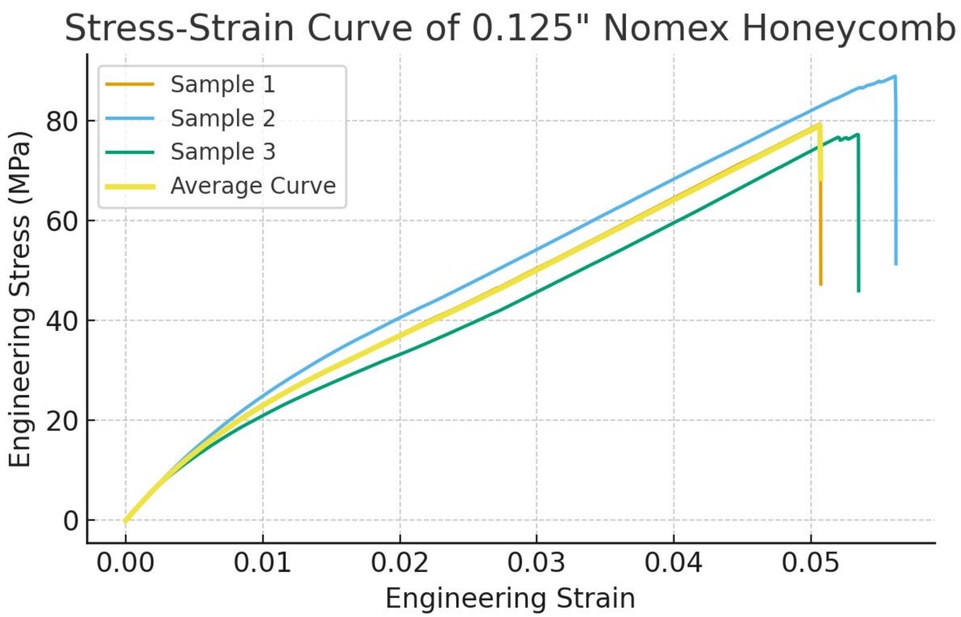
Sample	Thickness *(in.)*	Dogbone Mass *(g)*	Density *(g/cm^3)*	Young's Modulus *(MPa)*	Ultimate Tensile Strength *(MPa)*
Plain Carbon	0.033	3.9	1.295	18287.6	691.9
0.1" Nomex Honeycomb	0.14	8.36	0.654	2130.1	71.5
0.125" Nomex Honeycomb	0.17	8.5	0.548	2404.4	79.0
0.25" Nomex Honeycomb	0.296	9.67	0.358	1556.9	37.7
Rohacell-31	0.13	5.6	0.472	2313.9	140.2
3D Core PET 100	0.162	6.37	0.431	1872.2	87.9

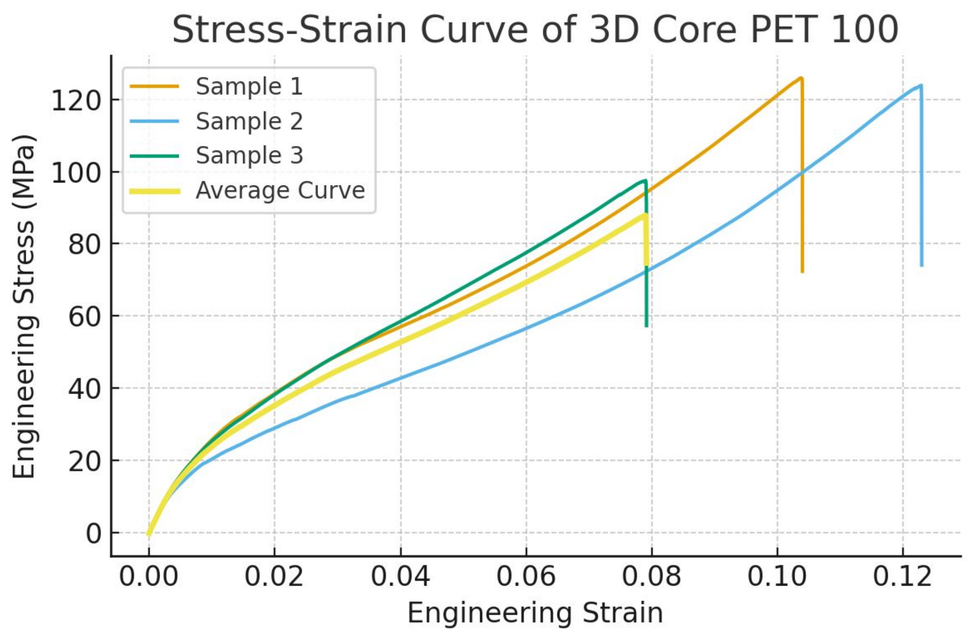
Composites - Tensile Testing

UC Berkeley



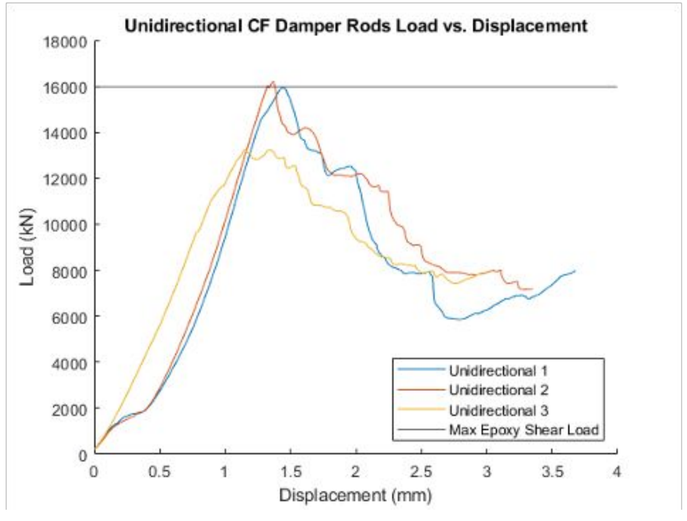


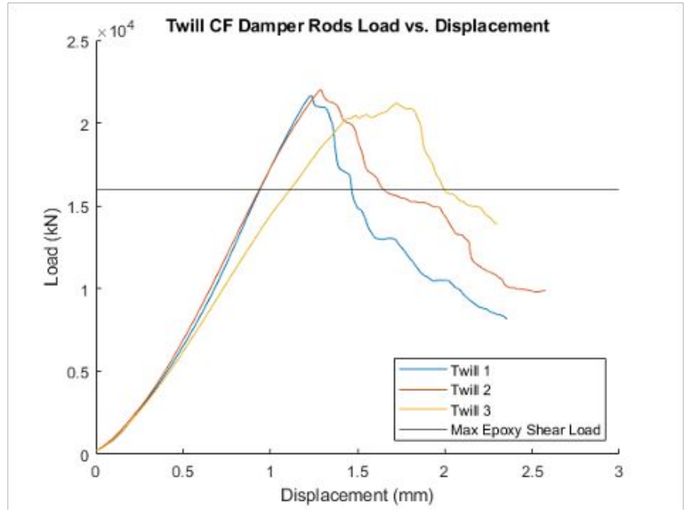


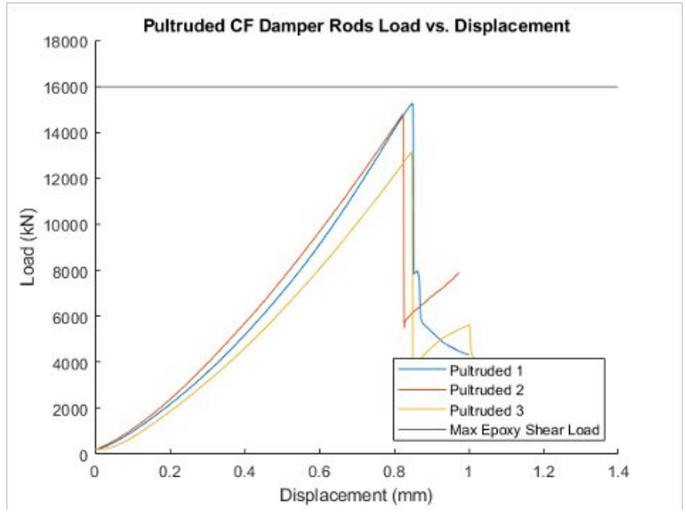


Carbon Fiber Compression Testing

UC Berkeley







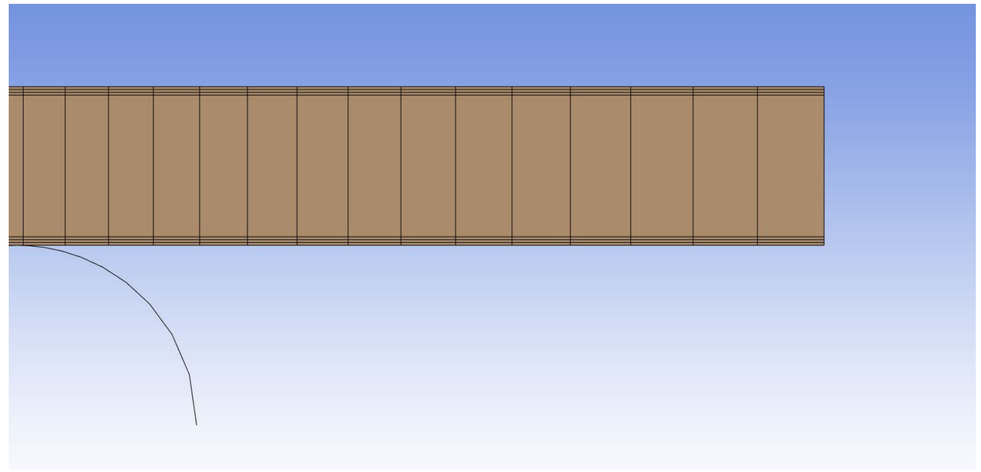
	Documented CF tube Elastic Modulus (GPa)	Tested Damper Rod Elastic Modulus - Averaged (GPa)	Calculated Min Failure Load (kN)	Tested Failure Load - Averaged (kN)
Unidirectional	102.0	22.1	53.3	14.9
Twill	95.8	24.4	63.1	21.2
Pultruded	131.0	27.3	117.6	13.1

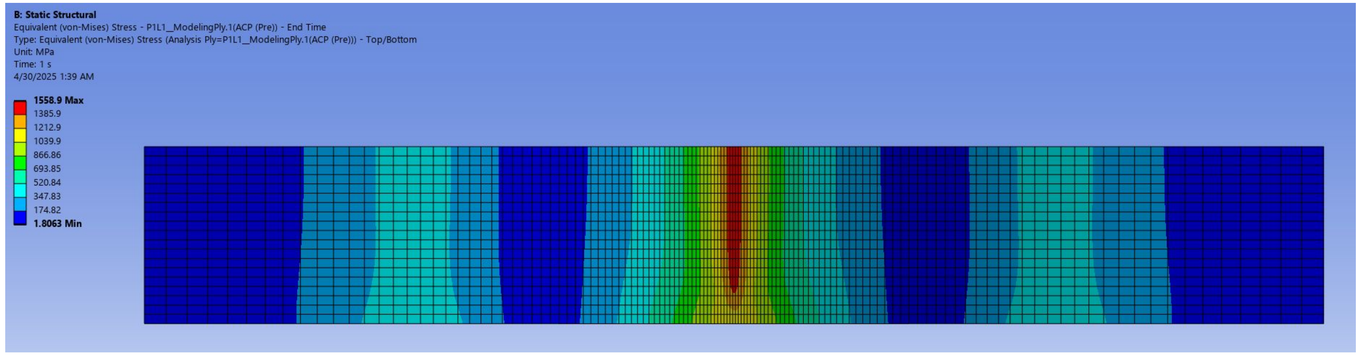
Calculated Epoxy Max Shear Load (kN)	16.2

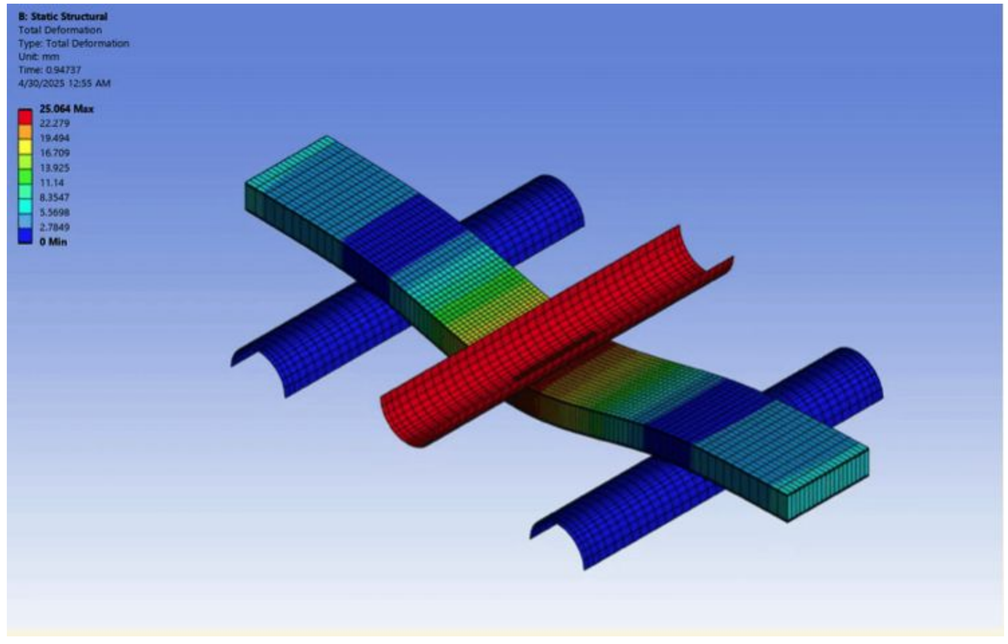
Composites - Sandwich FEA

UC Berkeley

- Derived lamina material properties from technical datasheets
- Modeled Ansys ACP to develop composite sandwich panel
- Recorded max stress, deflection
- Determined failure mode via ACP Post
    - Max Stress
    - Min Strain
    - Tsai-Hill
    - Tsai-Wu





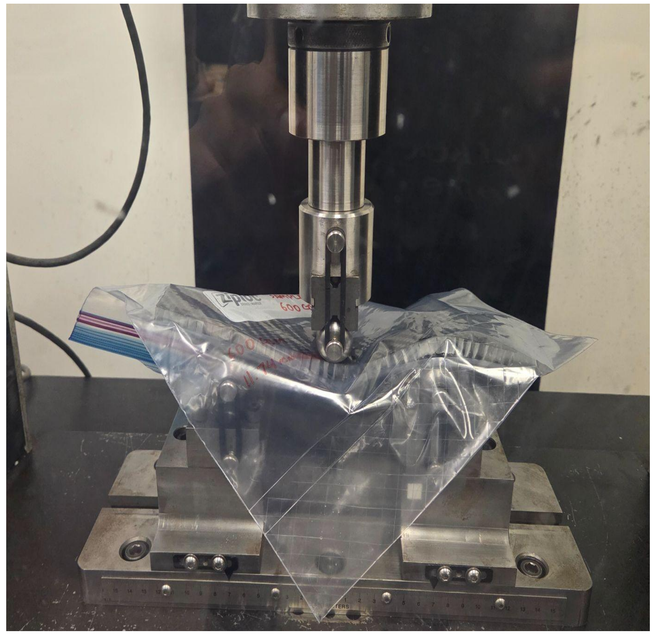


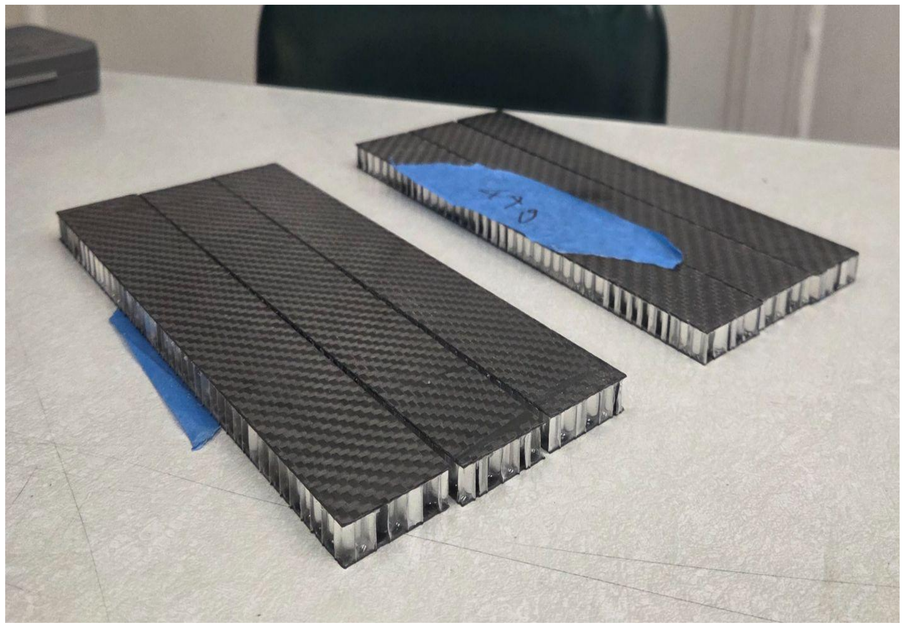
Sandwich Panel 3-point Bending

Sandwich Panel 3-point Bending

UC Berkeley

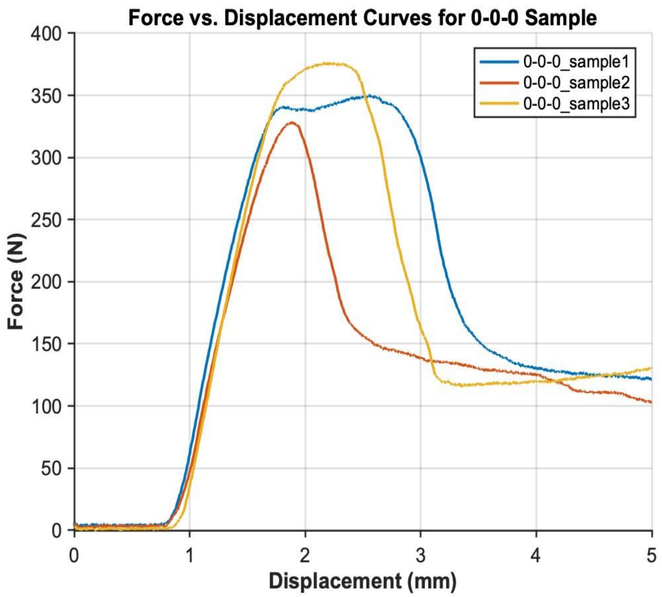
- **Dimensions:** 254 mm x 38.10 mm x 11.50 mm
- **Span Length:** 14.0 mm
- **Displacement Rate:** 4.0 mm/s
- 3-point bending test on Instron
- Most common failure mode was delamination of layup on honeycomb (matrix resin shears)

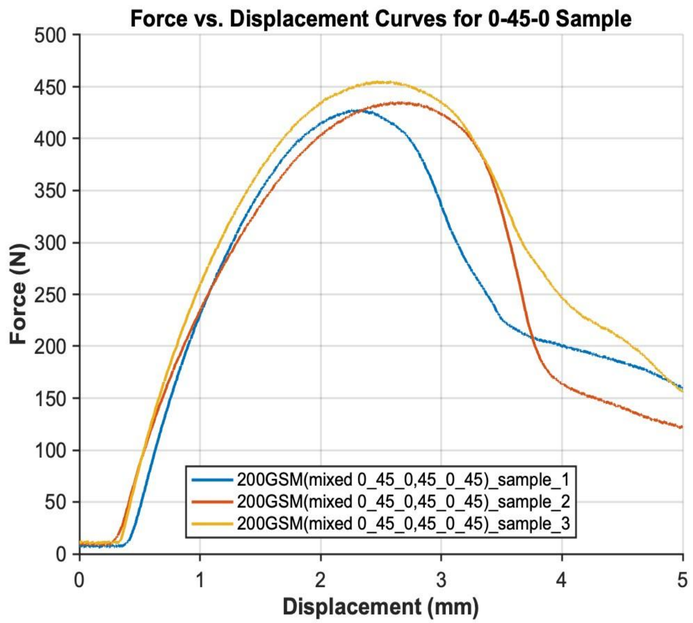




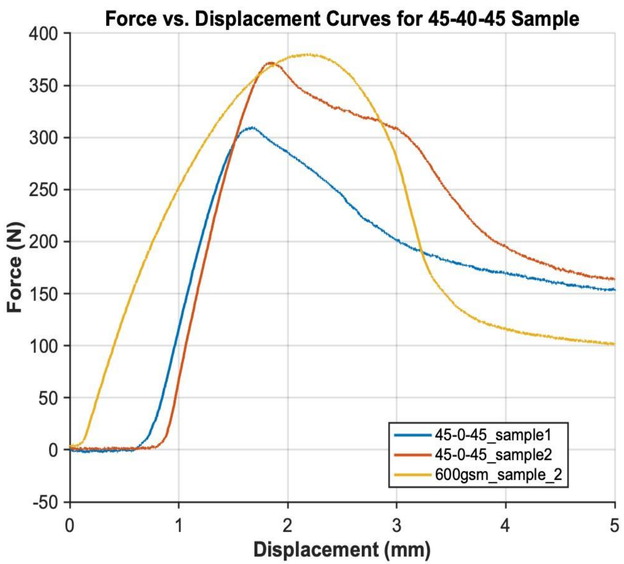
Sandwich Panel 3-point Bending

UC Berkeley





Top Layer Orientation	Bottom Layer Orientation	Flexural Modulus (MPa)
0-0-0	0-0-0	174.48
0-45-0	0-45-0	174.05
45-0-45	45-0-45	163.51
0-45-0	45-0-45	168.78



- From these results, **0-0-0 is the best lamina orientation** because of the higher flexural modulus, but main contributor is the fiber orientation on the surface plies.

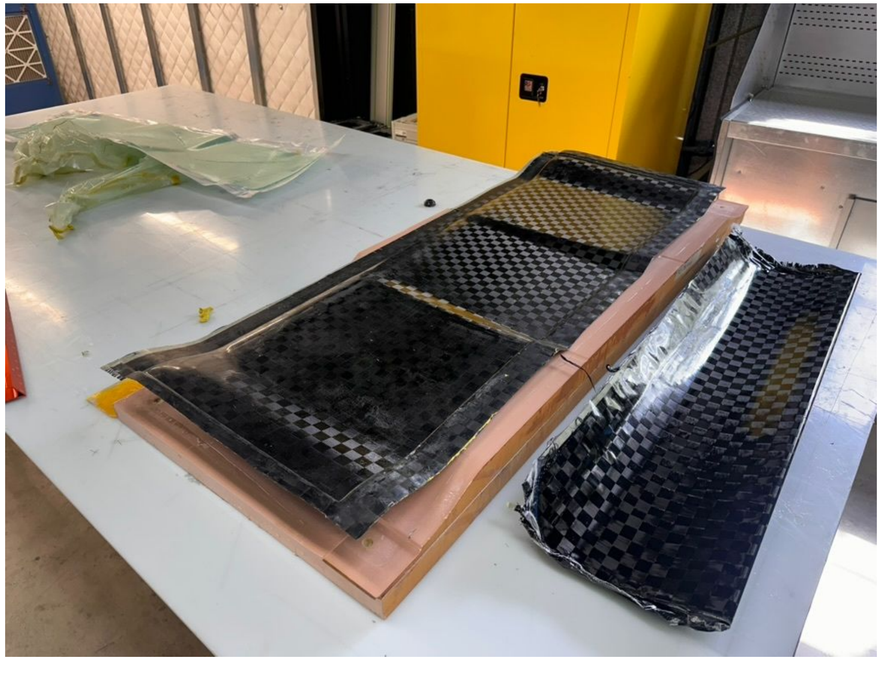
Things to Consider:

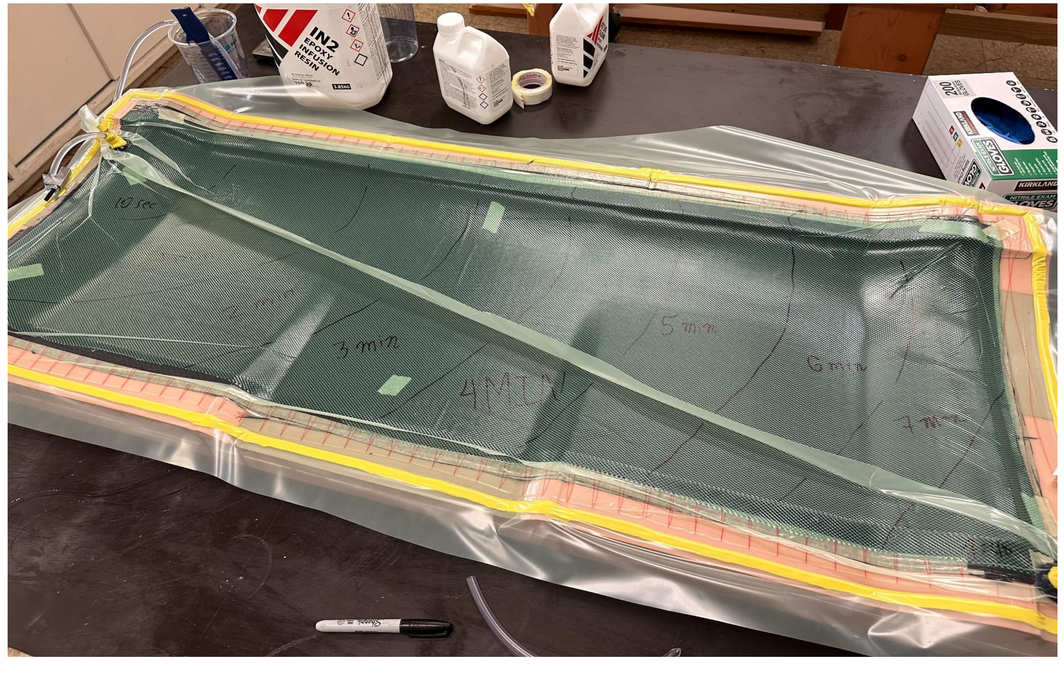
- Manufacturing defects and inconsistency
- Air voids or dry spots from poor resin infusion
- Lack of epoxy on both surfaces

Manufacturing

UC Berkeley

- Undertray, Bodywork, & Curved Wing Elements
  - Molds machined out of PBLT tooling board using CNC router, sealed using Easy Composites XCR Coating Resin to achieve seal and optimal surface finish.
  - Carbon Fiber part is created using resin infusion layup
  - Parts are dremeled to exact size using mold datums.

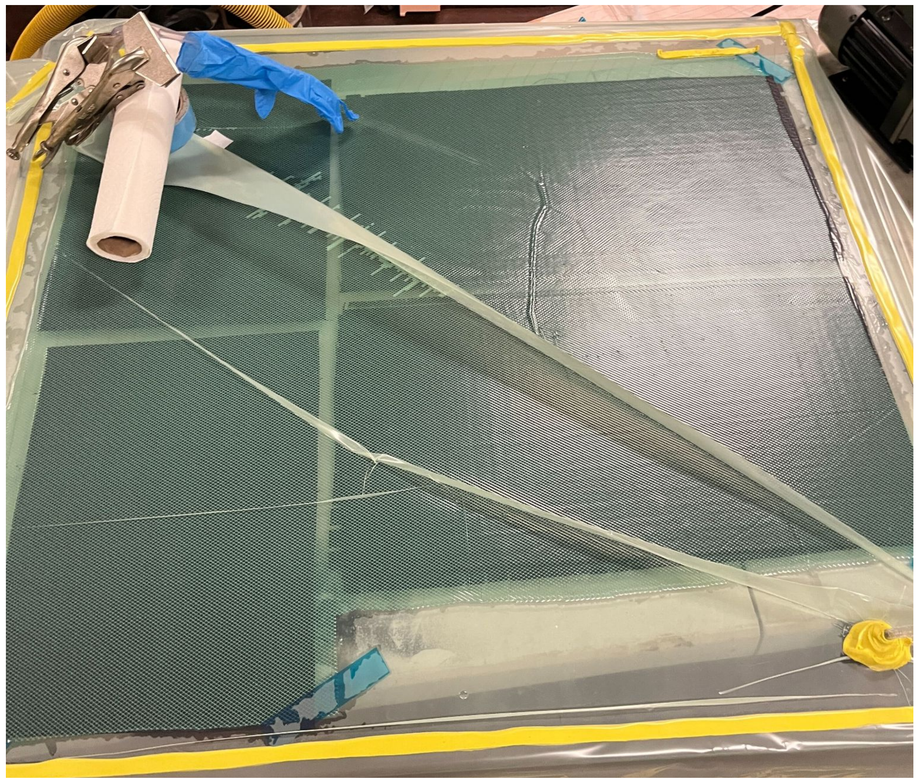


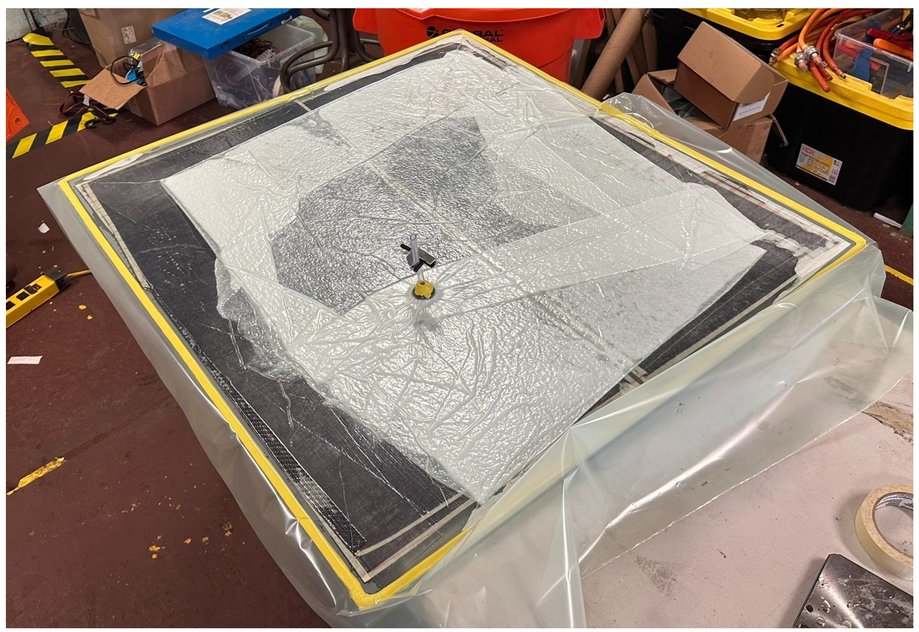


Manufacturing

UC Berkeley

- Flat panels
  - Combination of resin infusion and wet layups completed on glass panes to achieve excellent weight, surface finish, and strength.
  - Core materials are glued between flat panels using additional resin, strengthening final part.
  - Final parts are cut to size using waterjet or CNC Router.

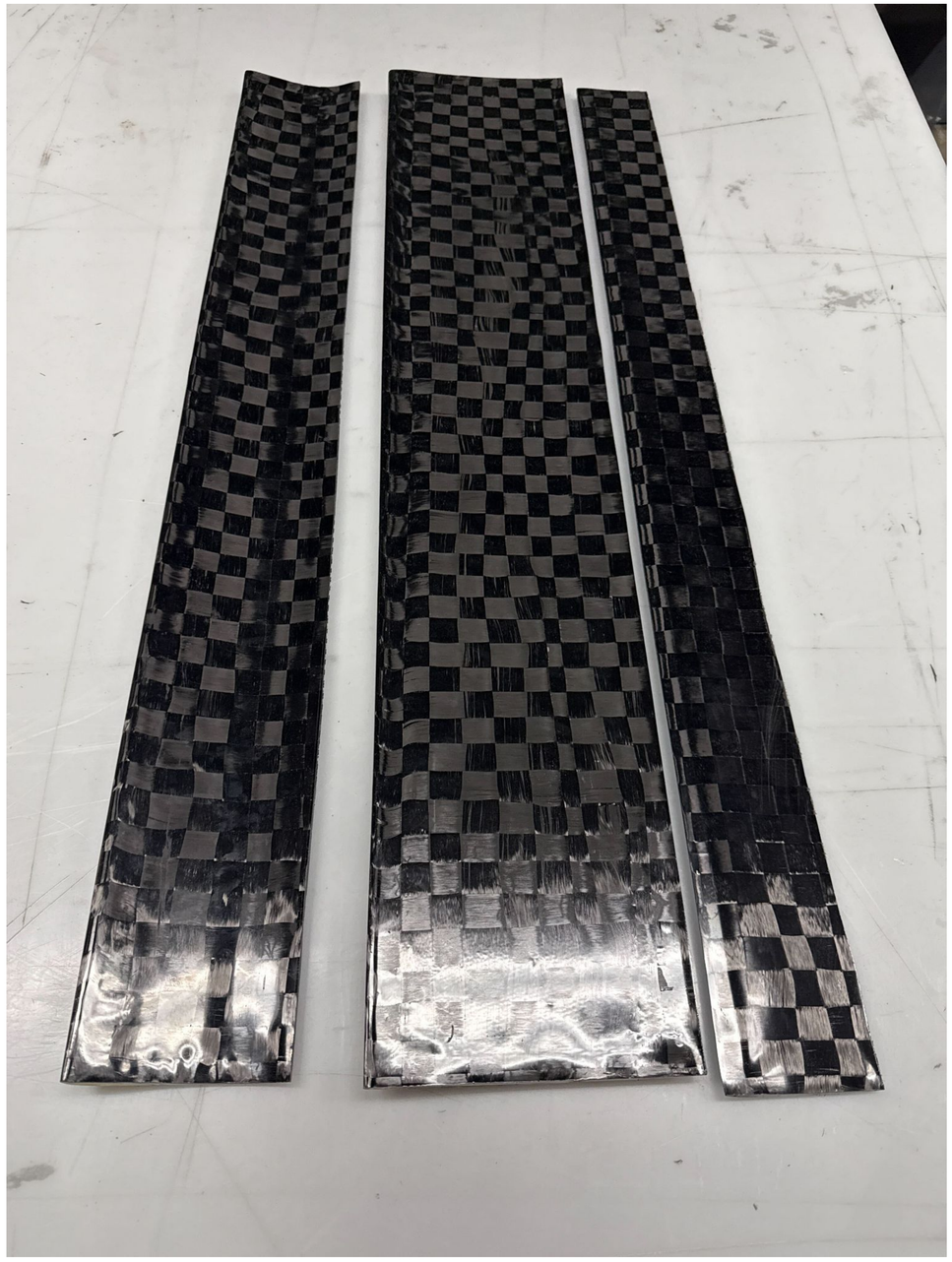


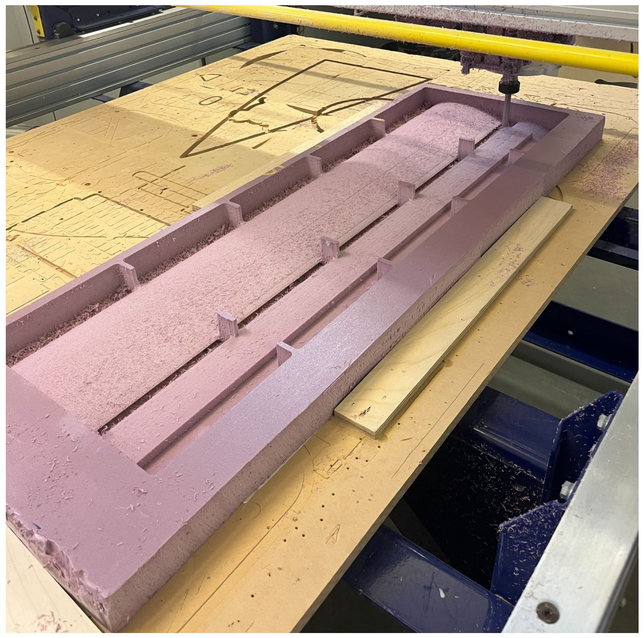


Manufacturing

UC Berkeley

- Straight Wing Elements
  - Foam Cores machined out of XPS foam cores, endplate ribs are glued onto each end.
  - Carbon Fiber is coated with resin using wet layup, then wrapped around foam core, covered in mylar, and placed under vacuum to achieve high quality surface finish and minimize weight.
  - Decreases manufacturing time per element by >6 hours by reducing machining and number of layups.





Testing & Validation Plans

UC Berkeley

- Drag
  - Coastdown testing
- Flow attachment
  - Tufts
  - Flow-vis: Kerosene + Chalk
- Downforce
  - Linear Potentiometer: non-linear compression rate, coordinate with Dynamics - translation of force to non-linear spring compression
  - Aero Rake: array of pitot tubes to measure airflow velocity, pressure, etc. to validate CFD results

Testing Plan

UC Berkeley

1. Get full aero map on heave, dive, roll, yaw
2. Validate on-vehicle aero components dimensions & position against CAD, coordinate with Dynamics alignment jig
3. Obtain real life aero metrics through small-scaled & full-scale wind tunnel tests & dynamic test
   a. Downforce vs. speed, Drag vs. speed, Cl, Cd & COP
4. Validate flow health through photo/video capture during dynamic test
   a. Flow visualization (chalk fluid)
   b. Flow attachment (tuft)

Fundamental Equations

UC Berkeley

$$F_D = \frac{1}{2}\rho v^2 C_D A$$

$$L = C_l \times \frac{1}{2} \times \rho \times V^2 \times A$$